Can the Public Sector Deliver Adaptive Learning Technology at Scale? Experimental Evidence from India*

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Abstract

The Government of Andhra Pradesh has scaled a personalized adaptive learning (PAL) math program to 135,000 students across 444 government middle schools in 17 districts. To estimate causal impacts, we embed a cluster-randomized trial in 120 additional schools, randomly assigning 60 to receive the same PAL program and 60 to control. Two years of PAL exposure raise endline math ability by 0.39 SD—equivalent to 1.8 additional years of learning in AP government schools. Gains are larger for girls (+0.46 vs. +0.33 SD), for the lowest-baseline quintile (+0.48 SD), and in smaller sections where tablet scarcity does not bind. Using the 30-tablet-per-school cap as an instrument for per-student usage, we estimate a causal return of 0.028 SD per hour of PAL. At $52 per student over two years, PAL delivers 0.94 LAYS per $100, within the range reported for adaptive-learning programs in other settings. At program scale, 30 tablets are shared across three subjects and four grades, and program students average 5.5 hours of math PAL. Linearly extrapolating the IV caps the program-scale effect at $+0.15$ SD; the within-treatment dose-response (descriptive, not causally identified) is flatter at low usage, suggesting smaller gains. We offer this dose-response as input to the government's allocation of scarce tablet-time across subjects and students.

JEL Codes: I21, I25, O15, C93.

Keywords: Personalized Adaptive Learning, EdTech, Scale, Government, India, Math Education.

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I. Introduction

A fundamental challenge in education is that students arrive at each grade with widely different skill levels but receive a single, common lesson. This dispersion is especially wide in low- and middle-income school systems: in rural Andhra Pradesh, the setting of this study, only 48% of grade-8 students can perform basic division — a grade-3 skill — even as their assigned curriculum has moved on to algebra and geometry (ASER Centre, 2024). A teacher cannot resolve this mismatch by varying instruction student-by-student in real time, and the same lesson is too easy for some students and too hard for most. One response is to replace part of regular math instruction with structured supervised practice on a tablet-based learning platform that gates content to each student's diagnosed level. Such programs — often grouped under the label personalised adaptive learning (PAL) or computer-aided learning (CAL) — have produced large learning gains in NGO-run efficacy trials (Muralidharan and Ganimian, 2019; de Barros and Ganimian, 2024). This raises the question central to our paper: whether such programs, whose largest measured effects have come from dedicated NGO-run trials, can deliver comparable gains when funded, staffed, and operated by a state government using its existing schools and teachers.

We evaluate this in the Government of Andhra Pradesh's PAL program, designed and operated by a private educational technology firm, ConveGenius, under state contract. The state has rolled the program out to 444 government middle schools serving approximately 135,000 students in 17 of its 26 districts, and in early 2025 launched a separate $\sim$700-school expansion of high-intensity adaptive learning into government residential schools. To estimate causal impacts, we add 120 additional government middle schools to the program and embed a cluster-randomized trial within them, stratifying across 8 districts and 21 mandal-cluster groupings and randomly assigning 60 to receive the same PAL program and 60 to control. Treatment schools received a 30-tablet computer lab and were required to replace two of their eight weekly math classes with PAL sessions across two academic years (2023–24 and 2024–25); a randomly selected 80-school subset was reassessed a year later (February–March 2026) to test the persistence of effects. Implementation was funded through state and central (Samagra Shiksha) government education budgets and operated by the AP government's regular school administration. Our primary endline, administered on tablets to 14,215 students at the end of the second treatment year (February–March 2025), is a 72-item IRT-scored math assessment spanning grades 2 through 9.

This paper contributes three causal estimates to the literature on adaptive learning at government scale:

an intent-to-treat (ITT) effect identified by school-level randomization, a per-hour return to PAL usage identified using the 30-tablet-per-school cap as an instrument for per-student tablet-time, and a within-school dose-response curve linking cumulative usage to endline ability. The first two are causally identified; the third is descriptive within the treatment arm and provides the curve we use to extrapolate from the trial's high-dose treatment students to the program's operating scale.

The ITT effect on endline math ability is $+0.39$ SD ($p < 0.001$), equivalent to approximately 1.8 additional years of learning relative to the cross-grade progression observed in control schools, or 0.49 Learning-Adjusted Years of Schooling (LAYS) at the Filmer–Angrist benchmark of 0.80 SD per year (Angrist et al., 2020; Filmer et al., 2018).[1] At $52 per student over the two-year intervention, this translates to 0.94 LAYS per $100 — within the range reported for adaptive-learning interventions across all contexts (0.24–2.66) and exceeding government-delivered Teaching at the Right Level in India (0.82). The effect is robust across three control specifications, survives Bonferroni and Benjamini–Hochberg multiple-testing corrections on the primary subgroup interaction, and is confirmed by randomization inference.

Learning gains are heterogeneous along three margins. Girls gain $+0.46$ SD against $+0.33$ SD for boys ($p = 0.001$), a difference that survives multiple-testing corrections and reflects differences in usage quantity rather than per-hour productivity: girls accumulate approximately two additional hours of PAL over two years, concentrated in active practice (questions) rather than passive content (videos). Effects decline across the baseline ability distribution, with students in the lowest pre-PAL quintile gaining $+0.48$ SD and those in the highest gaining $+0.31$ SD. Effects are uniformly large (0.60–0.65 SD) in small schools where section enrollment is below the 30-tablet cap and exhibit a declining gradient across grades in larger schools where sections exceed the cap — a mechanism: students in under-supplied sections receive less PAL exposure.

Our 2SLS estimate of the causal return to an hour of PAL usage is 0.028 SD ($p = 0.005$), approximately three times the OLS estimate — complier selection toward the steep region of the within-school dose-response. The within-treatment shape is flat below 10 hours of two-year usage, steep between 10 and 30, and flattening above 30. RCT treatment students accumulate 34 hours of math PAL on average, squarely in the steep region, while students in the state's 444-school rollout accumulate 5.5 hours, below the activation region. Linearly extrapolating the IV to program-scale usage yields an upper bound of $+0.15$ SD on the program-scale

[1] Translating into the ASER 2024 classification used to characterise foundational-skill deficits in rural Andhra Pradesh, the share of pooled grade 7–8 students with full grade-3 arithmetic mastery (subtraction and division) rises from 54.8% in control schools to 61.5% in treatment schools, a 6.7-percentage-point gain larger in magnitude than the 4-percentage-point decline ASER documents for AP grade 8 between 2022 and 2024 (52% $\rightarrow$ 48%). See Appendix A.

effect; integrating the within-treatment shape over program usage projects a near-zero effect (descriptive, not causally identified). Both readings place program-scale effects well below the RCT's 0.39 SD and point toward reallocation of tablet-time as the relevant policy margin.

Our Year-3 follow-up on a randomly selected 80-school subset of the RCT, assessed in February–March 2026, finds the Year-2 treatment-control premium attenuating by approximately half over the third academic year, with treatment students still above controls on raw percent-correct metrics for identical items. The attenuation is concentrated in schools where the program ceased to be actively delivered: a substantial subset of treatment schools recorded effectively zero PAL hours in 2025–26, coinciding with the AP government's redirection of state monitoring bandwidth toward the residential expansion and the original 444-school rollout's growth to approximately 1,200 schools. In schools that continued delivery, new Grade-6 entrants in the original treatment arm exhibit a within-school dose-response slope of $+0.030$ SD per hour — matching the Year-2 IV LATE of $+0.028$ SD per hour almost exactly, and indicating that the per-hour return is preserved in a fresh cohort one year beyond the experimental window.

What mechanism produces our $+0.39$ SD effect? Our platform is best described as structured curriculum-aligned computer-assisted learning with two algorithmic levers on remediation: a per-chapter diagnostic gate, and a cross-chapter taper keyed to completion history. Both levers adapt to recent student behaviour, not to measured ability. Pipeline metrics are flat across baseline ability quintiles — $r = 0.11$ between the platform's diagnostic and within-school baseline percentile; 9.7 vs. 9.3 below-grade modules per chapter for the lowest and highest baseline quintiles. Whatever produces the effect, it is not real-time per-student personalisation; the platform does not deliver that.

Three components of the intervention plausibly drive the effect. The first is structured math practice time: treatment students accumulate 34 hours of PAL against zero in controls, and our IV LATE for per-hour return is $+0.028$ SD ($p = 0.005$). The second is the implementation design that produces those hours: treatment schools dedicated their 30-tablet stock to math and were required to replace two of their eight weekly math periods with PAL, while the parallel 444-school rollout, sharing the same 30-tablet stock across three subjects and four grades on a discretionary timetable, delivers 5.5 hours per student against the RCT's 34. The third is the platform's diagnostic-based dosage of below-grade content: 27% of module-level events in our treatment schools sit below the student's enrolled grade, adding a bottom-loaded gain reaching weaker students — consistent with the heterogeneity we document. The Year-3 evidence indicates that sustaining this design depends on continued state monitoring: when central monitoring capacity shifted to other rollouts, a substantial subset of treatment schools collapsed to zero usage.

Each of the three components has parallel evidence in recent in-school adaptive-learning evaluations. de Barros and Ganimian (2024) bound the contribution of granular personalisation directly: in 15 model schools in Rajasthan, holding the Mindspark platform fixed and varying only access to below-grade exercises produces a precisely-estimated null average effect after nine months ($+0.05\sigma$, 95% CI $[-0.02, +0.13]$), with a $+0.22\sigma$ gain confined to the bottom quartile of baseline performers. Since our platform's adaptation is coarser than Mindspark's, this bound applies a fortiori. Muralidharan and Singh (2025) document a stable per-hour dose-response in their Rajasthan in-school Mindspark evaluation that matches the per-hour return we estimate; in their Year 3, when lab-in-charge staffing was halved but state-team monitoring of dashboards continued, usage held — the converse of our Year-3 collapse, in which monitoring was withdrawn rather than maintained. Oreopoulos et al. (2026) isolate the value of structured in-school implementation: adding a dedicated lab-in-charge to Uttar Pradesh boarding schools that already had Khan Academy — a non-adaptive platform — raised math achievement by $+0.44\sigma$ in 31 weeks, attributed to organisational structure rather than content.

Beyond the adaptive-learning literature, this paper contributes to a body of work on state capacity and technology-enabled government service delivery in low- and middle-income countries (Muralidharan et al., 2016, 2021; Okunogbe and Pouliquen, 2022; Beg, 2022). That literature has documented technology's contribution to procurement, disbursement, and identity verification — the administrative back-office of government. We extend it to classroom delivery: the operational machinery the AP government built — device infrastructure, real-time monitoring dashboards, structured teacher and headmaster training, and a school–district–state escalation chain for technical support — is the technology stack required to deliver an adaptive curriculum into each instructional period.

The remainder of the paper is organised as follows. Section II describes the Andhra Pradesh education system and the 30-tablet allocation rule that underpins both the program and our identification strategy. Section III describes the design, instrument, and estimation. Section IV documents PAL usage, estimates the causal return to an hour of usage, and characterises the within-school dose-response. Section V reports the ITT effect, heterogeneity, and robustness. Section VI reports cost-effectiveness. Section VII extrapolates to the state's program rollout. Section VIII discusses the Year-3 follow-up and the mechanism interpretation. Section IX concludes.

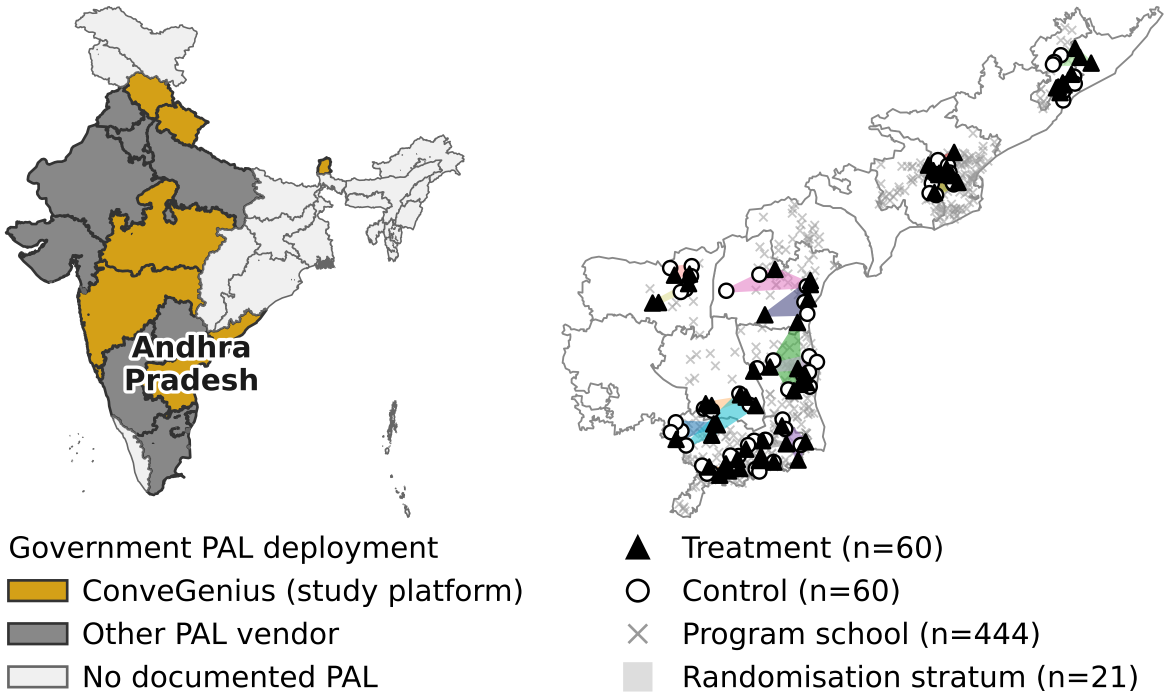

FIGURE 1. RCT SCHOOLS WITHIN THE ANDHRA PRADESH PAL ROLLOUT

Notes: **Panel A:** Indian states colour-coded by PAL deployment status. Gold = states where ConveGenius operates (the PAL platform of this study). Dark grey = states with another documented PAL vendor. Light grey = no documented government PAL deployment. **Panel B:** Andhra Pradesh with all district boundaries. The 120 RCT schools (60 treatment = filled triangles; 60 control = open circles) sit within the 444-school government PAL rollout (grey × markers) covering 18 of AP's 26 districts. The 21 randomisation strata, formed by grouping adjacent mandals within each of the 8 RCT districts, are shown as translucent polygons in distinct colors.

II. Background

A. Program rollout and study sample

The Government of Andhra Pradesh deployed Personalized Adaptive Learning (PAL), a tablet-based math program, in partnership with the educational-technology firm ConveGenius. Launched in 2019, suspended during the COVID-19 pandemic, and restarted in 2022, the programme reached 444 government middle schools across 17 of the state's 26 districts by 2022 and has since scaled to roughly 1,500 schools statewide. We partnered with the state to evaluate the programme in 120 randomly selected schools added to the rollout — 60 assigned to receive PAL and 60 retained as controls, stratified across 8 districts and 21 mandal-cluster groups (Figure 1) — with implementation in the treatment arm beginning in academic year 2023–24. The main results in this paper come from these 60 evaluation schools; we discuss external validity to the broader 444-school program rollout (the subset with ConveGenius usage records) in Section VII.

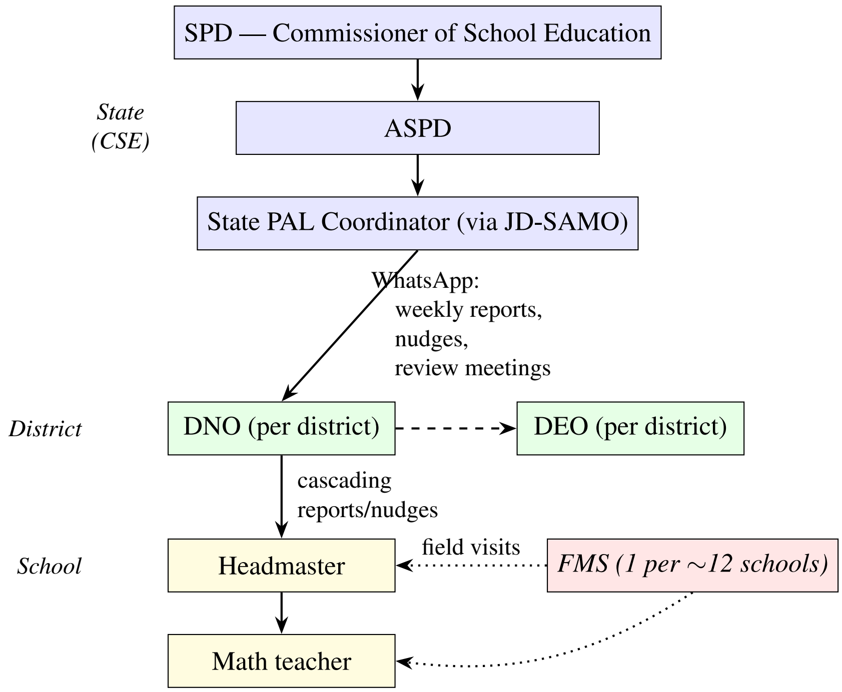

FIGURE 2. THE SCHOOL–DISTRICT–STATE ESCALATION CHAIN FOR AP PAL

Notes: Solid arrows: direct line of authority. Dashed: lateral district-level coordination. Dotted: technical-support contractor interface. SPD = State Project Director. ASPD = Additional SPD. JD-SAMO = Joint Director, State Academic Monitoring. DNO = District Nodal Officer. DEO = District Educational Officer. HM = Headmaster. FMS = Field Management Staff (contracted through ConveGenius). Funded through state and central (Samagra Shiksha) education budgets and operated by the AP government's regular school administration.

B. State implementation infrastructure

Each treatment school sat at the bottom of a four-tier chain that decided whether the PAL lab actually ran each week (Figure 2). At the top, the State Project Director (SPD) and the Additional SPD (ASPD) at the Commissionerate of School Education set policy; under them, a State PAL Coordinator handled day-to-day operational traffic via the Joint-Director-SAMO wing. Each district had a District Nodal Officer (DNO) for PAL who cascaded state instructions to the schools in their district and coordinated laterally with the District Educational Officer (DEO). At the school, the Headmaster (HM) oversaw delivery and the math teacher ran the lab session itself.

The chain operated through three concrete channels (Figure 3, top lane). **First**, a WhatsApp broadcast group from the State PAL Coordinator to all DNOs — used across our two-year observation window to share weekly school- and district-level usage rankings, to issue direct nudges to increase usage, and for operational pushes such as data syncing, orientation sessions, tablet maintenance guidance, and field-visit logistics. **Second**, scheduled review meetings: state-coordinator-led district-by-district sessions with HMs,

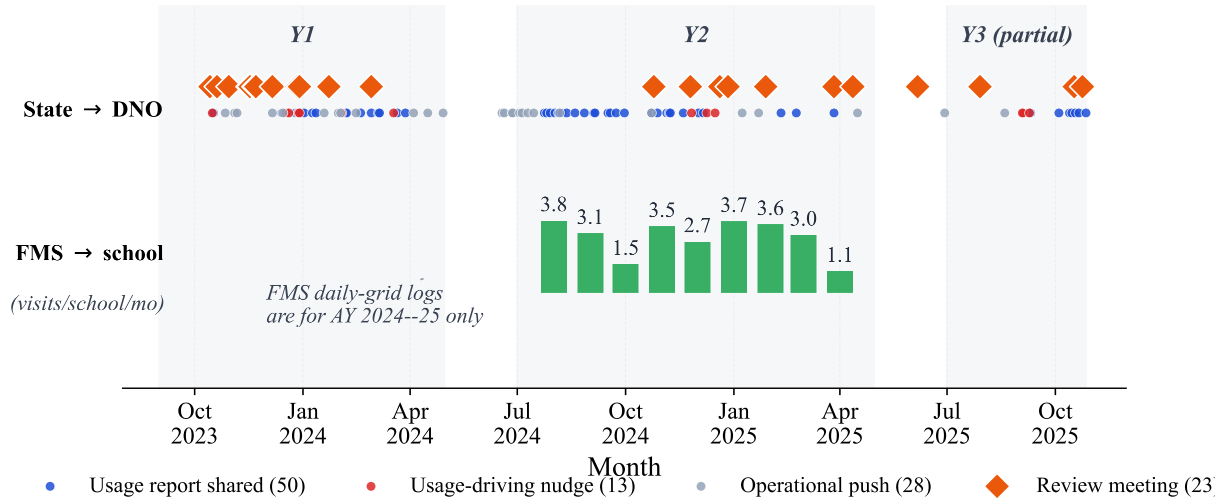

FIGURE 3. STATE IMPLEMENTATION INFRASTRUCTURE OPERATING ACROSS TIME, OCTOBER 2023 – OCTOBER 2025

Notes: **Top lane:** state-to-DNO communication traffic from the State PAL Coordinator's WhatsApp log and a parallel review-meetings record, classified by content as usage report shared (blue), usage-driving nudge (red), operational push such as data sync, training, tablet maintenance, or field-visit logistics (grey), and scheduled review meeting (orange diamond). **Bottom lane:** mean monthly Field Management Staff (FMS, the technical-support contractor) field visits per RCT-T school, from daily-grid visit logs available for AY 2024–25 only. Light shading marks AY-active windows. The Y3 column is partial through 27 October 2025.

math teachers, and DNOs in the program's first weeks; subsequent in-person summons of low-performing DNOs to the Commissionerate; and SPD-led review meetings with the broader DNO group. **Third**, real-time monitoring dashboards: per-school usage metrics that the State PAL Coordinator and DNOs could query directly, with the underlying telemetry coming from the ConveGenius platform itself. All 28 RCT-T HMs in our Y3 survey report receiving these state communications periodically, and 71 % report having seen the DIL school-level usage report card.

Alongside the government chain, a Field Management Staff (FMS) workforce contracted through ConveGenius operated at the school level (Figure 3, bottom lane). One FMS member was assigned to a cluster of approximately 12 schools; in AY 2024–25 they visited each treatment school an average of one to four times per month to handle technical issues with tablets and software, sync data from the offline platform, and in some cases flag low usage to the HM. The FMS is not a tier of the government chain — it is a parallel technical-support workforce that interfaces only at the school. Its existence is what allows the rest of the infrastructure to assume that tablets work and that data reaches the dashboard.

Figure 3 shows this infrastructure operating at a sustained cadence across academic years 2023–24 and 2024–25, and at a fraction of that cadence in the partial Y3 window for which transcribed monitoring data is available. The State PAL Coordinator's agenda over this period also took on a ∼700-school KGBV residential expansion that launched in January 2025; we return to both shifts and their implications in §VIII.

C. School-level delivery and the PAL software

In each treatment school, the government established a PAL lab: a designated classroom equipped with 30 tablets, charging stations, and earphones. The tablet count was fixed at 30 regardless of school or section enrollment. In the status quo, students had eight 40-minute periods of math instruction per week. Schools were instructed to convert two of these periods to PAL sessions, targeting approximately 80 minutes of software use per week. During PAL periods, the math teacher took students to the lab and supervised while they worked on the software independently. In sections with more than 30 students, tablets had to be shared: teachers were asked to split the section into batches that alternated between PAL and regular instruction, though in practice we also observed students being paired on a single tablet or taking turns. Across the 60 treatment schools, 44.7% of the 300 class sections exceeded the 30-tablet count; the median oversized section held 37 students (a 1.23:1 student-to-tablet ratio), and 59.6% of treatment students were in a sharing-required section (Appendix Figure C1).

Math teachers reported spending less time on remedial instruction and practice problems during regular class periods, instead focusing on teaching key concepts in the classroom and directing students to use PAL for practice on that chapter.

Personalized Adaptive Learning (PAL) is a tablet-based math software developed by ConveGenius, an Indian education technology company. The software delivers curriculum-aligned content organised into chapters that mirror the state mathematics textbook; remediation extends up to two grade levels below the student's enrolled grade. Across 8.78 million module-level events in the 60 treatment schools, 27% of events are below the student's enrolled grade, with an average depth of 1.43 grade levels below. The lowest content grade delivered is grade 4, seen only by grade-6 students (for whom it is 2 grades below enrolled); for grade-9 students, the deepest remediation available is grade 7. We test whether this two-grade limit binds in a pre-registered within-school experiment reported in a companion paper (Cupito et al., 2026, in preparation). Per the vendor's intended use model, students work individually on tablets, watching instructional videos, answering practice problems, and completing assessments. The software operates offline; teachers sync usage data to the server approximately once per week.

Using module-level platform data (8.78 million learning events across 12,019 students in the 60 treatment schools), we reconstruct the instructional pipeline that students experience. Each chapter follows a standardised sequence of six stages (Figure C2 in Appendix A): (1) a diagnostic test of 2–3 questions on prerequisite content; (2) below-grade instructional videos; (3) below-grade remedial practice; (4) a baseline assessment at enrolled-grade level; (5) at-grade videos paired with formative knowledge checks; and (6) a chapter endline assessment. Within any given chapter, the module sequence is largely uniform: the top sequence covers 13–36% of students in high-N chapters and the top five cover 38–54%. In the program's standard configuration in AP, students proceed through chapters in a fixed curriculum order (chapter locking). For the RCT schools, chapter locking was disabled, allowing students some flexibility in chapter selection by students or teachers; the cross-chapter order variation we observe is therefore an RCT-specific feature, not a software default.

The platform has two adaptive features, both operating on binary thresholds rather than smooth gradients. First, a *diagnostic-triggered remediation* at the start of each chapter: students scoring below 80% on the 2–3-question diagnostic receive the full below-grade video and remedial-practice sequence; those scoring $\geq$80% skip remediation (only 2% of chapter journeys). Since 97% of diagnostic scores fall below 80% (mean 20%), virtually all students receive the remediation pipeline. Second, a *completion-based carry-forward* across chapters: after a completed chapter (one where the student reached the endline), the next chapter begins with 28% below-grade content on average; after a non-completed chapter, 48% — a 20 percentage-point gap ($N = 234{,}810$). This completion carry-forward, not diagnostic improvement, drives the declining grade-gap pattern within each academic year: diagnostic scores remain flat at $\sim$20% throughout, while below-grade content share falls as students accumulate completed chapters. The completion state resets at the start of each academic year, producing the grade-gap reset visible at the Year 2 boundary (Figure C3 in Appendix A).

Neither adaptive feature differentiates by actual student ability. Pipeline metrics are effectively flat across baseline ability quintiles: students in the lowest quintile (Q1, by within-school $\times$ grade baseline percentile) receive 9.7 below-grade modules per chapter on average, while those in the highest quintile (Q5) receive 9.3. Total modules (34.5 vs. 36.2), knowledge-check rounds (2.0 vs. 1.9), and chapter completion rates (62% vs. 68%) differ by similarly small amounts across ability quintiles (Figure C4 in Appendix A visualises this: content-grade progression overlaps almost completely for bottom and top ability terciles in both years). The platform's within-system adaptive signals (diagnostic, baseline, KC performance) correlate only weakly with independently measured ability ($r = 0.11$ between first-ever diagnostic score and within-school $\times$ grade baseline percentile). PAL is best described as structured, curriculum-aligned computer-assisted learning

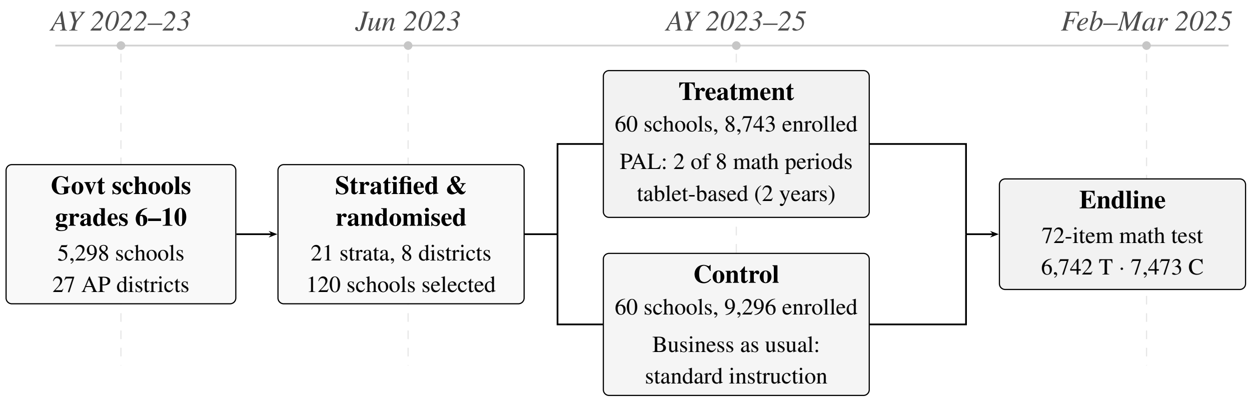

FIGURE 4. CONSORT FLOW DIAGRAM

Notes: Government schools with grades 6–10 in Andhra Pradesh (5,298 schools across 27 districts). Eight districts were selected as study sites and stratified into 21 geographic strata based on mandal clusters within each district. 120 schools were randomly selected within strata and assigned to treatment or control. In treatment schools, PAL (Personalized Adaptive Learning) replaced 2 of 8 weekly math periods with tablet-based adaptive learning for 2 academic years (2023–24 and 2024–25). Enrollment counts are from government administrative records (AY 2024–25).

with two algorithmic levers on remediation: a per-chapter diagnostic gate and a cross-chapter taper keyed to completion history. Both levers adapt to recent student behaviour but neither personalises on measured ability. We retain the program's name throughout; the learning gains we report are plausibly attributable both to additional structured math practice time and to the algorithm's pacing of remediation across chapters.

III. Methods

A. Experimental Design and Study Sample

Our study is a cluster-randomized controlled trial conducted in government schools in the Indian state of Andhra Pradesh. The government of Andhra Pradesh identified eight districts in which to implement the PAL program. Within these districts, schools were grouped into 21 geographic strata based on mandal clusters within each district. From a pool of eligible co-educational government schools serving grades 6–10, 120 schools were selected and stratified-randomly assigned to treatment (60 schools) or control (60 schools).[2] A separate 20-school within-school randomised experiment testing deeper remediation, conducted in program schools outside our 120 evaluation schools, is reported in a companion paper (Cupito et al., 2026, in preparation). Figure 1 maps the 120 RCT schools within the broader government PAL rollout. Table 1 presents balance on school and student characteristics. Treatment and control schools are statistically

[2] Three size-based filters were applied to the eligible pool before randomisation, ensuring feasibility of the 30-tablet-per-school intervention: (i) schools with any grade 6–9 section exceeding 60 students were excluded (at most a 2:1 student-tablet ratio at maximum congestion); (ii) schools with more than 20 total sections across grades 6–9 were excluded; (iii) schools in the top decile of the pupil-teacher ratio were excluded. These filters also truncate the support of the congestion instrument used in Section C.

balanced across 15 school-level characteristics in Panel A (joint $F = 1.14$, $p = 0.336$), spanning enrollment, staffing, and ICT infrastructure, and across nine student-level characteristics in Panel B (joint $F = 1.07$, $p = 0.387$), including grade composition, gender, age, parental education, and SES. The one exception is the pre-intervention government math score (Panel B, last row), where treatment students score 2.6 points lower — but this test is underpowered and compositionally biased because data coverage is only 62% and asymmetric across arms (see §B and Appendix D).

Treatment schools received the PAL intervention, which replaced two of eight weekly math periods with tablet-based instruction. Control schools continued with their regular math curriculum. The intervention ran for two academic years: AY 2023–24 and AY 2024–25.

A total of 18,039 students enrolled in grades 6–9 across the 120 study schools in AY 2024–25 comprise our study sample. Students' exposure to PAL varied by cohort: those in grades 7–9 at endline received two full years of the program, while grade 6 students (a new cohort entering in AY 2024–25) and grade 10 students (who aged out after AY 2023–24) received one year. We collected baseline data from government assessments administered in AY 2022–23, prior to the start of the intervention. Endline assessments took place in February and March 2025. Figure 4 presents the CONSORT flow diagram.

Of the 18,039 enrolled students, 14,215 (78.8%) completed the endline assessment—6,742 in treatment schools and 7,473 in control schools. Attrition was 3.9 percentage points higher in treatment schools (23.7% versus 19.8%). However, the majority of students who did not take the endline assessment remained enrolled in school: 81% appeared in government assessment records for 2024–25, and 70% had non-zero math scores. We present formal attrition analysis and Lee bounds in Appendix B.

B. Data

Our main sources of data come from independent assessments at endline conducted by a private survey firm; administrative data from the state government; and student usage and learning data collected through the software by ConveGenius.

We conducted math assessments at endline to capture grade-level and earlier-grade learning. The instrument comprised 72 multiple-choice items spanning five broad math topics—algebra, geometry, measurement, statistics, and number systems—with questions drawn from the grade 2 through grade 9 curriculum. Each student answered items from grade 2 up to their enrolled grade, yielding between 50 and 72 items depending on grade level (50 for grade 6, 72 for grade 9). Using item response theory (IRT), we transform each student's performance into a single ability parameter, which we standardize to the control group mean and standard deviation. Following other work using this method to generate comparable scores across children and grades, we call this our IRT score (Muralidharan and Ganimian, 2019; Muralidharan and Singh, 2025). We describe the development and piloting of the instrument in Appendix G.

Baseline math ability, used in robustness checks and for heterogeneity by baseline ability, comes from pre-intervention government assessments. The source is cohort-specific: grade 7–9 endline cohorts use the AY 2022–23 SA1 (the year immediately before PAL began); the grade 6 endline cohort, who entered the study schools in AY 2024–25, uses the AY 2023–24 SA1 from their prior (grade 5) school. Overall coverage is 61%, but missingness is sharply asymmetric across arms (8% treatment versus 45% control), driven by differential government data extracts rather than student characteristics; for analyses that condition on baseline (Appendix Table B2 robustness column and the baseline-quintile heterogeneity in Table 4) we include a missing-baseline flag and interpret the resulting estimates with caution. Appendix F validates these government scores against our independent endline IRT and documents why they serve only as within-school ranked controls, not as a cross-school ability measure.

We administered the assessment on tablets using SurveyCTO, conducted by enumerators independent of the schools and intervention. The tablet platform served several data-quality objectives. First, mandatory responses combined with an explicit "Don't know" option eliminated missing values entirely. Second, randomized presentation of both topic order and question order across three test forms reduced opportunities for copying. Third, the platform recorded per-item response times, enabling ex post detection of guessing or rushing. Fourth, same-day data syncing allowed us to monitor school-level completion rates in real time and schedule efficient return visits to high-absence schools. Separately, to address the concern that treatment students' greater familiarity with tablets could bias performance, we conducted a pre-pilot confirming similar response times and accuracy on the device across treatment and control, and a Year-3 within-student sub-study that formally tests tablet-vs-paper measurement equivalence on G8 students; both find no differential mode advantage for treatment students (see Appendix B, §A).

C. Empirical Specification

Our main estimating equation is the intent-to-treat (ITT) specification:

$$Y_{isg} = \alpha + \beta\, T_s + \mathbf{X}_i'\gamma + \delta_g + \mu_m + \varepsilon_{isg} \tag{1}$$

TABLE 1—SAMPLE CHARACTERISTICS AND BALANCE

	AP (1)	Control (2)	Treatment (3)	Difference (4)	(SE) (5)	p-value (6)	N (7)
Panel A: School Characteristics [AP: $N = 5,298$ schools]							
Enrollment (grades 6–9)	217.1	160.0	155.1	-4.333	(17.848)	0.808	120
Total enrollment	309.1	234.6	223.2	-11.020	(22.825)	0.629	120
Urban	0.129	0.100	0.117	0.027	(0.055)	0.627	120
Has primary section	0.493	0.683	0.583	-0.111	(0.087)	0.203	120
Math teachers	2.1	2.0	1.9	-0.098	(0.153)	0.525	120
Physical ed. teachers	1.7	1.5	1.3	-0.205*	(0.109)	0.060	120
Total teachers	11.8	10.8	10.5	-0.286	(0.706)	0.685	120
Classrooms	10.8	9.4	8.9	-0.512	(0.629)	0.416	120
Functional computers	1.9	1.3	1.6	0.212	(0.610)	0.728	120
Projector	0.614	0.600	0.550	-0.051	(0.092)	0.584	120
Smart TV	0.681	0.767	0.650	-0.108	(0.076)	0.158	120
Digital board	0.286	0.267	0.283	0.020	(0.070)	0.773	120
Uses ICT tools	0.794	0.817	0.833	0.020	(0.073)	0.782	120
Digital content available	0.695	0.717	0.733	0.037	(0.079)	0.638	120
Generator/UPS	0.133	0.133	0.117	-0.017	(0.060)	0.780	120
Joint F-statistic		1.14		[$p = 0.336$]			120
Panel B: Student Characteristics [AP: $N = 1,150,232$ students]							
Grade 6	0.223	0.237	0.228	-0.011	(0.008)	0.168	14,215
Grade 7	0.241	0.235	0.237	0.006	(0.007)	0.414	14,215
Grade 8	0.255	0.248	0.262	0.013	(0.008)	0.109	14,215
Grade 9	0.281	0.280	0.272	-0.007	(0.009)	0.407	14,215
Female		0.492	0.479	-0.020	(0.019)	0.301	14,215
Age		12.30	12.35	0.039	(0.037)	0.293	14,124
Father education		2.14	2.07	-0.057	(0.059)	0.336	12,744
Mother education		2.10	2.07	-0.037	(0.045)	0.412	13,135
SES index (PCA)		-0.02	0.02	-0.030	(0.040)	0.454	14,215
Joint F-statistic		1.07		[$p = 0.387$]			12,267
Baseline math score (govt SA1)		35.23	32.10	-2.585**	(1.299)	0.047	10,238

Notes: Column (1) shows means for all 5,298 government schools with grades 6–10 in Andhra Pradesh (1,150,232 students in grades 6–9). School characteristics from government administrative records (AY 2023–24); ICT variables from UDISE+ 2024–25. Columns (2)–(3) show unconditional group means for the 120 randomised study schools. Column (4) and standard errors are from OLS of each variable on a treatment indicator and 21 randomization strata (mandal cluster) fixed effects. Panel A reports school-level characteristics (including enrollment in grades 6–9, the study grades) with HC1 robust standard errors ($N = 120$ schools). Panel B reports student-level characteristics for the grades 6–9 study sample, with standard errors clustered at the school level. Baseline math score (government SA1, 2022–23) is excluded from the joint F-test due to differential administrative data coverage (62%; see text). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

where Y_{isg} is the endline math score (IRT ability parameter, standardized to control group mean and SD) for student i in school s and grade g; T_s is a treatment indicator (1 = PAL school); $\mathbf{X}_i$ is a vector of student-level controls (SES index,[3] parental education, age, gender, and missing-value flags); δ_g are grade fixed effects; and μ_m are randomization strata (mandal cluster) fixed effects. Standard errors are clustered at the school level (120 clusters). The coefficient β is the average ITT effect of PAL on math learning. Baseline math ability (described in Section B) is not included in $\mathbf{X}_i$ for the headline specification because the missingness is sharply imbalanced across arms (8% in treatment versus 45% in control), driven by differential government data extracts; a missing-flag dummy is therefore itself treatment-correlated and would absorb causal variation. We report a robustness column adding baseline in Appendix Table B2, where the pooled ITT moves only from 0.390 to 0.393, indicating the headline is robust to including baseline despite this contamination.

To estimate the causal return to PAL usage, we exploit exogenous variation in dosage arising from tablet congestion. All treatment schools received 30 tablets regardless of enrollment, so students in larger sections share fewer tablets per student and receive less PAL time. We instrument for usage hours using the interaction of treatment status and section enrollment:

$$\text{First stage:} \quad H_{isg} = \pi_0 + \pi_1 (T_s \times S_{sg}) + \pi_2 T_s + \pi_3 S_{sg} + \mathbf{X}_i' \pi_4 + \delta_g + \mu_m + \nu_{isg} \tag{2}$$

$$\text{Second stage:} \quad Y_{isg} = \alpha + \lambda \hat{H}_{isg} + \phi T_s + \psi S_{sg} + \mathbf{X}_i' \gamma + \delta_g + \mu_m + \varepsilon_{isg} \tag{3}$$

where H_{isg} is total PAL usage hours and S_{sg} is section enrollment (the number of students in the student's class section). The exclusion restriction requires that the difference in treatment effects between larger and smaller sections is driven entirely by the difference in PAL usage between these sections — equivalently, if usage were equalised across section sizes, treatment-effect magnitudes would be the same in larger and smaller sections.

For heterogeneity analyses, we estimate equation (1) separately by subgroup or include interaction terms with treatment status.

IV. Implementation and Usage

Table 2 summarises PAL usage across both academic years. Schools were allocated one 80-minute PAL period per week, though the effective instructional window was approximately 60 minutes after accounting for student transit to the computer lab and device setup. PAL sessions were suspended during holidays,

[3] First principal component of nine household-asset indicators collected at endline: refrigerator, laptop, father's mobile phone, mother's mobile phone, air conditioner, washing machine, car, scooter, and bicycle.

TABLE 2—PAL USAGE STATISTICS (TREATMENT GROUP)

	Mean	Median	SD	P25	P75	*N*
Panel A: Overall Usage						
Total usage (hours, 2 years)	34.2	30.2	23.3	17.4	47.3	6,742
Year 1 usage (hours)	14.7	12.6	15.5	0.0	23.8	6,742
Year 2 usage (hours)	19.5	17.8	12.5	10.7	26.2	6,742
Video time (hours)	18.9	16.5	14.2	9.6	25.2	6,742
Question time (hours)	15.3	13.5	10.5	7.3	21.3	6,742
Months active (Y2)	6.6	7.0	1.9	6.0	8.0	6,742
Chapters attempted	23.7	24	11.6	14	31	6,530
Chapters completed	15.9	14	11.1	7	23	6,530

Panel B: Usage by Grade	Y1 hours	Y2 hours	Total	Months (Y2)	*N*
Grade 6	—	20.4	20.4	6.0	1,540
Grade 7	18.9	20.5	39.4	6.8	1,599
Grade 8	19.7	20.0	39.7	6.9	1,769
Grade 9	18.4	17.5	35.9	6.8	1,834

Notes: Sample restricted to treatment students who took the endline assessment and have any recorded PAL usage ($N = 6{,}742$ students in 60 schools). Panel A reports individual-level usage aggregated across both academic years (AY 2023–24 and 2024–25). Video time is passive instruction; question time is active practice. Chapters are curriculum units within the PAL platform. Panel B shows usage by endline grade. Grade 6 students entered the study in Year 2 only. Usage data from ConveGenius platform logs, capped at the endline assessment start date (February 17, 2025). Each school was allocated one 80-minute PAL period per week; accounting for school closures and setup time, two-year students achieved 89% of the calendar-adjusted effective target (Appendix Table H5). Topic-level usage breakdown is reported in the appendix.

formative and summative assessment weeks, and pre-exam revision periods—leaving 22 eligible weeks in Year 1 and 21 in Year 2 (Appendix Table H5). Against this calendar-adjusted effective target of 43 hours over two years, students in grades 7–9 averaged 38.2 hours (89%). Compliance was higher in Year 2 (92%) than Year 1 (86%), consistent with implementation ramp-up during the program's first year. Lab setup was staggered across schools, with 49 of 60 schools beginning in September 2023 and the remaining 11 in October. Figure 5 traces the resulting monthly usage profile across both academic years and the corresponding content grade-level gap.

Baseline math test scores and age are the strongest predictors of PAL usage, while SES is not (Appendix Table H4). Students spend slightly more time watching videos (54%) than answering questions (46%; Appendix Table C1). Eighty percent of treatment students report enjoying PAL "a lot," though 37% report language comprehension and video loading as implementation challenges (Appendix Table C2). Appendix A characterises the platform's instructional pipeline using module-level data, showing that the student experience

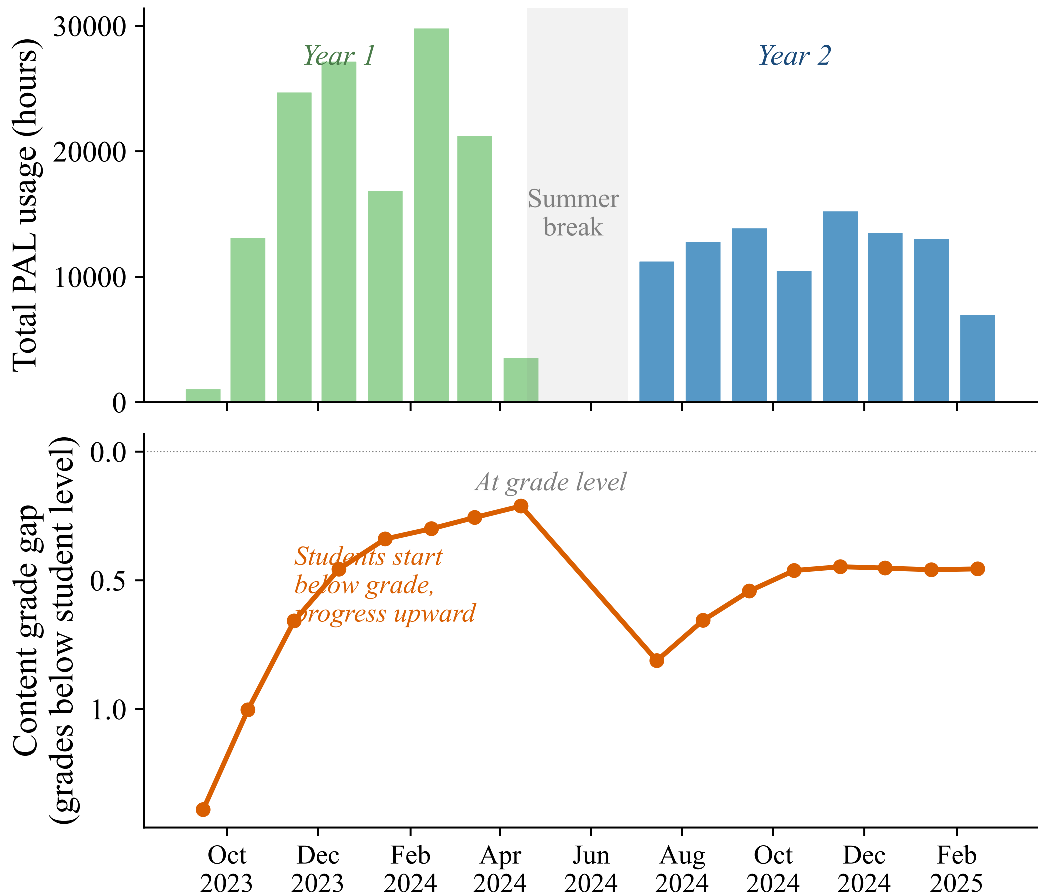

FIGURE 5. MONTHLY PAL USAGE AND CONTENT PROGRESSION

Notes: Top panel: total PAL usage hours per month across all treatment schools (green = AY 2023–24, blue = AY 2024–25). Usage ramps up from September, peaks in December–February, and drops during summer break (May–June). Bottom panel: average content grade gap (student grade minus content grade being studied). The system starts students approximately 1.3 grade levels below their enrolled grade, progressively moves them toward grade-level content, then resets at the start of the new academic year. Data from ConveGenius platform logs for 60 treatment schools.

is largely uniform across the baseline-test-score distribution.

V. Impact on Learning

A. ITT Estimates

Figure 6 plots kernel density estimates of endline math scores by treatment status and grade. The treatment distribution shifts visibly rightward for all four grades, with the largest shift at grade 7 and the smallest at grade 9. The shift is not confined to one tail of the distribution: treatment widens the distribution as well as shifting its mean, indicating that PAL generates variance in math-test-score gains across students.

Table 3 reports the corresponding regression estimates. The pooled ITT effect is 0.39 SD ($p < 0.001$).

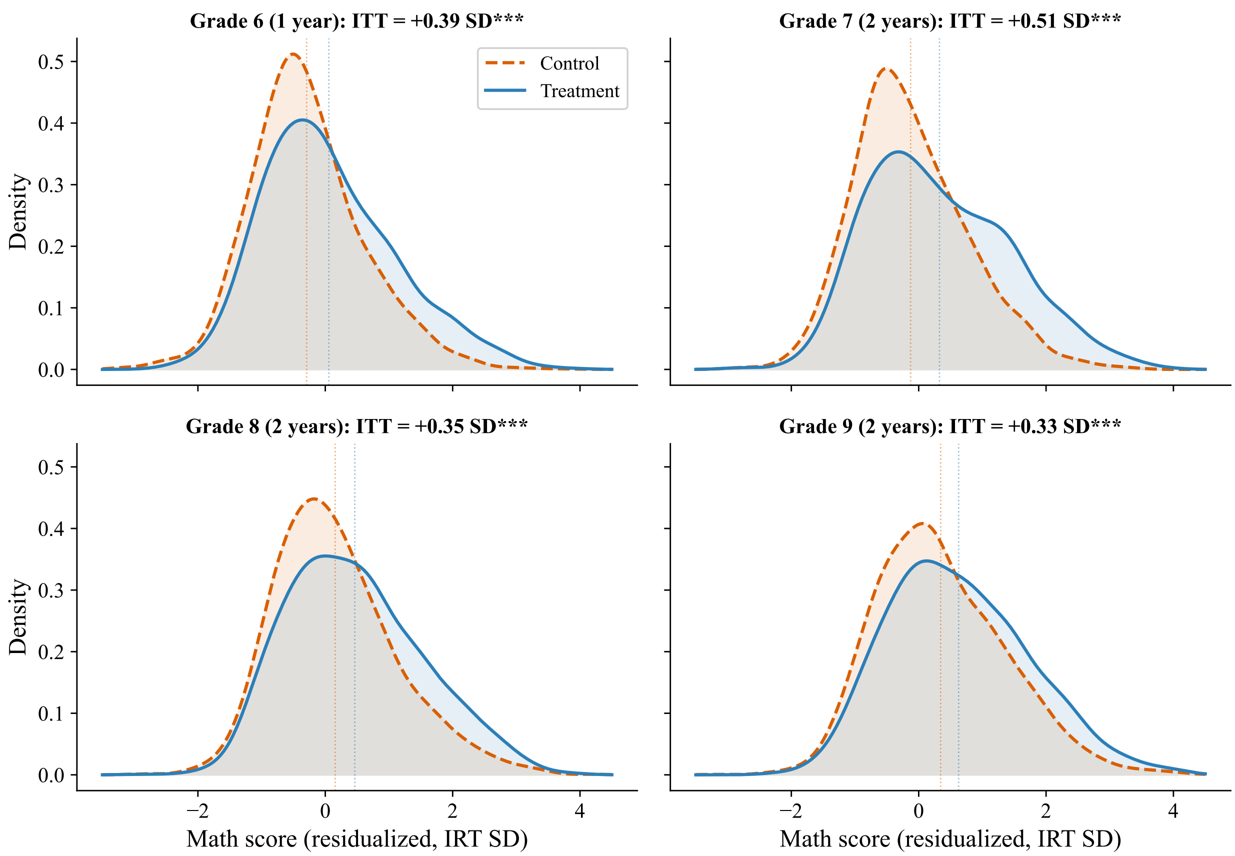

FIGURE 6. SCORE DISTRIBUTIONS BY GRADE: TREATMENT VS. CONTROL

Notes: Kernel density estimates of endline math scores by treatment status and grade. Scores are residualized for student covariates (SES index, father/mother education dummies, age, female, with missing-flag indicators) and mandal-cluster strata fixed effects, then re-centred at the grade-level mean to remain on the standardized IRT scale. Dotted vertical lines mark group means. Panel titles report the regression-adjusted ITT (matching Table 3) from OLS of *mathscore_z* on treatment, the same covariates, and strata fixed effects, with school-clustered standard errors. By Frisch–Waugh–Lovell, the T–C mean shift in each panel equals the panel-title ITT up to the residual correlation between treatment and the partialled-out covariates, which is approximately zero under random assignment. Grade 6 students had 1 year of PAL exposure; grades 7–9 had 2 years. Bandwidth = 0.25.

By grade, effects range from 0.33 SD in grade 9 to 0.51 SD in grade 7. The null hypothesis that effects are equal across grades is marginally rejected ($p = 0.091$), with the pattern suggesting a declining gradient from younger to older cohorts. We investigate this gradient — and its explanation through tablet congestion — in Section C.

These results are robust to alternative specifications and inference procedures. The treatment effect is stable across three control specifications (0.38–0.39 SD; Appendix Table B2). Randomization inference, which does not rely on asymptotic assumptions, confirms the main result (RI $p < 0.001$; Appendix Figure B1). Lee (2009) bounds, which trim the sample to address differential attrition between treatment and control (23.7% vs. 19.8%), remain large and positive (0.27–0.48 SD; Appendix Table B1). Finally, a within-

TABLE 3—INTENT-TO-TREAT ESTIMATES OF THE EFFECT OF PAL ON MATH LEARNING

	Pooled (1)	Grade 6 (2)	Grade 7 (3)	Grade 8 (4)	Grade 9 (5)
Treatment	0.390***	0.393***	0.509***	0.346***	0.327***
	(0.054)	(0.070)	(0.069)	(0.068)	(0.062)
Controls	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Grade FE	Yes				
Control mean	0.000	-0.338	-0.157	0.116	0.316
Observations	14,215	3,313	3,354	3,623	3,925
R^2	0.137	0.153	0.125	0.097	0.086
p-value (equality across grades)		0.091			

Notes: Dependent variable: IRT math ability parameter standardized to control group mean and SD. Each column reports OLS estimates of the treatment effect with 21 randomization strata (mandal cluster) fixed effects. Column (1) includes grade fixed effects (grade 6 = base). Controls: SES index, father education, mother education, age, gender, and missing flags for each imputed variable. Standard errors clustered at the school level (120 clusters) in parentheses. The p-value tests the null hypothesis that treatment effects are equal across grades (from a pooled interaction model). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

student tablet-vs-paper sub-study conducted on G8 students in Year 3 (Appendix B, §A) finds no significant mode × treatment interaction, ruling out tablet familiarity as a driver of the headline effect.

B. Heterogeneous Treatment Effects

Table 4 examines treatment effect heterogeneity along several dimensions.

Gender. Girls gain significantly more than boys (0.46 vs. 0.33 SD; $p = 0.001$). This gender interaction survives multiple hypothesis testing corrections (Bonferroni, Holm, and Benjamini–Hochberg; Appendix Table B3). The advantage is not explained by pre-existing differences in baseline ability or attendance, which are balanced across gender within treatment status. Rather, girls accumulate approximately 2 more hours of PAL usage than boys, driven by more time on active practice (questions) rather than passive content (videos; Appendix Figure D2). The gender-stratified IV analysis confirms that boys and girls have similar per-hour returns to PAL (Appendix Table E2), indicating the gap reflects differences in usage quantity rather than in the productivity of each hour. Appendix Table D1 provides the full regression results for this diagnosis.

Baseline math test scores. Treatment effects decline with baseline achievement: 0.48 SD for the lowest quintile and 0.31 SD for the highest ($p = 0.08$ for Q1 vs. Q5). PAL benefits students across the entire baseline-

TABLE 4—HETEROGENEOUS TREATMENT EFFECTS

	Treatment	(SE)	Control mean	N
Pooled	0.390***	(0.054)	0.000	14,215
Panel A: Gender				
Boys	0.329***	(0.056)	-0.063	7,289
Girls	0.464***	(0.059)	0.068	6,877
p (Boys = Girls)		0.001		
Panel B: Baseline Achievement				
Q1 (lowest)	0.483***	(0.071)	-0.410	2,483
Q2	0.353***	(0.063)	-0.207	1,897
Q3	0.330***	(0.075)	0.057	1,969
Q4	0.339***	(0.079)	0.341	1,936
Q5 (highest)	0.305***	(0.086)	0.929	2,020
Missing baseline	0.498***	(0.077)	-0.155	3,910
p (Q1 = Q5)		0.083		
p (joint)		0.126		
Panel C: SES				
Low SES (T1)	0.350***	(0.059)	-0.057	4,746
Mid SES (T2)	0.422***	(0.066)	0.039	5,182
High SES (T3)	0.381***	(0.060)	0.018	4,287
p (Low = High)		0.418		
Panel D: School Characteristics				
Small section ($\leq$33)	0.463***	(0.059)	-0.014	7,220
Large section (>33)	0.336***	(0.075)	0.014	6,995
p (Small = Large)		0.050		
Rural	0.456***	(0.057)	-0.018	11,615
Urban	0.520***	(0.151)	0.095	2,600
p (Rural = Urban)		0.075		

Notes: Dependent variable: IRT math ability parameter standardized to control group mean and SD. Each row reports the treatment coefficient from a separate OLS regression on the indicated subgroup, with 21 strata fixed effects, grade fixed effects, and controls (SES, parental education, age, gender, missing flags). Standard errors clustered at the school level in parentheses. Baseline achievement quintiles are within school × grade percentile ranks of government math SA1 (AY 2022–23). "Missing baseline" includes students without baseline data (28.8%). Baseline coverage differs by treatment arm for grades 6–7 due to a government data extraction artifact (Appendix Table D3); the treatment effect for this subgroup is not interpretable as a clean subgroup estimate. SES terciles are based on a PCA index of 9 household assets. Section enrollment is the number of students in the student's class section. p-values test equality of treatment effects between the indicated groups (from pooled interaction models). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

test-score range, but the lowest-baseline students gain the most. A conditional quantile analysis (Appendix Figure H8) reveals that within each baseline quintile, effects increase from the 10th to the 75th percentile of the endline distribution before flattening—suggesting that PAL creates within-group variance in outcomes in addition to the across-quintile gradient. The large treatment effect for the “missing baseline” subgroup (0.50 SD) reflects a data extraction artifact rather than a true subgroup effect: the government provided broader baseline data for treatment schools than for control schools in grades 6–7, creating compositional imbalance in this subgroup (Appendix Tables D3–D4 and Figure D3).

SES and school characteristics. Effects are uniform across SES terciles ($p = 0.45$; see also Appendix Table D2). Students in small sections (at or below 33 students, the approximate tablet supply) gain 0.46 SD compared with 0.34 SD in large sections ($p = 0.05$). Rural and urban schools show similar effects. The school-level heterogeneity by section size foreshadows the tablet congestion mechanism in Section C. Appendix Figure D1 provides a visual summary of heterogeneity across all four dimensions.

Topic-level effects. Treatment effects are uniform across all five math topics (0.31–0.38 SD; Appendix Table H7 and Figure H4), indicating that PAL improves foundational math skills broadly rather than producing topic-specific gains. Effects are slightly larger on harder items (Appendix Figure H5), suggesting that PAL helps students reach questions they would otherwise get wrong. Treatment students also outperform control students at every content grade level, with the largest gains on grade 4–6 content (Appendix Figure H6). PAL increases math enjoyment by 0.19 SD, with positive spillovers to other subjects, and reduces self-reported absences by 0.22 days over two weeks (Appendix Table H6).

C. The Grade Gradient and Tablet Congestion

The declining treatment effect across grades ($p = 0.091$; Table 3) could reflect several mechanisms: differential exposure (grade 6 had one year vs. two for grades 7–9), differences in instructional quality by grade, or a binding supply constraint on devices. Figure 7 distinguishes among these explanations by splitting schools at the median of total enrollment.

All treatment schools received exactly 30 tablets, regardless of enrollment. Appendix Figure C1 shows the full distribution of section sizes relative to this cap: 45% of treatment sections exceed 30 students and 60% of treatment students are in a sharing-required section. In small schools, section enrollment (students per grade-section) stays at or below 30 across all grades—every student gets a device during PAL periods (Panel A). In large schools, sections exceed the 30-tablet threshold and grow with grade (median 32 at grade 6, 40 at grade 9), requiring students to share devices or take turns.

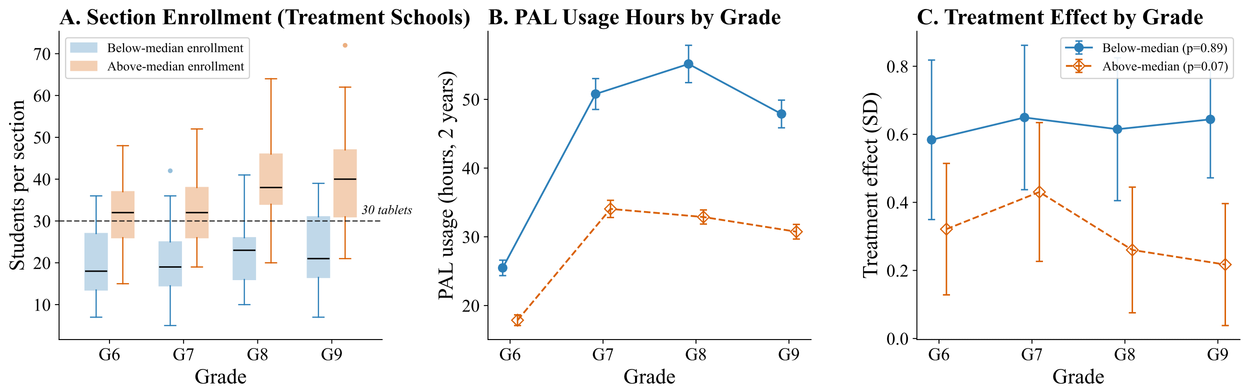

FIGURE 7. THE GRADE GRADIENT IN TREATMENT EFFECTS AND TABLET CONGESTION

Notes: Schools are split at the median of total enrollment in grades 6–9 (median = 138). *Panel A*: section enrollment by grade; the dashed line marks the 30-tablet supply. *Panel B*: mean PAL usage hours (2 years) by grade with 95% CIs. *Panel C*: ITT estimates on endline math scores. In small schools, effects are uniformly large across grades ($p = 0.85$); in large schools, effects decline with grade ($p = 0.03$). All regressions include controls, strata FE, and school-clustered SEs.

This physical constraint translates directly into usage. In small schools, PAL usage hours are relatively flat across grades (Panel B). In large schools, usage declines at higher grades where congestion is most severe. The treatment effect pattern mirrors this: in small schools, effects are uniformly large across all grades (0.60–0.65 SD) and statistically indistinguishable ($p = 0.89$; Panel C). In large schools, effects decline from 0.42 SD at grade 7 to 0.22 SD at grade 9, and the gradient is marginally significant ($p = 0.07$).

The parallel pattern across all three panels—flat in small schools, declining in large—is consistent with tablet congestion driving the grade gradient. When every student has a device, PAL produces large and uniform gains regardless of grade. The grade gradient in the pooled estimates reflects the fact that larger schools, where older students face the most congestion, pull down the average effect at higher grades. This mechanism has a direct policy implication: providing additional tablets to high-enrollment schools should eliminate the gradient and raise the average treatment effect toward the 0.60–0.65 SD observed in unconstrained settings.

D. Dose-Response

We formalise the congestion mechanism using an instrumental variables strategy. Because all treatment schools received 30 tablets regardless of enrollment, the interaction of treatment status and section enrollment generates exogenous variation in per-student PAL usage: students in larger sections mechanically receive less tablet time. We instrument for usage hours using this interaction (equations 2–3). The first-stage $F = 11.2$ exceeds the conventional Staiger-Stock (1997) rule-of-thumb of 10 and the Stock-Yogo (2005) 15%-maximal-bias threshold of 8.96, while falling below their 10%-bias threshold of 16.38; the reduced form

TABLE 5—DOSE-RESPONSE: IV ESTIMATES USING TABLET CONGESTION

	First Stage Usage (hrs) (1)	Reduced Form Math score (2)	OLS Math score (3)	2SLS Math score (4)
Treatment × Section enrollment	-0.370***	-0.0103***		
	(0.110)	(0.0036)		
PAL usage (hours)			0.0093***	0.0280***
			(0.0018)	(0.0097)
Treatment	48.3***	0.728***		-0.623
	(4.2)	(0.126)		(0.359)
Controls	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes
Grade FE	Yes	Yes	Yes	Yes
Control mean		0.001		0.001
First-stage F	11.2			11.2
Observations	14,123	14,123	6,665	14,123

Notes: Dependent variable in columns (2)–(4): IRT math score standardized to control mean and SD. Column (1): first stage regresses PAL usage hours on treatment × section enrollment and controls. Column (2): reduced form regresses math score on the same instrument. Column (3): OLS of math score on usage hours (treatment group only; biased by selection into usage). Column (4): 2SLS using treatment × section enrollment as instrument for usage hours. The instrument exploits the fact that all treatment schools received 30 tablets regardless of enrollment; larger sections share more students per tablet, reducing per-student PAL time. Section enrollment does not affect control student scores ($\beta = 0.001$, $p = 0.75$; see Figure 7). All regressions include controls, 21 strata FE, and grade FE. Standard errors clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

is consistent with the first stage (Appendix Figure E1); section enrollment has no effect on control student scores ($\beta = 0.001$, $p = 0.75$; Appendix Figure B2), supporting the exclusion restriction.

Table 5 reports the formal estimates. The 2SLS estimate of the causal return to PAL usage is 0.028 SD per hour ($p = 0.005$), approximately three times the OLS estimate of 0.009 SD per hour. The difference is consistent with a complier interpretation: the IV identifies the local average treatment effect for students whose usage is constrained by device scarcity, who have higher marginal returns than the average user. Replicating the IV separately by grade and by gender yields broadly consistent per-hour estimates (0.013–0.029 SD per hour; Appendix Table E2).

Figure 8 characterises the within-school shape of the dose-response. The relationship is S-shaped: approximately flat below 10 hours, steepest between 10 and 30 hours (slopes of 0.019 and 0.013 SD per hour), and flattening to 0.005 SD per hour above 30 hours. The IV LATE of 0.028 SD per hour sits above every segment slope on the curve, consistent with complier selection toward the steep region rather than with a structural per-hour production function. Scaling 0.028 SD linearly to the 36 two-year usage hours

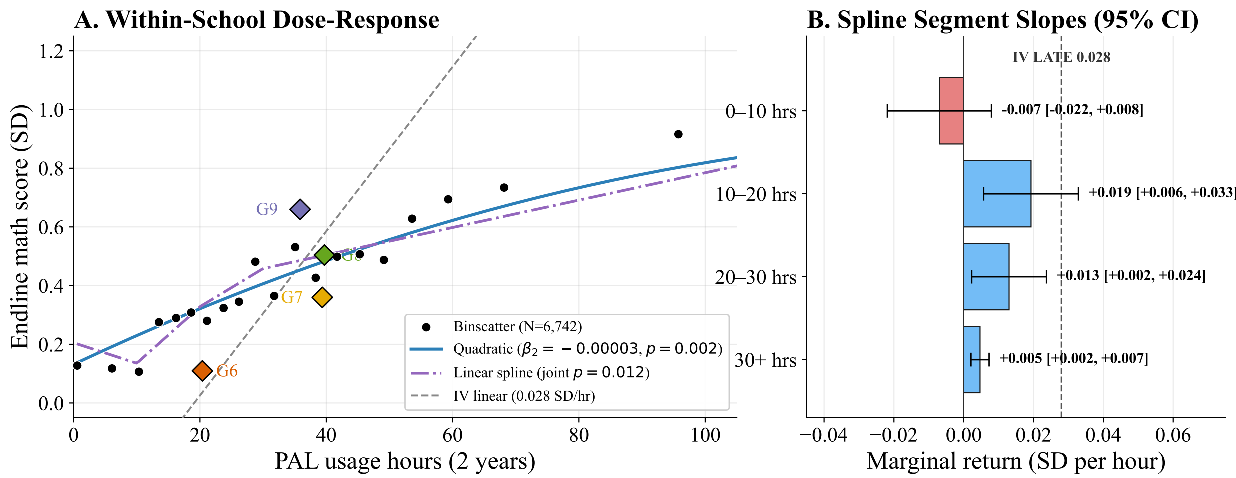

FIGURE 8. DOSE-RESPONSE SHAPE AND THE IV LATE

Notes: Treatment students only, within-school (school fixed effects), school-clustered standard errors. *Panel A*: binscatter of endline math scores by PAL usage hours (20 equal-frequency bins) with quadratic ($\beta_2 = -0.00003$, $p = 0.002$) and linear-spline (knots at 10, 20, 30 hours; joint $p = 0.012$) fits, conditional on covariates. The dashed grey line is the IV linear prediction $0.028 \times$ hrs anchored at the treatment-sample mean. Diamonds mark the raw mean score and mean hours for each grade at endline. *Panel B*: slope of the linear spline by hours range, with 95% CIs (school-clustered). The dashed grey reference marks the IV LATE of 0.028 SD per hour.

observed in grade 9 would predict a 1.0 SD effect, well above the 0.33 SD we estimate — and more generally, grade mean hours (the diamonds in Panel A) bracket the 10–40 hour range that spans both the steep learning zone and the flattening one. The pooled grade gradient therefore reflects more than differences in exposure alone; tablet congestion (Section C) is one contributor, and grade-specific variation in content difficulty and measurement are plausible others that we do not attempt to separate here. Appendix E reports the shape comparison across specifications.

E. Effects on Government Test Scores

Table 6 examines whether treatment effects appear on government-administered math assessments, which are set and graded by individual schools. Effects are positive and significant for several grade-year combinations: grade 6 in Year 2 (0.35 and 0.27 SD on SA1 and SA2), grade 8 in Year 1 SA1 (0.24 SD), and grade 8 in Year 2 SA1 (0.36 SD). Other cells are positive but not statistically significant. Effects tend to be stronger on SA1 (mid-year) than SA2 (end-of-year) assessments, potentially reflecting teachers' awareness of PAL content alignment with mid-year topics.

The grade 10 board exam is the one government outcome graded externally to the schools students attended (unlike the FA/SA assessments, which are school-graded and carry substantial grading heterogeneity across schools; Appendix F). For the cohort with one year of PAL exposure in grade 9, the math board

TABLE 6—EFFECTS ON GOVERNMENT MATH ASSESSMENT SCORES

	Year 1 (AY 2023–24)		Year 2 (AY 2024–25)	
	SA1	SA2	SA1	SA2
	(1)	(2)	(3)	(4)
Grade 6			0.341***	0.268*
			(0.113)	(0.138)
Grade 7	0.138	0.148	0.181*	-0.013
	(0.100)	(0.119)	(0.098)	(0.130)
Grade 8	0.238***		0.364***	0.092
	(0.082)		(0.115)	(0.153)
Grade 9	0.025	0.131	0.118	0.018
	(0.087)	(0.104)	(0.098)	(0.101)
Grade 10†		-0.108		-0.136**
		(0.095)		(0.067)
Controls	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes

Notes: Dependent variable: government math assessment score, standardized within grade to control mean and SD. SA1 and SA2 are mid-year and end-of-year summative assessments administered and graded by schools. Grades 7–9 at endline had PAL for both years; grade 6 had PAL in Year 2 only (new cohort entering AY 2024–25); grade 10 had PAL in Year 1 only (as grade 9 in AY 2023–24, then aged out). †Grade 10 Year 2 column shows the state board exam (externally graded), not SA2. Grade 10 regressions include strata FE only (no student-level controls available). Cells left blank where treatment or control coverage is below 500 students. All other regressions include demographic controls, strata FE, and school-clustered standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

exam shows a negative point estimate (-0.14 SD, $p = 0.04$), significant only with strata fixed effects; no student-level controls are available for this cohort (Appendix Table F1). An investigation of whether the effect reflects PAL displacing exam preparation finds the opposite pattern: it is concentrated among low-usage students, while high-usage students are indistinguishable from control (Appendix Table F2). We report the estimate because the external-grading structure makes it the cleanest government cross-school measure, while the specification sensitivity cautions against treating the negative direction as a settled finding.

Government school assessments have well-documented limitations as measures of absolute learning. The convergent validity between Year 2 SA2 scores and our independently administered endline is modest at the student level ($r = 0.33$) and negative at the school level ($r = -0.13$; Appendix Figure F1), consistent with school-specific grading standards introducing noise. The positive effects we observe on these assessments should therefore be interpreted as broadly consistent with the main IRT results, rather than as precise estimates of treatment effect magnitude.

VI. Cost Effectiveness

The PAL program is funded through state and central (Samagra Shiksha) government education budgets and implemented by the Andhra Pradesh state government's regular school administration. We calculate the government's cost of delivering the program using administrative expenditure data from the 524-school combined sample where state-deployed PAL is in use (444 program schools, 60 RCT treatment schools, and 20 remediation schools). In the first year (AY 2023–24), the per-student cost was \$33 (INR 2,861), which includes one-time lab setup costs: 30 tablets, charging infrastructure, earphones, and classroom preparation. In the second year (AY 2024–25), the per-student cost fell to \$19 (INR 1,616), reflecting recurring costs only: software licensing, Field Management Staff, government monitoring, and equipment replacements. The total cost of delivering PAL over the two-year study period was therefore \$52 per student (\$33 + \$19), or \$26 per student-year.[4] Costs are annualized assuming a 3-year tablet depreciation schedule with annual replacement of headphones and SD cards; software licensing and vendor service rates were set at program inception in 2018 and have remained nominally constant in state contracts since.

Hardware is the single largest cost component, accounting for approximately 36% of annualized expenditure. This cost is effectively recurring: a January 2026 inventory across the 524 schools where state-deployed tablets are in use found that only 57% of tablets remained functional, with 29% of schools reporting zero working tablets. Tablet replacement is driven by both physical degradation (a four-year deployment window for the program rollout cohort, with a two-year COVID-related shutdown from 2020 to 2022) and evolving software requirements—the program has already transitioned from 8-inch to 10-inch devices at higher unit cost. Software licensing accounts for only 5% of costs, though current pricing reflects a volume discount across 524 schools that may not persist. Monitoring and implementation support—Field Management Staff visits, state dashboards, and district follow-up—account for 11%.

To place these costs in context, we express learning gains in two complementary metrics. First, using our own data: in control schools, the cross-grade learning gradient is 0.22 SD per grade, meaning the treatment effect of 0.39 SD is equivalent to approximately 1.8 additional years of learning in the Andhra Pradesh government school system. Second, to enable comparison with the broader literature, we convert to

[4] All costs in 2024–25 prices, converted at \$1 = INR 86. Per-student figures assume 204 students per school, the average across the 80 research schools.

Learning-Adjusted Years of Schooling (LAYS), which expresses gains relative to a high-performance global benchmark of 0.8 SD per year (Angrist et al., 2020; Filmer et al., 2018):

$$\text{LAYS} = \frac{\hat{\beta}}{\delta_h} = \frac{0.39}{0.80} = 0.49$$

The PAL intervention produced learning equivalent to just under half a year of frontier-quality schooling. Given a total program cost of \$52 per student over two years, this translates to:

$$\text{LAYS per \$100} = \frac{0.49}{0.52} \times 1 = 0.94$$

PAL's cost-effectiveness of 0.94 LAYS per \$100 falls within the range reported for adaptive and computer-assisted learning programs (0.24–2.66 LAYS per \$100) and exceeds government-led Teaching at the Right Level in India (0.82 LAYS per \$100) (Angrist et al., 2020). The Global Education Evidence Advisory Panel classified adaptive learning software as "Promising but Limited Evidence," noting that no study had demonstrated cost-effectiveness at scale in the public sector (Global Education Evidence Advisory Panel, 2023). Our estimates suggest that government-delivered adaptive learning can be cost-effective, though it is not among the cheapest interventions per unit of learning gained—the cost is driven by hardware, a general infrastructure constraint rather than a feature specific to the intervention design.

VII. External Validity

Our RCT sits within an active government rollout. The Government of Andhra Pradesh launched PAL in 2019 in a phased design; following a COVID-related shutdown from 2020 to 2022, the program restarted and reached 444 schools across 17 of the state's 26 districts by 2022. By the 2024–25 school year the program serves approximately 135,000 students in grades 6–9 in these schools. The 60 evaluation schools were added to this rollout in AY 2023–24. The external-validity question is therefore not whether a hypothetical scale-up would produce the same effect as our experimental estimate, but what the dose-response characterised in Section D implies for a program that is already operating at scale under different delivery conditions. This section documents the structural dose gap between RCT and program implementation, integrates the within-treatment dose-response over the program's usage distribution, and draws implications for how scarce tablet-time is and could be allocated.

TABLE 7—PAL USAGE: RCT TREATMENT SCHOOLS VS. PROGRAM SCHOOLS

	RCT Treatment (60 schools)	Program (all)	Program (>0 working tabs)
Schools	60	444	375
Students (active on PAL)	6,532	134,090	110,458
Mean enrollment (grades 6–9)	155	326	319
Working tablets (reported, mean)	27.1	19.0	22.5
Peak monthly active students per grade (mean)	26.5	66.5	65.1
Student-tablet ratio per grade-cohort (mean)	1.2	4.4	4.4
Math usage, hours (mean)	35.3	6.9	7.1
Math usage, hours (median)	31.1	5.5	5.7
Months active (mean)	6.9	5.2	5.3

Notes: RCT treatment schools used PAL for math only. Program schools used PAL for math, Telugu, and English (3 subjects). Usage statistics are for math only in both groups, summed across AY 2023–24 and AY 2024–25 (2-year cumulative). "Active" students are those with any recorded PAL usage across either academic year. "Working tablets (reported)" is from government/CG infrastructure records. "Peak monthly active students per grade" is computed per school-grade pair: the maximum, across months of AY 2024–25, of the count of students in that grade with any usage that month; averaged across school-grade pairs. Grade-level disaggregation avoids pooling four cohorts who use the same tablets at different periods. "Student-tablet ratio per grade-cohort" is the mean of grade-cohort enrollment / school working tablets, computed across (school $\times$ grade) cells. A PAL period serves one grade-cohort sharing the school's tablet allocation; the ratio exceeds 1 when the cohort is larger than the working tablet count. For program schools we use student-grade counts as the cohort proxy (section-level enrollment data are not available in program records). Program schools were part of the government's PAL rollout (17 of 26 AP districts) beginning in 2019. RCT schools began implementation in AY 2023–24. All schools—both program and RCT treatment—have a 30-tablet cap regardless of enrollment.

Table 7 compares implementation across the 60 RCT treatment schools and the 444 program schools for which ConveGenius usage data are available. Program students receive roughly one-fifth the per-student math-PAL dose of RCT students—6.9 versus 35.3 hours of cumulative math usage over AY 2023–25—and the gap is structural rather than operational. Program schools have more than twice the grade-6–9 enrollment of RCT schools (326 vs. 155) under the same 30-tablet cap, producing a per-grade-cohort student-to-tablet ratio of 4.4:1 in program schools versus 1.2:1 in the RCT. Program PAL infrastructure splits weekly tablet-time across three subjects: math is allocated two of the four weekly PAL periods, with one each for Telugu and English. The RCT intervention was math-only with two weekly periods, so each tablet supplies roughly half the math time per student in program schools that it does in the RCT. The structural gap is reinforced by tablet age and condition: program-school tablets were deployed at program inception (2019–2022) and have not been systematically replaced, while RCT-school tablets were deployed in 2023 with inventory re-checked in 2024 before the experimental year began; the lower mean working-tablet count in program schools (19.0 vs. 27.1) is consistent with this deployment-age difference. Implementation is otherwise comparable across the two groups: program students attend PAL sessions of similar size to the RCT (median 20 vs. 17 students

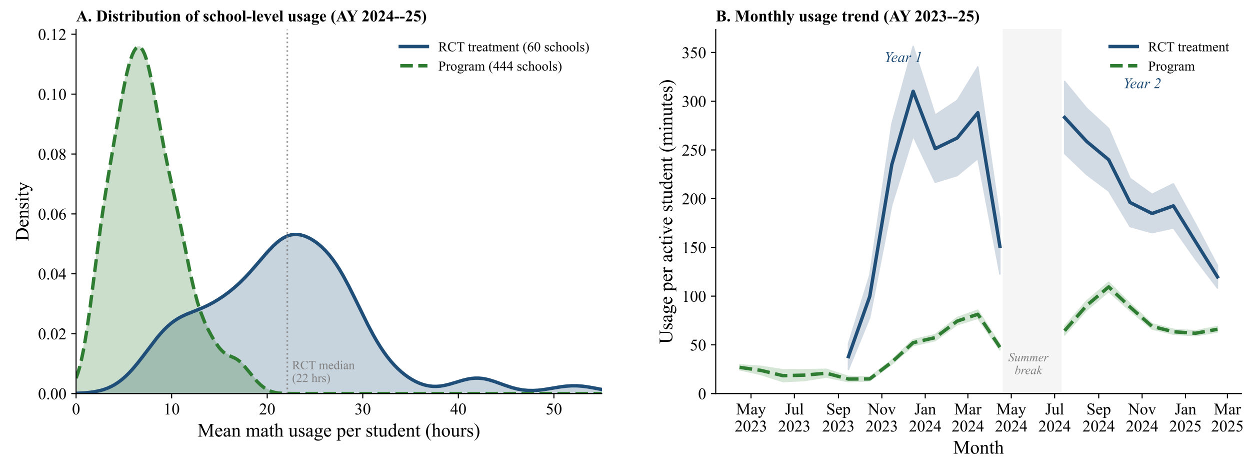

FIGURE 9. USAGE GAP BETWEEN RCT TREATMENT AND PROGRAM SCHOOLS

Notes: Panel A: kernel density of school-level mean math usage in AY 2024–25, RCT treatment schools (blue) and program schools (green). *Panel B*: monthly mean math usage per active student across both academic years, with 95% CIs (shaded) computed from school-level means. Summer break is shaded. Parallel seasonal pattern with a persistent level gap.

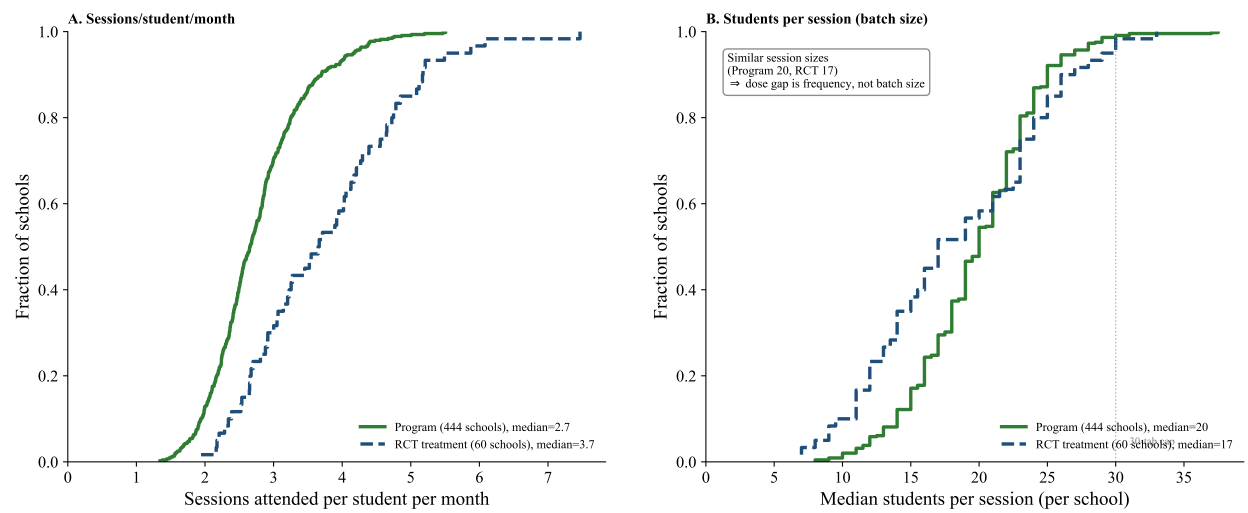

FIGURE 10. SESSION-LEVEL COMPARISON: PROGRAM VS. RCT TREATMENT SCHOOLS

Notes: ECDFs derived from day-level ConveGenius usage data, both arms in AY 2024–25. A session is defined as a (school $\times$ grade $\times$ day) combination with at least 5 active math students. *Panel A*: ECDFs of sessions attended per student per month, computed as school-level means across 444 program schools (green solid) and 60 RCT treatment schools (blue dashed). Program median = 2.7, RCT median = 3.7. *Panel B*: ECDFs of median students per session at the school level. Program median = 20, RCT median = 17; session sizes are essentially equivalent across arms. Together the two panels show that the dose gap is driven by session frequency (Panel A) rather than by batch size (Panel B).

per session; Figure 10, Panel B), engage regularly (2.7 vs. 3.7 sessions per active student per month; Panel A), and show the same seasonal engagement arc within and across academic years (Figure 9, Panel B). The bottleneck is in the tablet-time allocation rather than in operational capacity to deliver PAL.

This structural dose gap places program students in a different region of the dose-response curve than RCT students. Our causal estimator—the 2SLS LATE of 0.028 SD per hour from Table 5—linearly extrapolated

TABLE 8—PREDICTED PROGRAM-SCHOOL EFFECT: LINEAR-IV UPPER BOUND AND WITHIN-TREATMENT-SPLINE LOWER BOUND

Method	Predicted $\mathbb{E}[\tau]$ (SD)
Linear IV $\times$ $\mathbb{E}[H_{\text{program}}]$ (upper bound)	$+0.155$
3-knot spline (strata FE)	$+0.006$
3-knot spline (school FE)	-0.018
5-hr-knot spline (strata FE)	-0.089
5-hr-knot spline (school FE)	-0.093
95% CI (preferred spec)	$[-0.086, +0.047]$

Linear-IV upper bound applies the 2SLS coefficient from Table 5 (0.028 SD per hour) to the program-school mean of 2-year usage ($\bar{H} = 5.5$ hrs; $N = 245,690$ students). Spline specs use the within-treatment piecewise-linear dose-response from Table E1, integrated student-by-student over the program usage distribution. 5-hr knot specs are robustness at finer resolution. School-FE specs absorb between-school composition. The 95% CI uses Monte Carlo draws of the spline coefficients from their joint sampling distribution. Low-dose slopes (0–5 or 0–10 hrs) likely reflect selection on unobservables (students with low usage differ from those with higher usage on engagement and underlying ability), so the spline-based estimates are lower bounds. The true structural effect of program-level dose at scale is not identified without a program-vs-no-program counterfactual.

to the program mean of 5.5 hours implies an expected effect of approximately $+0.15$ SD at scale. This extrapolation requires two assumptions: constant per-hour returns across the 0–40 hour range, and LATE $=$ ATE. The within-treatment shape in Figure 8 shows returns that are approximately flat below 10 hours, steep between 10 and 30 (≈ 0.015 SD per hour), and flattening above 30, suggesting linear extrapolation overstates returns in the 0–10 region where program students sit. Integrating that within-treatment shape over the program usage distribution yields a substantially smaller projected ATE of approximately -0.02 SD (95% Monte Carlo CI $[-0.09, +0.05]$; Table 8, method in Appendix E), though this projection lacks causal identification: the within-treatment slope in the 0–10 hour region is confounded by selection on unobservables among low-usage treatment students whose 0–5 hours of recorded use systematically differ on engagement and unobserved academic preparation from their higher-usage peers. Taken together, the causally identified linear extrapolation puts an upper bound of $+0.15$ SD on the program-scale effect, while the within-treatment shape—descriptive, not causal—suggests returns in the 0–5.5 hour range are likely small. RCT students averaged 34 hours, squarely in the steep-return region, rationalising the pooled 0.39 SD ITT; program students at 5.5 hours are below the activation region where returns accrue under either reading.

A Year-3 endline assessment administered in February–March 2026 provides direct empirical evidence on this projection (Eble et al., 2026, in preparation). Three samples were assessed on a parallel Math Master

Challenge instrument and put on a common IRT scale through 45 anchor items: a random 80-school subset of the parent RCT, a 40-school subsample of the 444-school program rollout, and 40 Kasturba Gandhi Balika Vidyalaya (KGBV) residential schools that received PAL under a separate state expansion in early 2025. The three samples occupy three distinct regions of the dose-response curve. Program-school students average 0.64 math PAL hours over AY 2025–26, with 84% recording zero hours, and post a pooled Year-3 θ_z statistically indistinguishable from the RCT-control mean — exactly what the integrated within-treatment shape projects at near-zero exposure. Among new Grade-6 entrants in the original RCT treatment arm, the within-school dose-response slope on Year-3 hours is $+0.030$ SD per hour, matching the Year-2 IV LATE of $+0.028$ SD per hour almost exactly and demonstrating that the per-hour return is preserved in a fresh cohort one year after the experimental window. KGBV girls average 13.6 hours per student, sit in the steep region of the dose-response curve, and recover the same concave within-school shape with a marginal-return-zero plateau at approximately 41 hours. The cross-sample pooled comparisons are descriptive — the program panel selects schools that returned endline data, KGBV selects on residential eligibility, and prior cohort exposure differs across modes — but the per-hour return and the dose-response shape both reproduce across these distinct delivery contexts.

The proximate driver of the program-school dose collapse is hardware. Three inventory snapshots taken across the 444 program schools document a nine-month attrition rate of approximately 19% in mean working tablets per school between April 2025 and January 2026; by the January 2026 snapshot 34% of program schools have zero working tablets, and 83 schools have moved from a working inventory to zero working tablets over the nine-month window (Figure 11). Because the original 2019–2022 deployment has not been systematically replaced, the dose collapse documented above reflects accumulating hardware attrition layered on top of the structural three-subject, four-grade allocation across a 30-tablet cap. The structural dose gap on its own — the 4.4:1 student-to-tablet ratio reported in Table 7, with math allocated only two of four weekly PAL periods — would predict roughly half the per-student math hours of the math-only RCT configuration even at full hardware availability; the hardware attrition then compresses delivery further, into the near-zero region observed at Year-3.

Read as a governance question, the scaling problem is therefore not whether PAL "works at scale"—an ambiguous question given the shape of the identified returns—but how a government with a fixed tablet stock should allocate tablet-time across subjects and students. The current configuration spreads tablet-time thinly: 135,000 students receive approximately 5 hours of math PAL per year each. Both the causal linear-IV

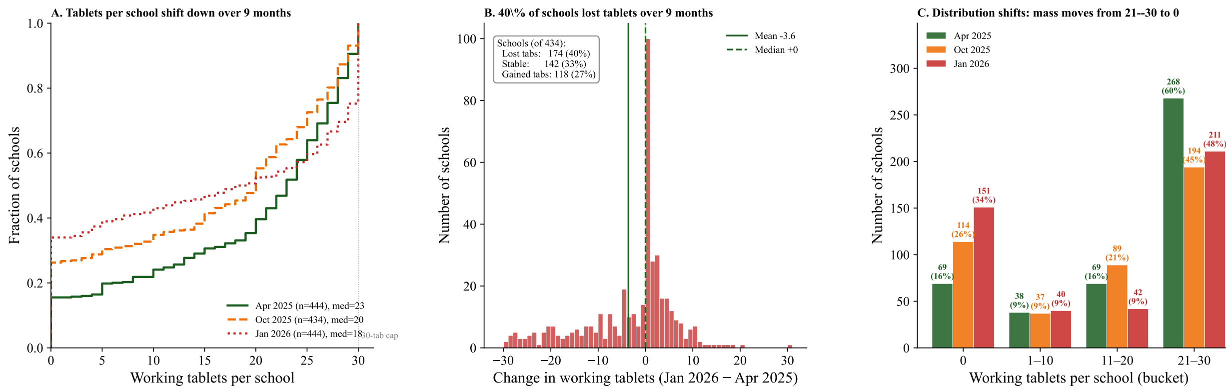

FIGURE 11. WORKING TABLET INVENTORY ACROSS 444 PROGRAM SCHOOLS, 2025–2026

Notes: Three inventory snapshots: April 2025 (end of AY 2024–25), October 2025 (start of AY 2025–26), and January 2026 (mid AY 2025–26). *Panel A*: empirical CDF of working tablets per school at each snapshot. *Panel B*: per-school change between April 2025 and January 2026. *Panel C*: share of schools with zero working tablets at each snapshot. Mean attrition over the nine-month window is -3.6 tablets per school (-19%).

extrapolation and the within-treatment descriptive shape place program-scale gains well below the RCT's 0.39 SD, with the within-treatment shape in particular suggesting returns near zero in the 0–10 hour region where program students sit. An allocation that concentrated a larger share of tablet-time on math—approaching the RCT's math-only design, which places students in the steep 10–30 hour region—would translate the same physical infrastructure into substantially more math learning per student at the cost of reduced dosage on Telugu and English PAL. Reducing section sizes through additional tablets would shift the margin through a different mechanism. The welfare-maximising allocation cannot be identified without data on returns to Telugu and English PAL and without a program-vs-no-program counterfactual. What our evidence does identify is the shape of returns in the 10–40 hour range of math exposure, against which a government can read its own marginal-cost information when choosing an allocation. What it shows is that the current allocation sits below the activation region of the math dose-response, and that the RCT's 0.39 SD gain cannot be projected to program scale without first moving students into that region.

VIII. Discussion

This paper contributes three causal estimates to the literature on adaptive learning at government scale, and a reading of how they fit together. The first, identified by school-level randomisation, is a pooled intent-to-treat effect of $+0.39$ SD on math achievement after two years of state-delivered exposure. The second, identified by the interaction of treatment status with class-section enrollment under a binding 30-tablet-per-school supply constraint, is a per-hour causal return of 0.028 SD. The third, descriptive within the treatment arm, is the shape of the dose-response curve linking cumulative usage to endline ability.

Our +0.39 SD treatment effect compares favorably to recent adaptive-learning evaluations in low- and middle-income countries while operating under markedly lighter delivery scaffolding. The Rajasthan in-school adaptation of Mindspark produced +0.22 SD in math over 18 months (Muralidharan and Singh, 2025); the Khan Academy field experiment in 83 Uttar Pradesh boarding schools produced +0.44 SD over 31 weeks (Oreopoulos et al., 2026). Our 2SLS estimate is of comparable per-hour magnitude to the descriptive within-treatment returns those papers report. We document the largest learning gain for an adaptive learning programme delivered by a state government with no external research-team subsidy, supported by approximately one external field coordinator per twelve schools — substantially lighter scaffolding than either comparison study.

We argue that two well-established learning-process mechanisms jointly account for the size, shape, and heterogeneity of our effect. The first is the Teaching-at-the-Right-Level intuition that students whose foundations are weakest gain additional benefit from explicit below-grade remediation (Banerjee et al., 2007, 2016); this mechanism accounts for the bottom-quintile premium in our heterogeneity — +0.48 SD in the lowest pre-PAL math quintile against +0.31 SD in the highest — with the lowest-baseline students gaining most where the platform's near-universal diagnostic gate exposes them to substantial below-grade content. The second is Vygotsky's Zone of Proximal Development (Vygotsky, 1978), in which learning advances when content sits just above current independent ability with appropriate scaffolding rather than after every prerequisite has been mastered; this mechanism accounts for the bulk of effects across the rest of the distribution, where the majority of platform module-level events sit at the student's enrolled grade and produce learning even for students whose foundational skills are visibly incomplete, provided the platform supplies scaffolding through the diagnostic gate and the completion-based remediation taper. The first describes *which* students need explicit remediation; the second describes *how* grade-level instruction still produces learning when prerequisites are not fully mastered. Our results support a weakened version of the Teaching-at-the-Right-Level rationale — foundational gaps still matter for the lowest-baseline students — but reject a strong version that conditions all progress on prerequisite mastery.

Two item-level patterns describe where on the test the treatment effect manifests. First, the gain is somewhat larger on harder items than on easier ones, consistent with the program helping students reach material they would otherwise leave incorrect rather than reinforcing material they already had right. Second, treatment students outperform controls at every content grade level, with the largest differential on grade 4–6 content.

We test directly whether the platform's two-grade-below-enrolled remediation cap binds. A pre-registered within-school experiment embedded in 20 high-usage treatment schools across two academic years randomly assigned students in grades 6–9 either to the standard platform configuration or to a variant that allowed remediation up to five grade levels below enrolled grade (Cupito et al., 2026, in preparation). The dose contrast is meaningful: students in the deeper-cap arm logged roughly two additional hours per year on content the standard configuration does not render. The aggregate learning effect on the endline math assessment is small (+0.05 SD, $p = 0.09$), concentrated in the Grade 6 cohort and the lowest pre-PAL math tercile, and the Grade 6 gain registers specifically on items one to two grades below enrolled grade rather than at the deepest rendered content level. Deeper remediation has bottom-loaded returns where prerequisite gaps are deepest but does not detectably help students whose foundations are already adequate to access at-grade content with platform-supplied scaffolding. Grade 6 students gained from deeper remediation after one academic year of the experimental contrast, while the Grades 7–9 cohorts show no differential gain after two academic years. The standard configuration's diagnostic gate and completion-based remediation taper, given two academic years of exposure, deliver enough below-grade content to close the foundational-gap channel for the modal middle-school student. The Grade 6 concentration is over-determined: this cohort has the shortest cumulative experimental exposure and the largest deeper-remediation dose under the experimental contrast, and our data are consistent with each as a contributing channel without fully adjudicating among them.

Our Year-3 follow-up on a randomly selected 80-school subset of the impact sample finds the Year-2 treatment-control premium attenuating by approximately half over the third academic year (Eble et al., 2026, in preparation). Treatment students remain above controls on raw percent-correct metrics for identical items, so the pattern reflects narrowing year-on-year gains rather than forgetting of prior gains. The same small state team that monitored implementation in the 60 evaluation schools has, over the same period, managed the original 444-school rollout (which has since grown to approximately 1,200 schools across 26 districts) and, in early 2025, launched a high-intensity adaptive-learning expansion to roughly 700 government residential schools. With limited central monitoring bandwidth committed to the new expansion, a substantial subset of the original treatment schools delivered effectively zero PAL in the third year. For new Grade-6 entrants in schools that did continue to deliver the program, the within-school dose-response slope reproduces the Year-2 per-hour estimate almost exactly. The Year-3 attenuation is therefore not evidence that the program, the schools, or the teachers have failed — it is evidence that, without active state-team monitoring, headmasters and teachers do not yet sustain implementation autonomously, a transition that, for an intervention this new in this delivery modality, the system has not yet completed.

The per-student two-year delivery cost of $52 places this programme within the published range of adaptive learning interventions on a cost-effectiveness basis. At program scale, however, students average 5.5 hours of math PAL per year rather than the ~ 17 hours per year accumulated in our evaluation schools, and the cross-sample evidence reported in Section VII shows that the per-hour return is preserved across delivery contexts that successfully move students into the activation region of the dose-response. The marginal policy investment that would produce additional learning is therefore reallocation toward more engaged hours per student rather than more sophisticated personalisation: smaller per-tablet ratios, longer cumulative exposure within the activation-to-saturation band of the dose-response, and more reliable hardware delivery in schools where the program currently produces little usage. The Kasturba Gandhi Balika Vidyalaya residential schools reported in Section VII demonstrate that high-dose PAL delivery within the AP government system is operationally feasible. The harder margin is sustaining that dose in regular government schools, where shared tablet stock across subjects, structurally larger cohorts, and accumulating hardware attrition together compress per-student exposure to a level below the activation region of the dose-response curve.

IX. Conclusion

In this paper, we report the results of a large randomized controlled experiment evaluating the impacts of a personalized adaptive learning program on math achievement of middle school students in Andhra Pradesh, India. We find that the program had large, sustained effects on math learning, particularly for girls and early grade students. We find some evidence of salutary effects on performance on government exams, and we show that this is an intervention that is highly cost-effective and compares favorably to others in the literature. As a whole, the paper shows that technology can be used to deliver high-fidelity learning interventions, even when administered within government systems by government teachers.

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Appendices

TABLE A1—ASER-EQUIVALENT CLASSIFICATION, CONTROL VS TREATMENT ENDLINE

	Subtract, not divide	Cannot subtract	Both ($\geq$ G3 arithmetic)	n
Y2 grade 7–8 control endline	16.4%	28.8%	54.8%	3,609
Y2 grade 7–8 **treatment** endline (2yr PAL)	13.0%	25.5%	**61.5%**	3,368
Treatment − Control	*−3.5 pp*	*−3.3 pp*	*+**6.7 pp***	—
ASER 2024 AP grade 8 (ASER Centre, 2024)	30.9%	14.2%	48%	—

Notes: Subtraction anchor: g2q3 (12−4 word problem). Division anchor: any of g3q3 (250 ÷ 10) or g3q5 (65 ÷ 5). ASER "11–99" floor refers to students who can recognise 11–99 but not subtract.

A. Translating the ITT into the ASER 2024 Classification

This appendix maps the items in our Math Master Challenge to the foundational-skill classification used by the ASER survey, replicates the classification on our control endline, and re-expresses the two-year ITT in the same categorical language used by ASER and by national policy audiences.

ASER Centre (2024) reports for rural Andhra Pradesh grade 8 that 48% of students can perform basic division (down from 52% in 2022), 30.9% can subtract but cannot divide, and 14.2% are stuck at the "recognise 11–99" floor. We replicate the classification on our control endline samples to anchor our control arm's foundational-skill profile against this published reference, and translate the two-year ITT into the same categorical language.

Anchor items. We identify items in our test by *operation content* rather than item-grade label, since calibrated IRT difficulty in our instrument is not monotonic in the question's grade tag. The subtraction anchor (g2q3, "Lata has 12 chocolates and gives 4 to her friend") is a word problem without borrowing. The division anchors (g3q3, 250 ÷ 10; g3q5, 65 ÷ 5) are clean two-digit-by-one-digit divisions without remainders, materially easier than ASER's three-digit-by-one-digit-with-remainder protocol.

Headline. Two years of PAL raise the share of treated grade 7–8 students with full grade-3 arithmetic mastery from 54.8% in control schools to 61.5% in treatment schools, a 6.7-percentage-point gain that more than reverses the 4-percentage-point decline in basic-division mastery ASER documents for rural AP grade 8 between 2022 and 2024 (52% → 48%). The shift breaks down into roughly equal parts: −3.3 pp move off the "cannot subtract" floor, and −3.5 pp clear the subtraction-to-division cliff—both flowing into the "both-operations" bucket.

Why control levels differ from ASER. Our control "both-operations" rate (54.8%) is 6.8 pp higher than ASER's "can divide" rate (48%), consistent with our division anchor being easier than ASER's. Our higher "cannot subtract" rate (28.8% vs 14.2% at ASER's 11–99 floor) is consistent with two contrasts: our 100%-government-school sample lacks the higher-performing rural private-school enrollees ASER's household-based survey includes, and our subtraction item is a word problem requiring reading comprehension while ASER's is bare arithmetic.

TABLE B1—ENDLINE ATTRITION ANALYSIS AND LEE BOUNDS

	Pooled	Grade 6	Grade 7	Grade 8	Grade 9
Panel A: Attrition Rates					
Treatment	0.237	0.221	0.210	0.234	0.275
Control	0.198	0.186	0.201	0.185	0.216
Difference	0.033^{**}	0.035	0.009	0.049	0.059
	(0.013)				
N (enrolled)	18,039	4,117	4,186	4,567	5,169
Panel B: Differential Attrition Predictors					
	Covariate effect	Treatment × Covariate	p (interaction)	N	
Female	-0.0405^{***}	-0.0279^{**}	0.042	18,039	
Age	0.0233^{***}	0.0120^{*}	0.073	18,039	
Urban	-0.0081	0.0902^{**}	0.015	18,039	
Baseline math (SA1)	-0.0030^{***}	0.0008	0.186	10,638	
Panel C: Lee Bounds on Pooled ITT					
Original ITT			0.390^{***} (0.054)		
Lower bound			0.270^{***} (0.047)		
Upper bound			0.481^{***} (0.051)		
Trim proportion			0.049 (332 students)		

Notes: Panel A: attrition rates from the enrollment frame (18,039 students in 120 schools) to endline assessment completion. Difference from OLS of attrition indicator on treatment with strata FE and school-clustered SE. Panel B: each row regresses the attrition indicator on treatment, the baseline covariate, and their interaction (plus strata FE). The interaction tests whether attrition is differentially predicted by the covariate across treatment and control. Panel C: Lee (2009) bounds trim the treatment group outcome distribution to equalize attrition rates. The lower bound trims the top of the treatment distribution; the upper bound trims the bottom. Both bounds being positive rules out attrition bias as an explanation for the main treatment effect. All regressions include strata FE. Panel C includes demographic controls and grade FE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B. Robustness of the Main ITT

This appendix confirms that the 0.39 SD headline estimate survives differential attrition, alternative control sets, multiple-testing corrections, non-parametric inference, and the congestion IV's exclusion restriction.

Table B1 presents attrition analysis and Lee (2009) bounds. Differential attrition is modest (3.3 percentage points, treatment higher) but statistically significant. Lee bounds remain large and positive (0.27–0.48 SD), ruling out attrition bias as an explanation for the main treatment effect.

We assess the sensitivity of the main ITT estimate to alternative specifications and inference procedures. The treatment effect is stable across three control specifications (0.38–0.39 SD; Table B2). The gender interaction survives Bonferroni, Holm, and Benjamini-Hochberg corrections (Table B3). Randomization inference, which does not rely on asymptotic assumptions, confirms the main result (RI $p < 0.001$; Figure B1).

Figure B2 visualises the exclusion-restriction placebo for the congestion IV. We plot raw within-grade binned means of endline math scores against section enrollment, separately for treatment and control. Conditioning on grade, controls, and strata FE, section enrollment has a negative effect on T scores ($\beta = -0.007$, $p = 0.07$) but is flat for C ($\beta = +0.001$, $p = 0.75$). The flat control line is direct visual evidence that section size does not affect learning in the absence of PAL, supporting the exclusion restriction.

TABLE B2—SENSITIVITY OF ITT TO CONTROL SPECIFICATION

	No controls (1)	Demographics only (2)	Demographics + baseline (3)
Treatment	0.379***	0.390***	0.393***
	(0.054)	(0.054)	(0.055)
R^2	0.106	0.137	0.238
Observations	14,215	14,215	14,215

Notes: Dependent variable: IRT math score standardized to control. Column (1): treatment + grade FE + strata FE only. Column (2): adds demographic controls (SES, parental education, age, gender). Column (3): adds within-school×grade baseline ability percentile (28.8% missing, imputed at median with missing flag). The treatment effect is stable across specifications. All regressions include strata FE and school-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE B3—MULTIPLE HYPOTHESIS TESTING CORRECTIONS

Hypothesis	Unadjusted p	Bonferroni	Holm	BH (FDR)
Gender	0.0005***	0.0049	0.0049	0.0049
Grade 7 interaction	0.3680	1.0000	0.9208	0.3680
Grade 8 interaction	0.2733	1.0000	0.9208	0.3514
Grade 9 interaction	0.1236	1.0000	0.6181	0.2225
Baseline Q2 interaction	0.0064***	0.0580	0.0516	0.0290
Baseline Q3 interaction	0.2302	1.0000	0.9208	0.3453
Baseline Q4 interaction	0.0607*	0.5460	0.4247	0.1681
Baseline Q5 interaction	0.3494	1.0000	0.9208	0.3680
Urban	0.0747*	0.6723	0.4482	0.1681

Notes: Unadjusted p-values from treatment × subgroup interaction tests. Bonferroni and Holm control family-wise error rate. BH controls false discovery rate. All from regressions with controls, strata FE, grade FE, and school-clustered SE.

A. *Tablet vs. paper assessment-mode equivalence*

A natural concern with our tablet-administered endline is that PAL exposure may inflate measured ability through tablet familiarity rather than learning, biasing the headline ITT upward. We test this directly using a sub-study embedded in the Year-3 endline (AY 2025–26) that re-tested the *same* G8 RCT students on *the same items* using both modes. Each of 1,953 G8 students in our 80 RCT schools sat both a tablet (SurveyCTO) and a paper (OMR) Math Master Challenge in randomised order, with a counter-balanced crossover assigning opposite question sets to the two modes (Tab Set 1 students received OMR Set B and vice versa). Twenty-six "common" items appear on both modes for every student, providing within-student same-item measurement; 40 set-specific items contribute cross-form information through the IRT.

Table B4 reports treatment-effect specifications analogous to the main paper's Table 3, with mode (tablet vs. paper) and its interaction with treatment as the key terms. Panel A uses 2PL IRT θ from a pooled fit across all items administered in the dual-mode design; Panel B restricts the IRT to the items every student answered on both modes (a fully balanced design that does not rely on cross-form linking). Across all six (panel × specification) cells, the mode × treatment interaction lies between -0.09 and $+0.003$ SD and is

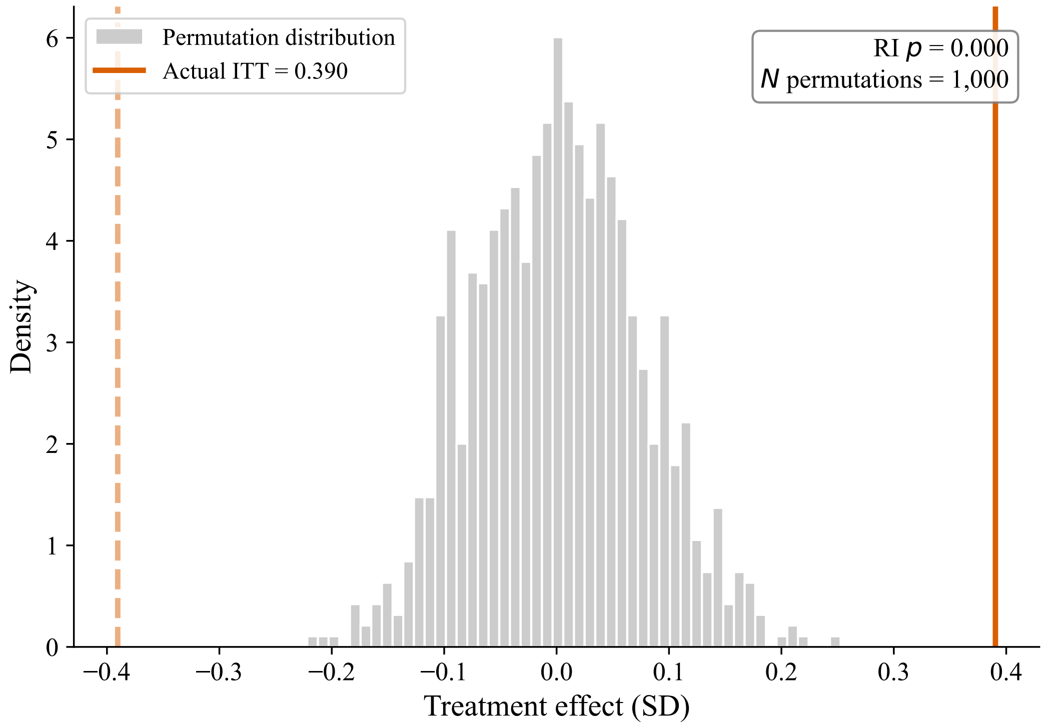

FIGURE B1. RANDOMIZATION INFERENCE: PERMUTATION DISTRIBUTION OF TREATMENT EFFECTS

Notes: Distribution of treatment effects from 1,000 random permutations of treatment assignment within strata (at the school level). The solid vertical line shows the actual ITT estimate. The RI p-value is the fraction of permuted effects at least as extreme (in absolute value) as the actual effect. This non-parametric test does not rely on asymptotic assumptions about the error distribution.

not statistically significant in any specification. The mode main effect is at most -0.09 SD, with $p \geq 0.13$ throughout.

A model-free verification on the common items reaches the same conclusion: mean percent correct is 50.4% on tablet and 50.7% on paper, with a within-student Pearson correlation of 0.83 on per-student common-items scores — indicating coherent measurement of the same latent ability across modes. The DiD interaction on percent correct is -1.32 pp (SE 0.91, $p = 0.15$), consistent in sign and SD-equivalent magnitude with Panel A and Panel B.

We conclude that mode of administration does not differentially advantage treatment students on the tablet endline. The tablet-familiarity channel cannot account for the headline $+0.39$ SD ITT.

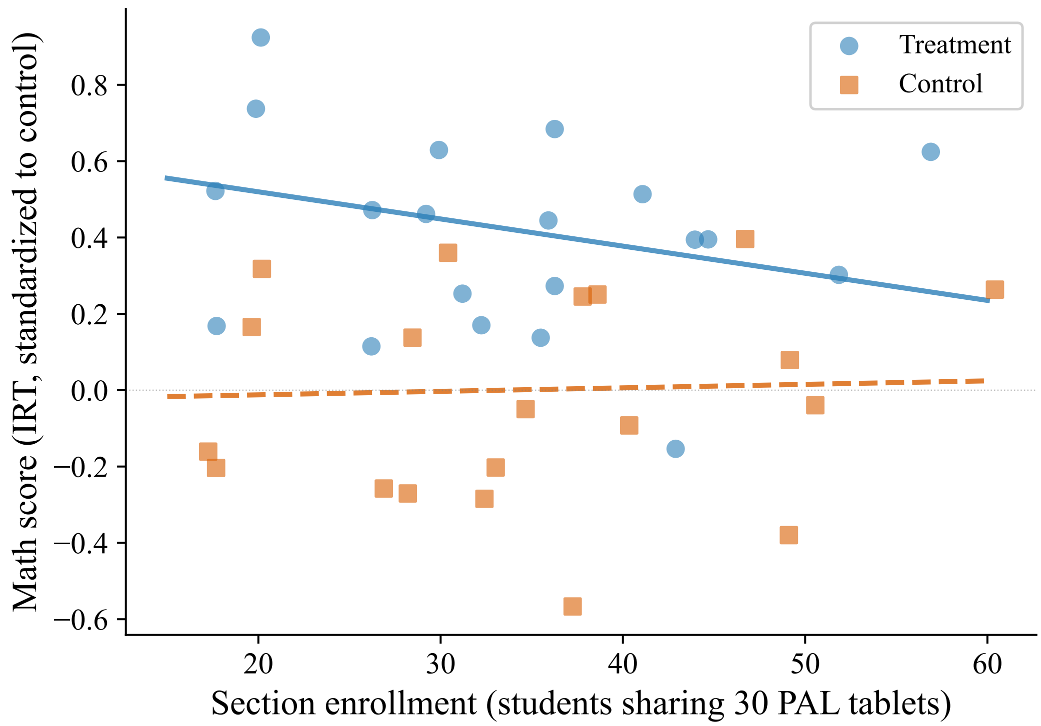

FIGURE B2. EXCLUSION-RESTRICTION PLACEBO: SECTION ENROLLMENT AND MATH SCORES, TREATMENT VS. CONTROL

Notes: Dots: raw within-grade bin means of endline IRT math scores (5 bins per grade, pooled across grades 6–9). Lines: regression-predicted slopes from full specifications with controls, strata FE, grade FE, and SE clustered at the school level. Treatment slope: $\beta = -0.007$, $p = 0.07$ (larger sections reduce T scores, consistent with congestion). Control slope: $\beta = +0.001$, $p = 0.75$ (flat, consistent with section size not affecting learning in the absence of PAL). The flat control line provides visual evidence for the exclusion-restriction placebo on the congestion IV (§D).

TABLE B4—G8 MODE COMPARISON: TABLET VS PAPER ASSESSMENT (Y3 RCT STUDENTS)

	(1) Model 1 (main)	(2) Model 1 (robust)	(3) Model 2 (1st exam)
Panel A. Primary specification: 2PL IRT on 67 items			
1.mode	0.006	0.006	-0.063
	(0.031)	(0.031)	(0.062)
1.mode × 1.treatment	-0.092	-0.094	-0.019
	(0.060)	(0.060)	(0.094)
R^2	0.914	0.913	0.095
Panel B. Robustness: 2PL IRT on 26 common items only			
1.mode	-0.038	-0.038	-0.094
	(0.030)	(0.030)	(0.062)
1.mode × 1.treatment	-0.070	-0.071	0.003
	(0.045)	(0.045)	(0.090)
R^2	0.922	0.922	0.087
N	3,906	3,886	1,953
Fixed effects	Student	Student	Mandal cluster

Notes: Outcome is θ standardised to the control × tablet reference cell (mean 0, SD 1). Panel A uses 2PL IRT θ from a pooled fit across all items administered in the dual-mode design; Panel B uses 2PL IRT θ from a fit restricted to the items every G8 RCT student answered on both modes (a fully balanced spec). Models 1 use student fixed effects; Model 2 (first exam) uses mandal-cluster strata FE plus covariates (gender, age, SES, parental education with missing flags). Standard errors clustered at school. A negative interaction would indicate that treatment students gain a relative tablet advantage. * p<0.10, ** p<0.05, *** p<0.01.

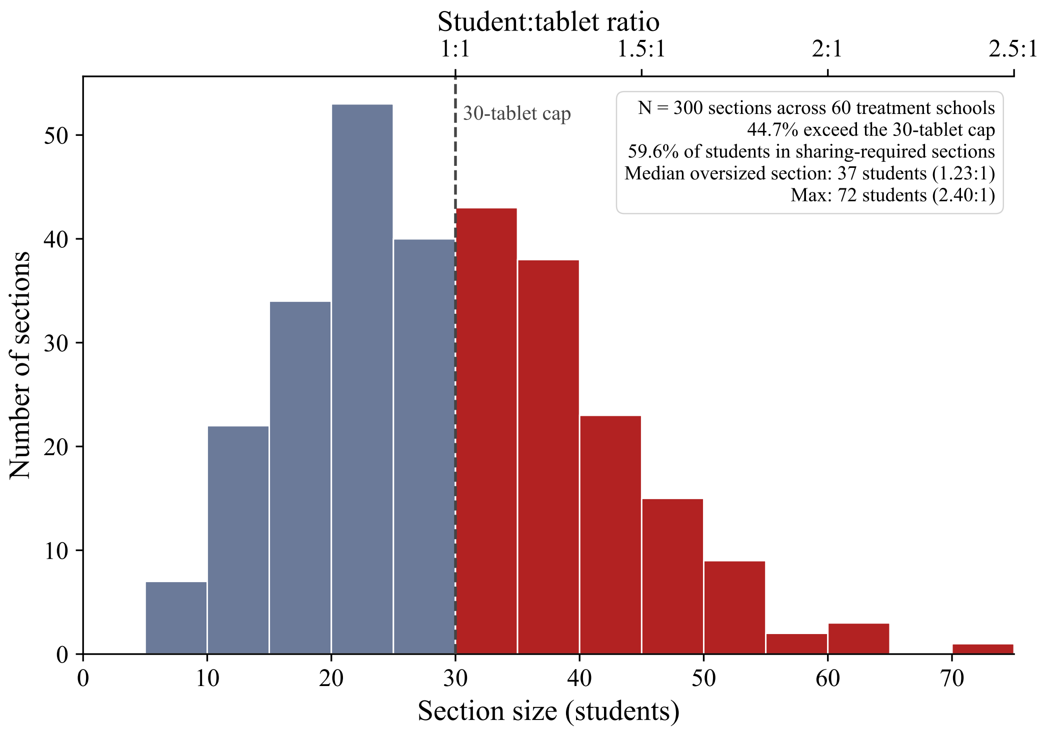

FIGURE C1. CLASS SECTION SIZES RELATIVE TO THE 30-TABLET CAP

Notes: Treatment schools only; $N = 300$ class sections across 60 schools, defined as unique school-grade-section combinations. Section enrolment was missing for 1.1% of treatment students; these were imputed using the mean section enrolment within each student's school-grade cell. Sections ≤ 30 students (blue-grey, 166 of 300) could accommodate every student on a dedicated tablet; sections > 30 (red, 134 of 300) required sharing. The top axis shows the implied student-to-tablet ratio. 59.6% of treatment students were in a sharing-required section.

C. Implementation: Tablets, Effort, and the Chapter Pipeline

This appendix documents how PAL is actually delivered — the 30-tablet cap and within-section sharing, the split between active practice and passive viewing, student-reported implementation challenges, and the standardised six-stage module sequence that proceeds uniformly across ability levels.

Figure C1 shows the distribution of class-section sizes across the 60 treatment schools. Because every treatment school received 30 tablets regardless of section enrollment, sections above 30 students faced a mechanical sharing constraint: 44.7% of the 300 treatment sections exceed this count, with the median oversized section holding 37 students (a 1.23:1 student-to-tablet ratio). 59.6% of treatment students were in a sharing-required section.

Students spend slightly more time watching videos (54%) than answering questions (46%; Table C1). Eighty percent of treatment students report enjoying PAL "a lot," though 37% report language comprehension and video loading as implementation challenges (Table C2).

A. PAL Instructional Pipeline

Using module-level platform data (8.78 million learning events across 12,019 students in the 60 treatment schools), we reconstruct the instructional pipeline that students experience within each chapter. Figure C2 shows the standardised six-stage sequence: a short diagnostic on prerequisite content, below-grade remediation (videos and practice), a baseline assessment at grade level, at-grade instruction with formative knowledge checks, and a chapter endline. The chapter sequence within each grade is fixed and identical for all students.

TABLE C1—ACTIVE VS. PASSIVE LEARNING TIME (TREATMENT GROUP ONLY)

	Overall	Grade 6	Grade 7	Grade 8	Grade 9
Total hours	34.2	20.4	39.4	39.7	35.9
Video hours	18.9	11.2	21.5	21.6	20.6
Question hours	15.3	9.2	17.9	18.1	15.3
% Video	53.6%	51.7%	52.6%	52.8%	56.9%
N	6,742	1,540	1,599	1,769	1,834

Notes: Treatment group only. Total hours is cumulative PAL usage across 2 academic years. Video hours = passive content viewing. Question hours = active practice (answering questions). Grade 6 had 1 year of PAL; grades 7–9 had 2 years.

TABLE C2—PAL EXPERIENCE (TREATMENT GROUP ONLY)

Panel A: Student experience (% reporting)	
Enjoy PAL lab "a lot"	80.3%
PAL helped understand math "a lot"	73.2%
Like watching PAL videos	87.5%
Prefer additional PAL period over math	66.3%
PAL less difficult than regular math	57.5%
Prefer English in PAL lab	52.1%
Always log in with own credentials	74.3%
Always get teacher help to log in	34.4%
PAL classes per week, mean (SD)	2.6 (1.4)
Panel B: Implementation quality (% reporting no problem)	
Video quality is good	88.4%
Knows how to use PAL	87.1%
Enough time allotted	84.1%
Enough tablets available	78.5%
Screen easy to see	77.3%
Lab not too noisy	74.5%
Headphones work	73.8%
No login problems	70.7%
Videos load fine	63.4%
Language understandable	62.8%

Notes: Treatment group only ($N \approx 6{,}740$). Panel A: percentage of students giving the top/positive response to each PAL experience question from the endline student survey (tablet-based, administered by independent enumerators). Panel B: percentage reporting no problem for each implementation dimension (binary: 1=good/no problem, 0=has problem). The most common implementation challenges are language comprehension (37% report problems) and video loading speed (37%).

Figure C3 shows the content grade progression over time, tracking five cohorts defined by their enrolled grades across the two study years. All cohorts begin approximately 1.4 grade levels below their enrolled

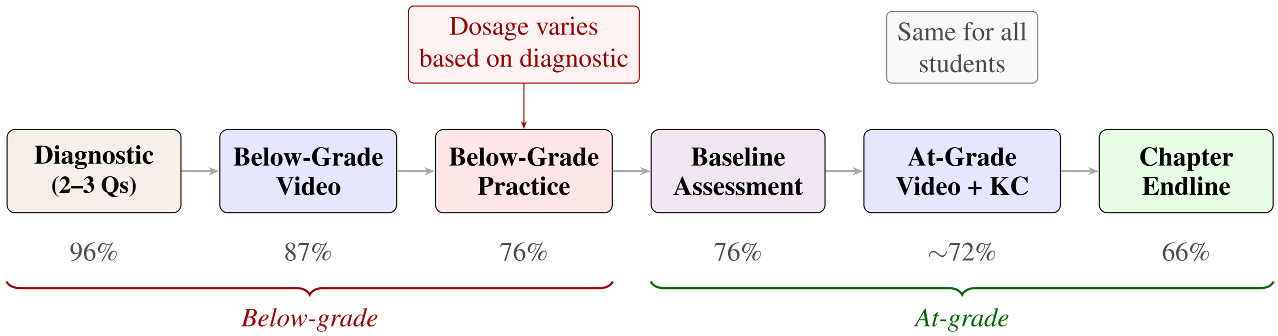

FIGURE C2. PAL CHAPTER PIPELINE

Notes: Standardised chapter pipeline based on 246,829 chapter journeys across 12,019 students. Percentages indicate the fraction of students reaching each stage. The below-grade practice stage (highlighted) is the primary locus of dosage variation; the at-grade stages are uniform across students.

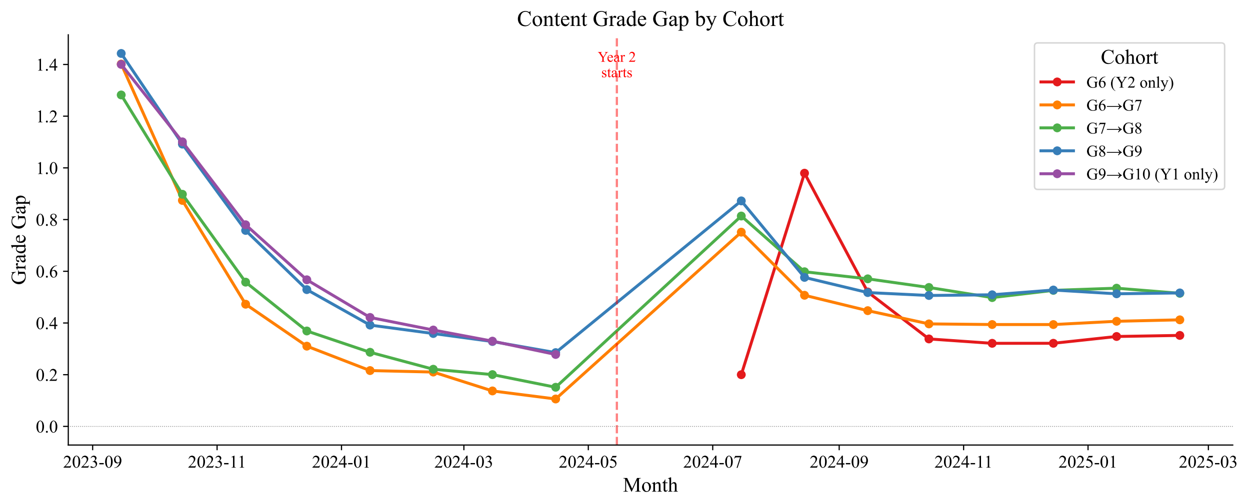

FIGURE C3. CONTENT GRADE GAP BY COHORT

Notes: Grade gap = student's enrolled grade − average content grade of modules used that month. Each line tracks a cohort across its years of PAL exposure: three cohorts (G6→G7, G7→G8, G8→G9) span both academic years; the G9→G10 cohort aged out after Year 1; the Grade 6 cohort entered in Year 2 only. $N = 10{,}920$ students across 60 treatment schools (excluding repeaters and mid-year dropouts).

grade and converge toward grade-level material over roughly four months. At the start of Year 2, the system resets continuing cohorts back to below-grade content regardless of Year 1 progress. The Grade 9→10 cohort appears only in Year 1 (students aged out after grade 9); the new Grade 6 cohort appears only in Year 2.

Table C3 shows that the student experience is largely uniform across the baseline ability distribution. Total usage hours, the video–practice split, and the share of below-grade content are all flat across baseline ability quintiles. The Q5–Q1 gap in video share is 3 percentage points; in below-grade content share, 4 percentage points. Figure C4 provides the most direct evidence: the content grade progression for the bottom and top terciles of baseline ability overlaps almost completely in both years. The system moves all students through the same content at the same speed.

TABLE C3—PAL USAGE BY BASELINE ABILITY

Baseline quintile	Total usage (hours)	Video (hours)	Practice (hours)	Video share (%)	Below-grade content (%)
Q1	34.1	19.2	15.0	56	36
Q2	34.3	19.2	15.0	56	35
Q3	30.8	17.3	13.5	56	35
Q4	35.6	19.6	16.0	55	34
Q5	35.9	19.2	16.7	53	33
All	34.2	18.9	15.3	55	35
Q5 − Q1	+1.8			-3	-4

Notes: Treatment students with non-missing baseline scores and PAL usage data ($N = 6,742$). Total usage is the sum of video and practice hours over two academic years (2023–24 and 2024–25). Below-grade content share is the average fraction of modules within each chapter journey that cover content below the student's enrolled grade.

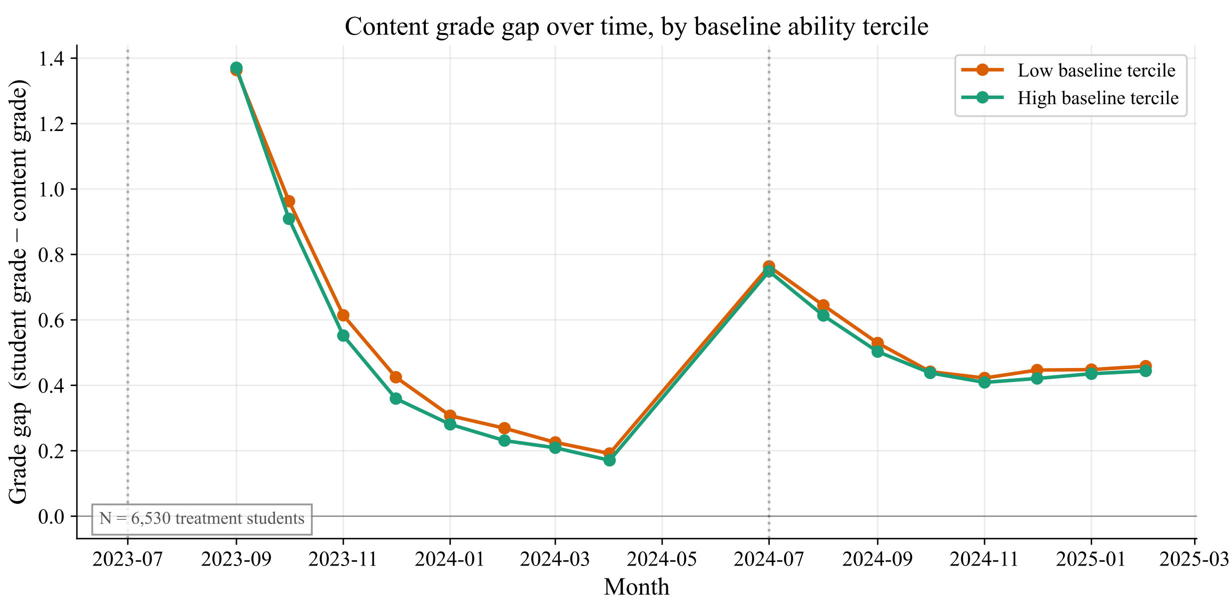

FIGURE C4. CONTENT GRADE PROGRESSION BY BASELINE ABILITY

Notes: Grade gap = student's enrolled grade − average content grade used that month. Bottom and top terciles defined by within school × grade percentile rank of government math score. Lines overlap almost completely in both years, indicating identical progression trajectories regardless of actual ability. N = 12,019 students across 60 treatment schools.

D. Heterogeneity: Gender, Baseline Ability, and Data Coverage

This appendix resolves why girls gain more than boys (usage quantity, not per-hour productivity), documents the across-quintile gradient in baseline ability, and rules out that the gradient is a data-coverage artefact of asymmetric baseline extracts.

We examine treatment effect heterogeneity along several dimensions. Girls gain more than boys (+0.46 vs +0.33 SD), and this advantage persists across all grades (Figure H1). Effects are uniform across SES terciles ($p = 0.62$; Table D2) but decline across baseline ability quintiles, with the largest gains for the lowest-baseline students (Figure D1, Panel C). Girls accumulate more PAL usage hours and report higher engagement than boys (Figure D2); Table D1 provides the full regression results for the gender gap diagnosis.

Baseline ability in our regressions is the student's within-school $\times$ grade percentile rank on the pre-PAL government math SA1 assessment, with a cohort-specific source: grade 7–9 at endline use AY 2022–23 SA1, while grade 6 at endline (new cohort, PAL exposure only in 2024–25) use AY 2023–24 SA1 — their grade-5 pre-PAL year. Using 2022–23 for grade 6 would measure them as grade 4 students two years pre-endline.

The large treatment effect for the "Missing baseline" subgroup in Table 4 (0.50 SD) reflects overlapping data-coverage issues, not a true subgroup effect, and these issues differ by cohort. For the 2-year cohort (grades 7–9 at endline), missingness reflects a **data-extraction asymmetry**: the AP government delivered AY 2022–23 data for the 524 program + treatment schools in one extract and the 60 control schools in another, with no overlap. In grade 7, this gives 87–89% coverage for treatment vs. 26–29% for control (Table D3, Panel A); in grades 8–9, coverage is balanced at 78–93% in both arms. For the 1-year cohort (grade 6 at endline), missingness is **structural and not closed by choice of baseline year**: grade 6 students are brand-new entrants to the study schools in 2024–25, so their pre-PAL records live in whatever government school they attended as grade 5. Treatment grade 6 students came from the 444 program schools (captured by the 524-school extract, $\sim$97% coverage), while control grade 6 students came from schools outside the 60-school control extract, yielding only $\sim$29% coverage. Switching the grade 6 baseline source from 2022–23 to 2023–24 slightly improves total coverage (55% $\rightarrow$ 61%) but does not close the treatment–control gap, because both years draw from the same asymmetric extracts.

A linear probability model confirms that missingness is predicted entirely by the treatment $\times$ grades 6–7 interaction and school category, not by student characteristics (Panel B). Restricting to grades 8–9—where coverage is balanced—the treatment effect among missing-baseline students is 0.46 SD; in grades 6–7, where coverage is asymmetric or structural, it is inflated to 0.56 SD (Table D4). The inflated pooled estimate reflects compositional imbalance in the missing-baseline subgroup, not a real effect on students without baseline.

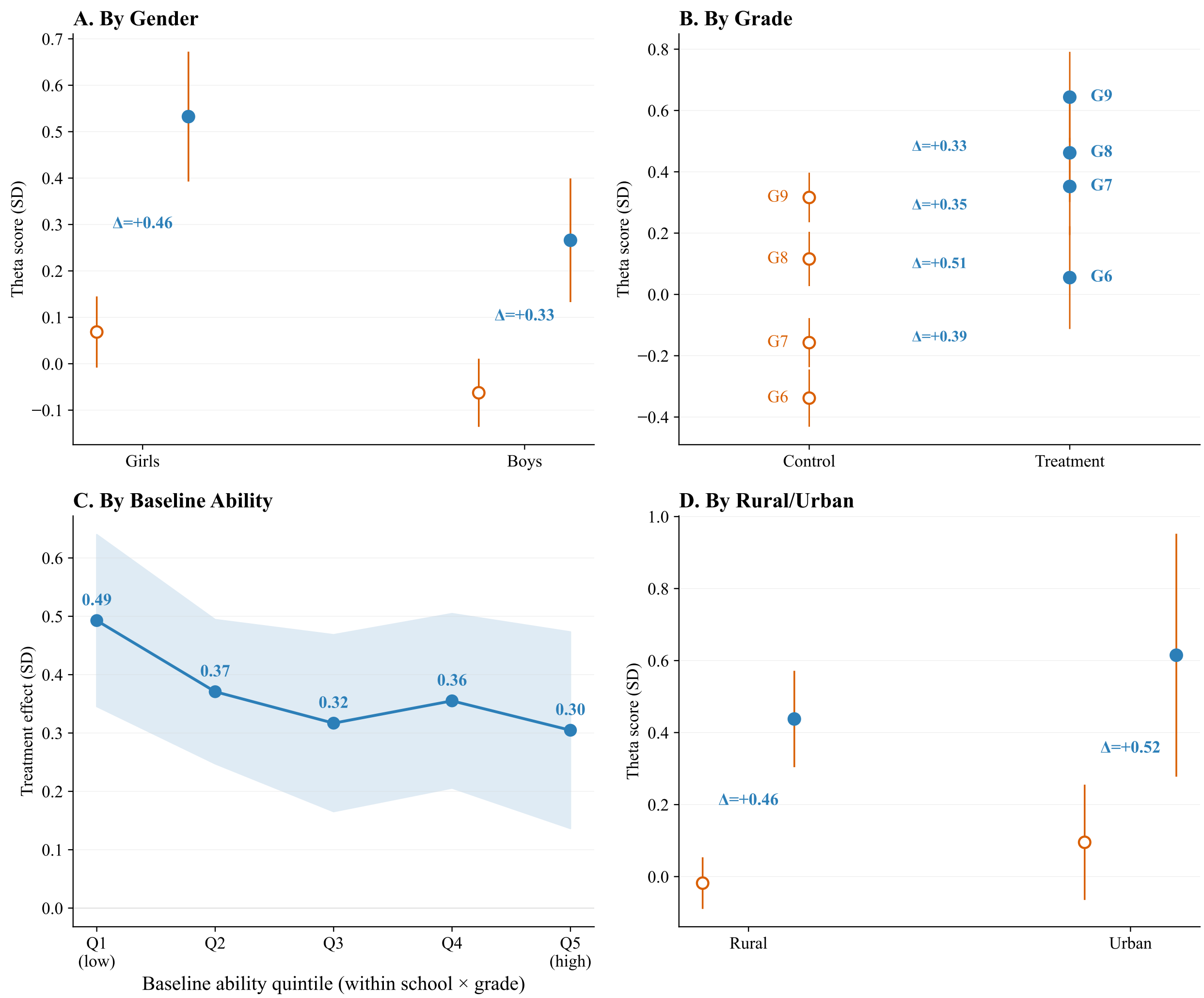

FIGURE D1. HETEROGENEOUS TREATMENT EFFECTS BY SUBGROUP

Notes: Open circles show control group means; filled circles show adjusted treatment group means (control mean + regression-adjusted treatment effect); whiskers show 95% school-clustered CIs. Δ annotations show the ITT estimate from regression with controls and strata FE. *Panel A*: Girls start slightly higher and gain more (+0.46 vs +0.33 SD). *Panel B*: Control scores rise with grade (older students know more) while treatment effects decline — consistent with the congestion mechanism (Figure 7). *Panel C*: Treatment effects (line with 95% CI band) decline across baseline ability quintiles (within school × grade rank), with the largest gains for the lowest-baseline students. *Panel D*: Rural and urban schools show similar effects. Panels A, C, D include grade FE; Panel B estimates are within-grade.

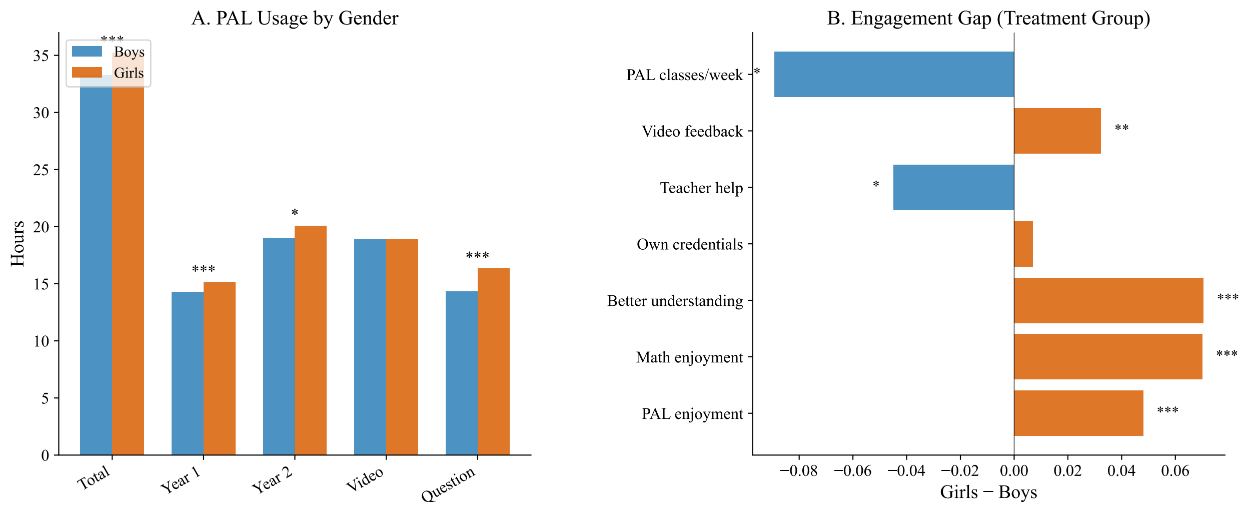

FIGURE D2. GENDER DIFFERENCES IN PAL USAGE AND ENGAGEMENT

Notes: Panel A: Mean PAL usage hours by gender (treatment group only). Girls accumulate more total hours than boys, driven primarily by more time on questions (active practice) rather than videos. *Panel B*: Gender gap in engagement measures (girls − boys, treatment group). Girls report higher PAL enjoyment, math enjoyment, and perceived understanding. Significance stars from regressions with strata FE and school-clustered SE.

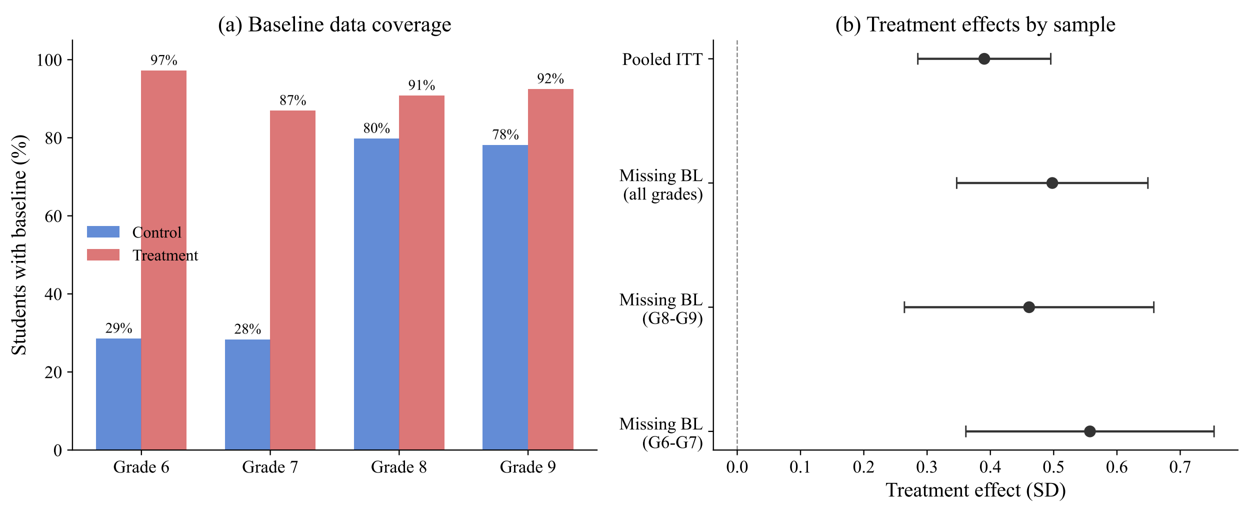

FIGURE D3. BASELINE DATA COVERAGE AND TREATMENT EFFECT DECOMPOSITION

Notes: Panel (a) shows the percentage of students with a valid pre-PAL government baseline math score (AY 2022–23 SA1 for grades 7–9 at endline; AY 2023–24 SA1 for grade 6) by treatment status and endline grade. The grade 7 imbalance is a data-extraction artifact (treatment and control data delivered in separate extracts without overlap); the grade 6 imbalance is structural (new-cohort students' pre-study schools are not in either extract). Panel (b) shows treatment effects for the pooled sample and for missing-baseline students overall, in grades 8–9 (balanced coverage), and in grades 6–7 (asymmetric coverage). Whiskers show 95% confidence intervals.

TABLE D1—GENDER DIAGNOSIS: WHY DO GIRLS GAIN MORE FROM PAL?

	Boys	Girls	Difference	SE
Panel A: Ruling out alternative explanations				
Baseline ability (observed, within school×grade pctile)				
Control group	0.452	0.532		
Treatment group	0.459	0.505		
Gender TE gap	T×F interaction			
Without baseline control	$+0.170^{***}$			(0.049)
With baseline control	$+0.172^{***}$			(0.047)
Attendance (days present, last 2 weeks)				
Control group	10.2	10.4		
Treatment group	10.5	10.6		
Treatment × Female			-0.180^{**}	(0.080)
School composition (T×F×Moderator)				
Enrollment (G6–9)	$+0.079^{*}$			(0.046)
Urban	+0.015			(0.136)
% female in school	+0.009			(0.063)
Tablets/student	+0.043			(0.047)
	Boys	Girls	Difference	SE
Panel B: Usage and engagement (treatment group)				
PAL usage (hours)				
Total hours (2 years)	33.3	35.2	$+2.0^{***}$	(0.7)
Year 1 hours	14.3	15.1	$+1.2^{***}$	(0.5)
Year 2 hours	19.0	20.1	$+0.7^{*}$	(0.4)
Video hours	18.9	18.9	-0.1	(0.4)
Question hours	14.3	16.3	$+2.0^{***}$	(0.4)
Engagement (higher = better)				
PAL enjoyment	3.74	3.79	$+0.05^{***}$	(0.02)
Math enjoyment	3.65	3.71	$+0.07^{***}$	(0.02)
Better understanding	3.66	3.72	$+0.07^{***}$	(0.02)
Own credentials	3.63	3.63	+0.01	(0.02)
Teacher help	3.04	2.97	-0.04^{*}	(0.03)
Video feedback	2.85	2.88	$+0.03^{**}$	(0.01)
PAL classes/week	2.59	2.51	-0.09^{*}	(0.05)
Challenges (total)	7.32	7.91	$+0.58^{***}$	(0.09)
	Boys TE	Girls TE	Interaction	SE
Panel C: Topic-specific treatment effects (standardized)				
Algebra	0.291	0.425	$+0.160^{***}$	(0.047)
Number systems	0.309	0.434	$+0.159^{***}$	(0.047)
Geometry	0.314	0.385	$+0.101^{**}$	(0.046)
Measurement	0.341	0.438	$+0.125^{**}$	(0.049)
Statistics	0.263	0.375	$+0.145^{***}$	(0.050)

Notes: Exploratory analysis investigating mechanisms behind the gender gap in treatment effects (Table 4, Panel A). Panel A tests whether baseline ability, attendance, or school characteristics explain the gap. Panel B compares PAL usage and engagement by gender within the treatment group. Panel C tests whether the gender advantage is concentrated in specific topics. All regressions include strata FE; standard errors clustered at school level. No multiple testing correction. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE D2—HETEROGENEOUS TREATMENT EFFECTS BY SES TERCILE

	Low SES (1)	Middle SES (2)	High SES (3)
Treatment	0.350^{***}	0.422^{***}	0.381^{***}
	(0.059)	(0.066)	(0.060)
Observations	4,746	5,182	4,287
p (equality)		0.619	

Notes: Each column estimates the ITT within a SES tercile. SES index is PCA PC1 of 9 asset indicators. All regressions include controls, strata FE, grade FE, and school-clustered SE. p (equality) tests that treatment effects are equal across terciles. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Baseline coverage by treatment status and grade

TABLE D3—BASELINE DATA AVAILABILITY: DIFFERENTIAL COVERAGE BY TREATMENT STATUS

	Control		Treatment	
	Has baseline	%	Has baseline	%
Grade 6	506/1,773	28.5%	1,497/1,540	97.2%
Grade 7	497/1,755	28.3%	1,390/1,599	86.9%
Grade 8	1,479/1,854	79.8%	1,606/1,769	90.8%
Grade 9	1,634/2,091	78.1%	1,696/1,834	92.5%

Panel B: Linear probability model — predictors of missing baseline

	Coefficient	(SE)
Treatment	-0.1243^{***}	(0.0264)
Grades 6–7	0.5462^{***}	(0.0428)
Treatment × Grades 6–7	-0.5054^{***}	(0.0470)
Has primary section (Cat. 6)	-0.1290^{***}	(0.0243)
Female	-0.0095	(0.0100)
SES index	0.0049	(0.0046)
Age	0.0204^{***}	(0.0034)
Observations	14,215	
R^2	0.376	
Strata FE	Yes	

Notes: Panel A shows the number and percentage of students with 2022–23 government baseline math scores (SA1) by treatment status and endline grade. The differential coverage for grades 6–7 reflects a data extraction artifact: the government provided broader baseline data for treatment-eligible schools than for control schools (see text). Panel B reports coefficients from a linear probability model with missing baseline as the dependent variable. Standard errors clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE D4—TREATMENT EFFECTS BY BASELINE DATA AVAILABILITY AND GRADE

	(1) Full sample	(2) Has baseline	(3) G8–G9 only	(4) G8–G9 has baseline	(5) Missing BL G8–G9	(6) Missing BL G6–G7
Treatment	0.390***	0.361***	0.336***	0.313***	0.461***	0.557***
	(0.054)	(0.059)	(0.059)	(0.062)	(0.101)	(0.100)
Control mean	0.000	0.126	0.222	0.242	0.146	-0.254
N	14,215	10,305	7,548	6,415	1,133	2,777

Notes: Each column reports the ITT treatment effect from OLS of standardized math scores on treatment assignment with controls (SES, parental education, age, gender, missing flags), strata fixed effects, and grade fixed effects appropriate to the sample. Columns (1)–(2) include all grades with full and baseline-restricted samples. Columns (3)–(4) restrict to grades 8–9, where baseline coverage is balanced across arms (∼80–93% in both treatment and control). Columns (5)–(6) isolate students missing baseline data: column (5) in grades 8–9 (balanced missingness), column (6) in grades 6–7 (where control students are disproportionately missing due to the data extraction artifact). Standard errors clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

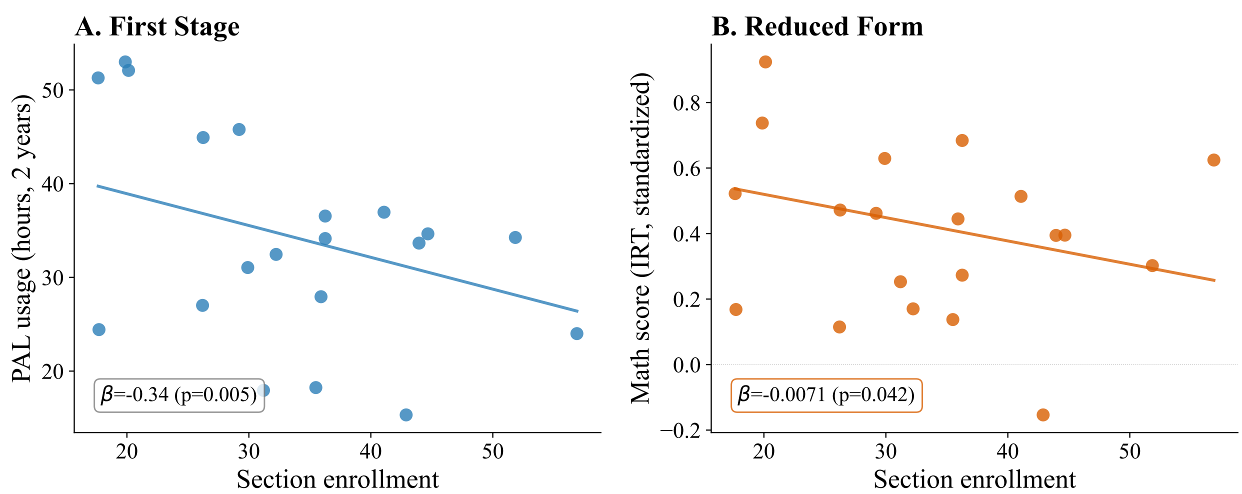

FIGURE E1. IV DIAGNOSTICS: FIRST STAGE AND REDUCED FORM

Notes: Treatment students only. *Panel A*: bin means of PAL usage hours by section enrollment, with a fitted regression line conditional on controls, grade FE, and strata FE (school-clustered SE). Each additional student in a section reduces PAL usage by 0.34 hours ($p = 0.005$). *Panel B*: bin means of endline math scores by section enrollment, same covariate structure. Each additional student reduces scores by 0.007 SD ($p = 0.042$). Bin means are formed within grade.

E. Dose-Response: IV Diagnostics, Shape Robustness, and Subgroup IV

This appendix supports the IV identification behind Table 5 and Figure 8: the first-stage and reduced-form scatters, the shape robustness of the dose-response curve under alternative fixed effects, and the subgroup IV replications by grade and gender. The method behind the program-scale extrapolation table in Section VII is documented as the final subsection.

First Stage and Reduced Form

Figure E1 visualises the first-stage and reduced-form relationships that identify the IV in Table 5. Panel A shows that each additional student in a section reduces PAL usage by 0.34 hours over two years ($p = 0.005$). Panel B shows that each additional student reduces endline math scores by 0.007 SD ($p = 0.04$). The ratio of these slopes delivers the 2SLS estimate reported in Table 5.

Shape of the Dose-Response

Figure 8 in the main text characterises the within-school shape of the dose-response. Table E1 supports that figure by reporting the pooled quadratic and linear-spline specifications under two alternative sets of fixed effects. The quadratic coefficient on hours^2 is negative across all six specifications; under 21 mandal-cluster strata FE the coefficient is significant at the 5% level ($p = 0.021$), and under 60 school FE (within-school identification only) precision tightens to $p = 0.002$. The linear spline's joint Wald test rejects linearity under school FE ($p = 0.012$). The shape finding — steep learning between 10 and 30 hours, flat above — is robust to including the one-year G6 cohort and to retaining students with 100+ hours of usage.

Program-School Effect Extrapolation Method

This subsection describes the method underlying Table 8 in Section VII. Let $Y(H)$ denote endline IRT math score as a function of 2-year PAL usage hours H, as estimated within the RCT treatment sample. The within-treatment dose-response in Table E1 (column 6, school FE) estimates a piecewise-linear $Y(\cdot)$ with

TABLE E1—SHAPE OF THE POOLED DOSE-RESPONSE

	Strata FE			School FE		
	Linear (1)	Quadratic (2)	Spline (3)	Linear (4)	Quadratic (5)	Spline (6)
Usage hours	0.0091***	0.0128***	-0.0023	0.0068***	0.0101***	-0.0070
	(0.0018)	(0.0029)	(0.0078)	(0.0012)	(0.0018)	(0.0076)
Usage hours^2 ($\times 10^3$)		-0.036**			-0.033***	
		(0.016)			(0.010)	
Slope change at 10 hrs			+0.0235**			+0.0263**
			(0.0115)			(0.0118)
Slope change at 20 hrs			-0.0061			-0.0063
			(0.0111)			(0.0109)
Slope change at 30 hrs			-0.0082			-0.0083
			(0.0074)			(0.0058)
Spline joint Wald p			0.086			0.012
Strata FE (21)	Y	Y	Y			
School FE (60)				Y	Y	Y
Controls + grade FE	Y	Y	Y	Y	Y	Y
Observations	6,742	6,742	6,742	6,742	6,742	6,742

Notes: Treatment-group students only. Dependent variable is endline IRT math score standardised to the control mean and SD. Columns (1)–(3) use the 21 mandal-cluster strata FE; columns (4)–(6) replace strata FE with 60 school FE, absorbing between-school composition. The quadratic coefficient on hours^2 is negative across all six specifications, indicating concavity in the pooled within-treatment dose-response. Precision tightens substantially under school FE. Spline knots at 10, 20, and 30 hours; rows report the change in slope at each knot. Standard errors clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

knots at 10, 20, and 30 hours. The predicted treatment effect for a program-school student with usage H_i is $\tau_i = Y(H_i) - Y(0)$, and the population-integrated predicted ATE at program scale is $E[\tau_i]$, averaged across the 245,690 program students with any recorded 2-year usage in AY 2023–25. The Monte Carlo 95% CI in Table 8 is constructed by drawing the four spline coefficients from their joint sampling distribution. Finer 5-hour knots produce more negative point estimates (-0.09 SD) because the finer parameterisation captures a sharply negative slope in the 0–5 hour region that the 3-knot specification smooths into the 0–10 average.

Subgroup Replications

Table E2 replicates the IV analysis from Table 5 separately by grade and by gender. Per-hour estimates are broadly consistent across subgroups (0.013–0.029 SD per hour), though statistical power is limited within any single subgroup. Girls accumulate approximately 2 more hours of PAL usage than boys over two years; per-hour returns are similar.

TABLE E2—IV ESTIMATES BY GRADE AND GENDER

	By Grade					By Gender	
	Grade 6	Grade 7	Grade 8	Grade 9	Pooled	Boys	Girls
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: First Stage (Dep. var.: PAL usage hours)							
Treatment × Section enrollment	-0.359***	-0.518**	-0.581***	-0.405***	-0.370***	-0.411***	-0.315***
	(0.106)	(0.247)	(0.158)	(0.135)	(0.110)	(0.133)	(0.101)
First-stage *F*	11.5	4.4	13.5	9.0	11.2	9.5	9.7
Panel B: Reduced Form (Dep. var.: Math score)							
Treatment × Section enrollment	-0.008	-0.006	-0.011**	-0.009**	-0.010***	-0.012***	-0.008*
	(0.006)	(0.007)	(0.006)	(0.005)	(0.004)	(0.004)	(0.004)
Panel C: IV Estimates							
IV (2SLS) estimate	0.0211	0.0125	0.0188	0.0230	0.0280	0.0293	0.0240
OLS estimate	0.016***	0.010***	0.006***	0.011***	0.009***	0.009***	0.009***
	(0.006)	(0.002)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
Panel D: Exclusion Restriction (Control group only)							
Section enrollment	-0.000	0.006**	-0.002	-0.000	0.001	0.005*	-0.003
	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grade FE	—	—	—	—	Yes	Yes	Yes
Mean usage, T group (hrs)					34.4	33.6	35.4
Observations	3,282	3,328	3,607	3,906	14,123	7,232	6,842

Notes: Each column replicates the IV analysis from Table 5 within a single grade (columns 1–4), pooled (column 5), or by gender (columns 6–7). The instrument is treatment × section enrollment: treatment schools received 30 tablets regardless of enrollment, so larger sections share more students per tablet. Panel A: first stage (section enrollment reduces PAL usage hours). Panel B: reduced form (section enrollment reduces math scores in treatment schools). Panel C: 2SLS and OLS estimates of the return per PAL hour. The 2SLS estimate is stable across grades (0.013–0.023 SD/hour) and similar for boys and girls, indicating the grade gradient in treatment effects (Table 3) reflects differences in usage quantity, not in the per-hour productivity of PAL. The gender gap likewise reflects quantity: girls use PAL ≈2 hours more than boys, but the per-hour return is similar. Panel D: exclusion restriction — section enrollment should not affect control student scores. This holds in all grades except grade 7 ($p = 0.01$). Grade 7 also has a weak first stage ($F = 4.4$); the IV estimate for grade 7 should be interpreted with caution. Grade 6 students had 1 year of PAL; grades 7–9 had 2 years. All regressions include controls and strata FE; pooled and gender columns also include grade FE. Standard errors clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F. Government School Assessments: Validity and the Grade 10 Board

This appendix explains why govt FA/SA scores are useful only as within-school rank controls (not cross-school level measures), reports the negative Grade 10 board exam estimates on the externally graded cohort, and runs a displacement test that rules out PAL-driven exam-prep crowd-out as the mechanism.

Government-administered school assessments (FA and SA exams) are set and graded by individual schools, with teachers proctoring their own students. These assessments have well-documented limitations as measures of absolute learning: teacher-proctored school exams inflate scores by approximately 16–20 percentage points relative to independently monitored tests (Singh, 2024), and grading standards vary across schools, making scores non-comparable.

Figure F1 decomposes the rank and level structure of Year 2 SA2 scores against independently administered endline IRT scores, in the control group only. Two patterns emerge. First, within any given school, govt scores preserve student rank: the median within-school correlation is $+0.47$ and is positive in 98% of 60 control schools (Panel A, grey lines). Teachers rank their own students accurately. Second, across schools the signal disappears: the school-level correlation is -0.14 (Panel B), and the intra-class correlation of govt scores (0.21) is roughly three times that of the endline IRT (0.068). This excess between-school variance is visible in Panel A's marginal kdensities — school-mean govt scores (top, dashed) span nearly as widely as individual students, while school-mean endline scores (right, dashed) collapse to a tight band near zero. Most cross-school variation in govt scores reflects grading heterogeneity rather than ability. These patterns justify the use of govt scores only as within-school ranked baseline controls, and motivate the externally graded Grade 10 board exam as the cleaner cross-school measure.

Table F1 reports Grade 10 board exam effects for the cohort that had one year of PAL exposure in grade 9. Unlike the school-graded FA/SA assessments discussed in the preceding sub-appendix, the board exam is administered and graded externally, removing the grading heterogeneity that dominates cross-school variation in govt scores (Figure F1). Point estimates on the six subjects are all negative (-0.08 to -0.15 SD), but results are sensitive to the inclusion of strata fixed effects: no subject is significant at 5% without strata FE. The uniform direction across non-math subjects is puzzling given PAL is a math-only intervention; one possible explanation is differential exam preparation displacement during grade 9. We report these estimates because the external-grading structure makes them the cleanest cross-school government measure, while the specification sensitivity cautions against treating the negative direction as a settled finding.

To investigate whether the negative board exam effects reflect displacement of exam preparation by PAL usage, Table F2 decomposes the treatment effect by Year 1 PAL usage intensity. If PAL displaced exam preparation, the negative effect should be largest among high-usage students who spent the most time on tablets instead of studying. The evidence points in the opposite direction: the negative effect is concentrated entirely among low-usage students (-0.37 SD composite, $p < 0.001$), while high-usage students are statistically indistinguishable from control ($+0.01$ SD, $p = 0.93$). Within the treatment group, each additional hour of PAL predicts higher board scores across all six subjects ($+0.015$ SD/hour, $p < 0.001$), though this gradient likely reflects selection—higher-ability students engage more with the platform—rather than a causal benefit of usage. Splitting at median school enrollment, the negative effect appears in small schools (-0.14 SD) but not large schools (-0.00 SD), the opposite of the grade 6–9 endline pattern where small schools show the largest positive effects. Taken together, these results indicate that the negative board exam effect is not attributable to PAL usage displacing exam preparation time. The most parsimonious interpretation is that the effect reflects specification sensitivity—it is significant only with strata fixed effects and no student-level controls are available for this cohort—combined with possible school-level confounders that are absorbed differently across strata.

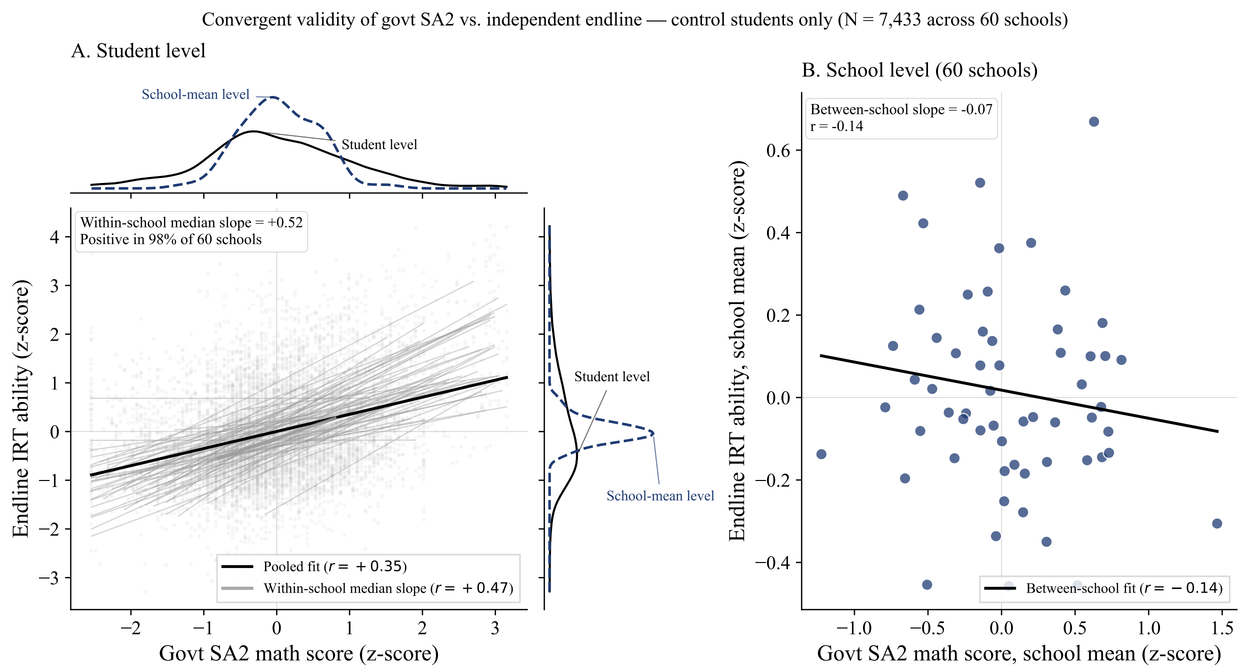

FIGURE F1. CONVERGENT VALIDITY OF GOVERNMENT SCHOOL ASSESSMENTS

Notes: Control students only ($N = 7{,}433$ across 60 schools). Outcome and regressor both z-scored to the control group's mean and SD. *Panel A*: student-level scatter of Year 2 govt SA2 math scores against independently administered endline IRT scores. Grey lines show OLS fits within each of the 60 control schools; the black line is the pooled student-level OLS fit. The grey fit at center of the panel marks the median within-school slope. Marginal kdensities on each axis compare the student-level distribution (solid black) with the school-mean distribution (dashed navy): schools differ moderately in mean govt score (top marginal) but barely in mean endline IRT (right marginal). *Panel B*: school-level means ($N = 60$) with OLS fit. Pooled student $r = +0.35$; within-school median $r = +0.47$ (positive in 98% of schools); between-school $r = -0.14$. The govt score's intra-class correlation (0.21) is $\approx 3\times$ that of the endline IRT (0.068); the excess between-school variance reflects grading heterogeneity rather than ability.

TABLE F1—GRADE 10 BOARD EXAM EFFECTS (1 YEAR PAL EXPOSURE)

	Math (1)	Science (2)	Social Sci. (3)	Lang 1 (4)	Lang 2 (5)	Lang 3 (6)
Panel A: With strata FE						
Treatment	-0.136**	-0.109*	-0.144**	-0.117*	-0.135*	-0.121*
	(0.067)	(0.065)	(0.067)	(0.070)	(0.071)	(0.070)
Panel B: Without strata FE						
Treatment	-0.138	-0.109	-0.152*	-0.139*	-0.121	-0.077
	(0.093)	(0.085)	(0.083)	(0.083)	(0.082)	(0.081)
Control mean (raw)	57.0	65.1	65.4	70.6	56.0	61.7
Observations	4,846	4,846	4,843	4,849	4,849	4,846

Notes: Grade 10 board exam scores for students who had PAL in grade 9 (AY 2023–24, 1 year exposure) and took the state board exam in 2024–25. Each column standardized to control mean and SD. Board exams are externally graded (state level). No student-level controls are available for this cohort (students not in endline sample). Standard errors clustered at the school level. Coverage: 98.4% treatment, 98.9% control. *Panel A* includes 21 strata fixed effects; *Panel B* shows results without strata FE. Point estimates are similar across panels (-0.08 to -0.15 SD), but standard errors are larger without strata FE (0.08–0.09 vs 0.07), and no subject is significant at 5% in Panel B. The negative coefficients across all six subjects suggest a possible effect, but interpretation requires caution: (i) this cohort had only 1 year of PAL exposure, (ii) the board exam was taken *after* students left treatment schools, (iii) the uniform direction across non-math subjects is puzzling given PAL is a math-only intervention, and (iv) results are sensitive to the inclusion of strata FE. One possible explanation is differential exam preparation: PAL may have displaced study time allocated to board exam review during grade 9. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE F2—GRADE 10 BOARD EXAM: INVESTIGATING THE ROLE OF PAL USAGE

	Math	Science	Social Sci.	Lang 1	Lang 2	Lang 3	Composite
Panel A: Treatment effect by Year 1 PAL usage (vs. control)							
Low usage (4 hrs, N_T=819)	-0.381***	-0.335***	-0.360***	-0.373***	-0.391***	-0.327***	-0.369***
	(0.080)	(0.086)	(0.080)	(0.085)	(0.095)	(0.088)	(0.076)
Mid usage (14 hrs, N_T=818)	-0.077	-0.061	-0.092	-0.069	-0.052	-0.024	-0.061
	(0.080)	(0.070)	(0.073)	(0.087)	(0.084)	(0.082)	(0.068)
High usage (27 hrs, N_T=819)	0.021	0.026	-0.023	0.077	-0.004	-0.059	0.006
	(0.076)	(0.079)	(0.085)	(0.079)	(0.083)	(0.085)	(0.069)
Panel B: Within-treatment usage gradient (OLS)							
PAL hours (Year 1)	0.0139***	0.0145***	0.0156***	0.0180***	0.0190***	0.0098***	0.0154***
	(0.0039)	(0.0035)	(0.0037)	(0.0034)	(0.0036)	(0.0027)	(0.0030)
Panel C: Treatment effect by school size							
Small schools (N_T=1,209, N_C=1,206)	-0.152						-0.138
	(0.099)						(0.086)
Large schools (N_T=1,208, N_C=1,223)	0.018						-0.003
	(0.102)						(0.106)

Notes: This table investigates whether the negative Grade 10 board exam effects (Table F1) are driven by students who used PAL intensively. *Panel A*: treatment students are split into Year 1 usage terciles and each tercile is compared to the full control group (separate regressions). The negative effect is concentrated among low-usage students (-0.37 SD composite, $p < 0.001$); high-usage students are indistinguishable from control ($+0.01$ SD, $p = 0.93$). *Panel B*: within the treatment group, each additional hour of PAL predicts *higher* board scores ($+0.015$ SD/hour, $p < 0.001$), across all six subjects including non-math. This gradient likely reflects selection (higher-ability students engage more) rather than a causal benefit of usage. *Panel C*: splitting at median enrollment, the negative effect is concentrated in small schools, the opposite of the endline pattern (Figure 7). Taken together, these results indicate that the negative board exam effect is not attributable to PAL usage displacing exam preparation. All regressions include strata FE and school-clustered SE. No student-level controls are available for this cohort. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G. Development of the Test Instrument

A. Item Development and Piloting

We developed the Math Master Challenge assessment to measure mathematics achievement across the wide range of ability levels in our sample. Items were drawn from three sources: (i) all questions from every chapter of the Andhra Pradesh state math textbooks for grades 2 through 9; (ii) grade-specific math items from the Young Lives Study (Barnett et al., 2013); and (iii) items from a previous evaluation of adaptive learning technology in India (Muralidharan and Ganimian, 2019). From this bank, we selected an initial set of items targeting 10 questions per grade for grades 2–7 and 6 questions per grade for grades 8–9, with two items per topic per grade across five broad mathematical domains: algebra, geometry, measurement, statistics, and number systems.

We piloted the instrument in five schools over the course of one week. Each day, we downloaded the pilot data and estimated a two-parameter logistic (2PL) IRT model, dropping items that exhibited poor discrimination (low slope parameters) or extreme difficulty, with the goal of constructing an instrument whose test information function was sufficiently wide to capture the range of learning levels in our sample.

B. Final Instrument

The final assessment comprises 72 unique multiple-choice items spanning the grade 2 through grade 9 curriculum. Each student answered items from grade 2 up to the content level of their enrolled grade, yielding between 50 items (grade 6 students) and 72 items (grade 9 students). The assessment was administered on tablets via SurveyCTO in both English and Telugu, with three randomized question orderings (set assignment uniform at $P = 0.33$ per student) to mitigate order and copying effects. Students completed the assessment in approximately 40–50 minutes on average.

Table G1 shows the distribution of items across content grades and topics. Items from grades 2–7 have 10 questions each, while grades 8–9 have 6 questions each. Number systems has the most items (23), reflecting its central role in the Indian mathematics curriculum from basic arithmetic through rational numbers.

TABLE G1—DISTRIBUTION OF ASSESSMENT ITEMS BY CONTENT GRADE AND TOPIC

Content grade	Algebra	Geometry	Measurement	Number Systems	Statistics	Total
Grade 2	1	2	1	5	1	10
Grade 3	2	2	2	3	1	10
Grade 4	2	2	2	3	1	10
Grade 5	2	2	2	3	1	10
Grade 6	0	2	2	4	2	10
Grade 7	1	2	2	3	2	10
Grade 8	2	1	1	1	1	6
Grade 9	2	1	1	1	1	6
Total	12	14	13	23	10	72

Notes: Topic classification follows the Indian NCERT curriculum convention, where Number Systems covers arithmetic on specific number types (whole numbers, fractions, decimals, integers) and Algebra begins with variables and unknowns. Students in grade k answer all items from grade 2 through grade k, yielding 50 items (grade 6) to 72 items (grade 9).

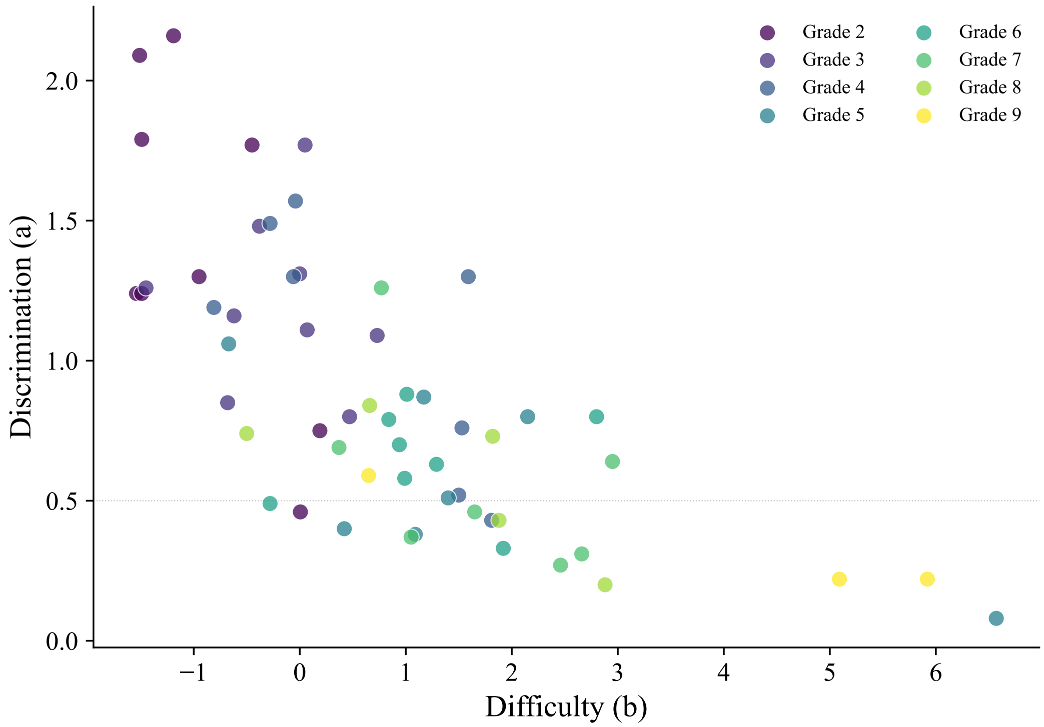

FIGURE G1. ITEM DISCRIMINATION AND DIFFICULTY PARAMETERS

Notes: Each point represents one of the 72 items on the Math Master Challenge, colored by the curriculum grade from which the item was drawn. Discrimination (a) reflects how well the item differentiates between students of different ability levels; difficulty (b) indicates the ability level at which a student has a 50% probability of answering correctly. Parameters estimated via a 2PL IRT model. The dashed line at $a = 0.5$ marks a common threshold for acceptable discrimination.

C. Psychometric Properties

We estimate a 2PL IRT model, which allows both discrimination (a_j) and difficulty (b_j) parameters to vary across items. Under this model, the probability that student i answers item j correctly is:

$$P(X_{ij} = 1 \mid \theta_i) = \frac{1}{1 + \exp\left(-a_j(\theta_i - b_j)\right)} \tag{4}$$

where θ_i is the latent ability of student i. We standardize the estimated ability parameters to the control group mean and standard deviation, following Muralidharan and Ganimian (2019) and Muralidharan and Singh (2025).

Across all 72 items, discrimination parameters range from 0.08 to 2.16 (mean = 0.88), and difficulty parameters range from -1.54 to 6.57 (mean = 0.84). Lower-grade items are easier and more discriminating (grade 2: mean $a = 1.42$, mean $b = -0.94$), while higher-grade items are harder with lower discrimination (grade 9: mean $a = 0.34$, mean $b = 3.89$), reflecting the intended difficulty progression across the curriculum. Figure G1 displays the joint distribution of discrimination and difficulty parameters, colored by content grade.

Internal consistency is high across all grade levels: Cronbach's α ranges from 0.90 (grade 6, 50 items) to 0.92 (grade 9, 72 items).

Figure G2 presents the test information function by enrolled grade. The instrument provides the most precise measurement in the ability range $\theta \in [-2, 1]$, with peak information near $\theta \approx -0.7$. The standard error of measurement remains below 0.35 across most of the ability distribution. Higher-grade students receive additional items that extend measurement precision into the upper ability range.

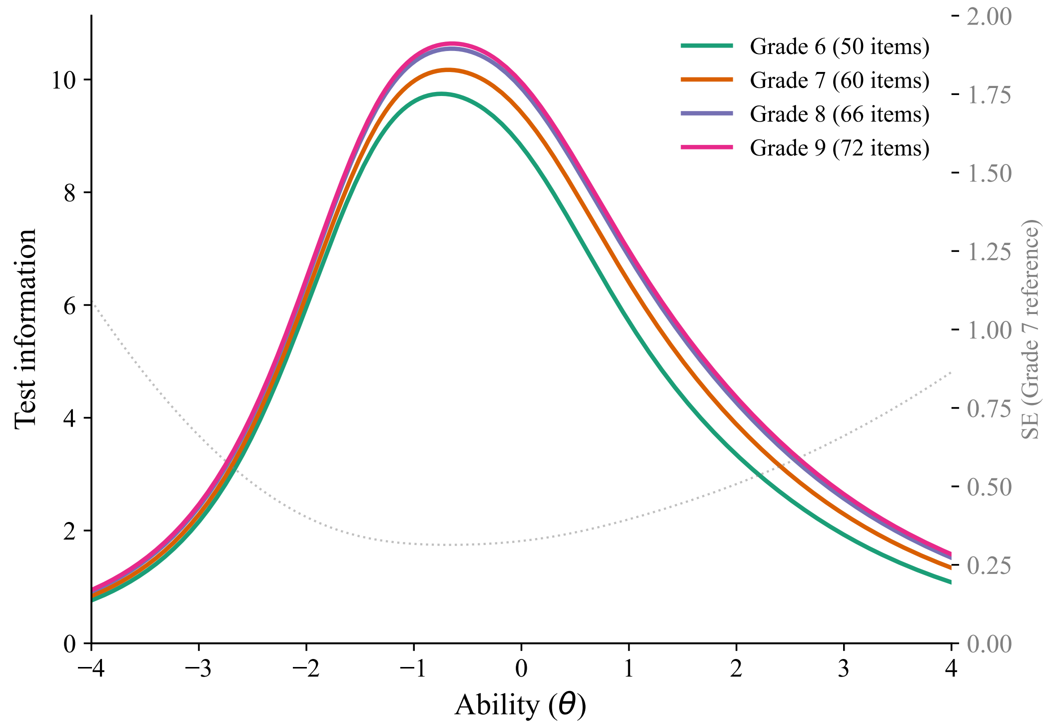

FIGURE G2. TEST INFORMATION FUNCTION BY GRADE

Notes: Test information $I(\theta) = \sum_j a_j^2 P_j(\theta)(1 - P_j(\theta))$ for each enrolled grade's item set. Higher information indicates more precise measurement at that ability level. The dashed gray line shows the corresponding standard error $\mathrm{SE}(\theta) = 1/\sqrt{I(\theta)}$ for grade 7 as a reference. Students in higher grades answer additional items from higher curriculum levels, extending the instrument's measurement range.

Supplementary Appendix

Additional tables and figures referenced from the main text and primary appendix.

TABLE H1—MATH SCORES BY CASTE CATEGORY (TREATMENT GROUP ONLY)

	SC	ST	BC-A/Other	BC-B/OBC
Mean IRT score (SD)	0.332	0.769	-0.047	0.430
	(1.066)	(1.211)	(1.059)	(1.150)
N	1,928	808	251	3,669
F-test (equality)		$F = 43.91$, $p = 0.000$		

Notes: Treatment group only. Caste information from government enrollment records, available for treatment students only. Scores are IRT math scores standardized to the overall control group mean and SD. Standard deviations in parentheses. SC = Scheduled Caste, ST = Scheduled Tribe, BC = Backward Class. F-test for equality of means across categories.

TABLE H2—HETEROGENEOUS TREATMENT EFFECTS BY URBAN/RURAL STATUS

	Rural (1)	Urban (2)
Treatment	0.456^{***}	0.520^{***}
	(0.057)	(0.151)
Observations	11,615	2,600
p (Rural = Urban)	0.075	

Notes: Each column estimates the ITT within rural or urban schools. p (Rural = Urban) tests equality of treatment effects from a pooled interaction model. All regressions include controls, strata FE, grade FE, and school-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H. Supplementary Results

A. Additional Heterogeneity

Tables H2 and H3 examine treatment effect heterogeneity by school urbanicity and infrastructure resources, respectively; effects are similar across rural and urban schools (Table H2) and across schools with above- versus below-median infrastructure resources (Table H3). Treatment-effect heterogeneity by caste cannot be estimated because caste records in government enrollment data are available for treatment students only; Table H1 therefore reports treatment-group mean math scores by caste category for descriptive context.

TABLE H3—TREATMENT EFFECT MODERATION BY SCHOOL RESOURCES

Moderator	Treatment × Moderator	SE	p
Enrollment (G6–9)	-0.134	(0.084)	0.109
Math teachers	-0.126	(0.078)	0.107
Total teachers	-0.143*	(0.077)	0.063
Classrooms	-0.043	(0.060)	0.474
Functional tablets	-0.066	(0.076)	0.380
Observations	14,215		

Notes: Each row shows the interaction of treatment with a standardized school-level moderator (mean 0, SD 1) from a separate regression. All regressions include treatment, the moderator, controls, strata FE, grade FE, and school-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

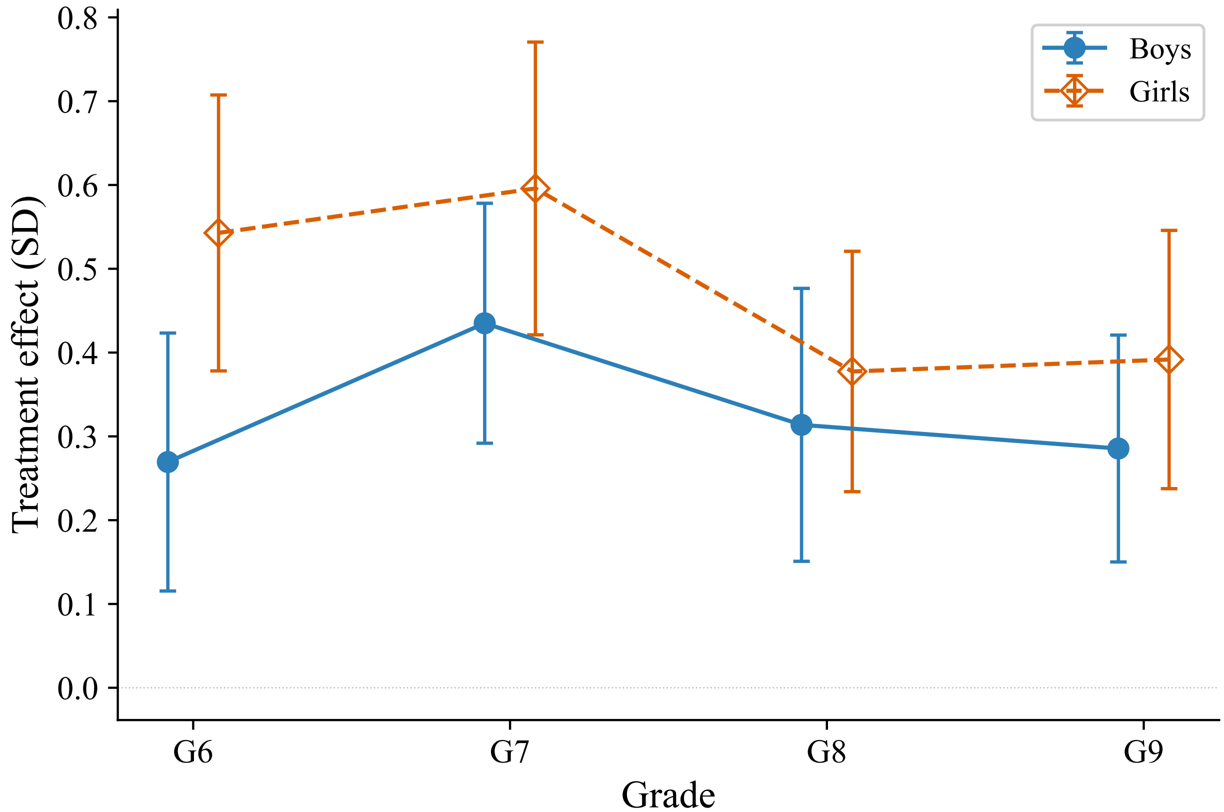

FIGURE H1. TREATMENT EFFECTS BY GENDER AND GRADE

Notes: ITT estimates of PAL treatment effects by gender and grade. Each point shows the treatment effect for a gender × grade subgroup, with 95% confidence intervals. Girls gain more than boys in every grade, with the largest gap at grade 6 (0.55 vs 0.27 SD). All regressions include demographic controls, strata FE, and school-clustered SE.

B. Gender Diagnosis

Figure H1 shows that girls gain more than boys in every grade, with the largest gap at grade 6. Figure H2 presents a comprehensive diagnostic of the gender gap in treatment effects, following a three-step narrative.

Step 1: Rule out. We test whether the gender gap can be explained by pre-existing differences. Baseline ability is balanced across gender within treatment status (Panel C). Attendance differences are small and work against girls, not in their favor (Panel B). School composition—enrollment, urbanicity, percent female, tablets per student—does not moderate the gender gap (Panel F). None of these channels account for the differential treatment effect.

Step 2: Identify. Girls spend approximately 2 more hours on PAL than boys, driven entirely by more time answering questions—active practice—rather than watching videos (Panel A). Girls also report higher enjoyment of both PAL and math more broadly (Panel D). The gender-stratified IV analysis (Table E2, columns 6–7) confirms that boys and girls have similar per-hour returns (≈0.009 SD/hour), indicating the gap reflects differences in usage quantity rather than per-hour productivity.

Step 3: Assess breadth. The gender advantage is broad-based across all five math topics rather than concentrated in specific areas (Panel E), consistent with a general engagement mechanism rather than

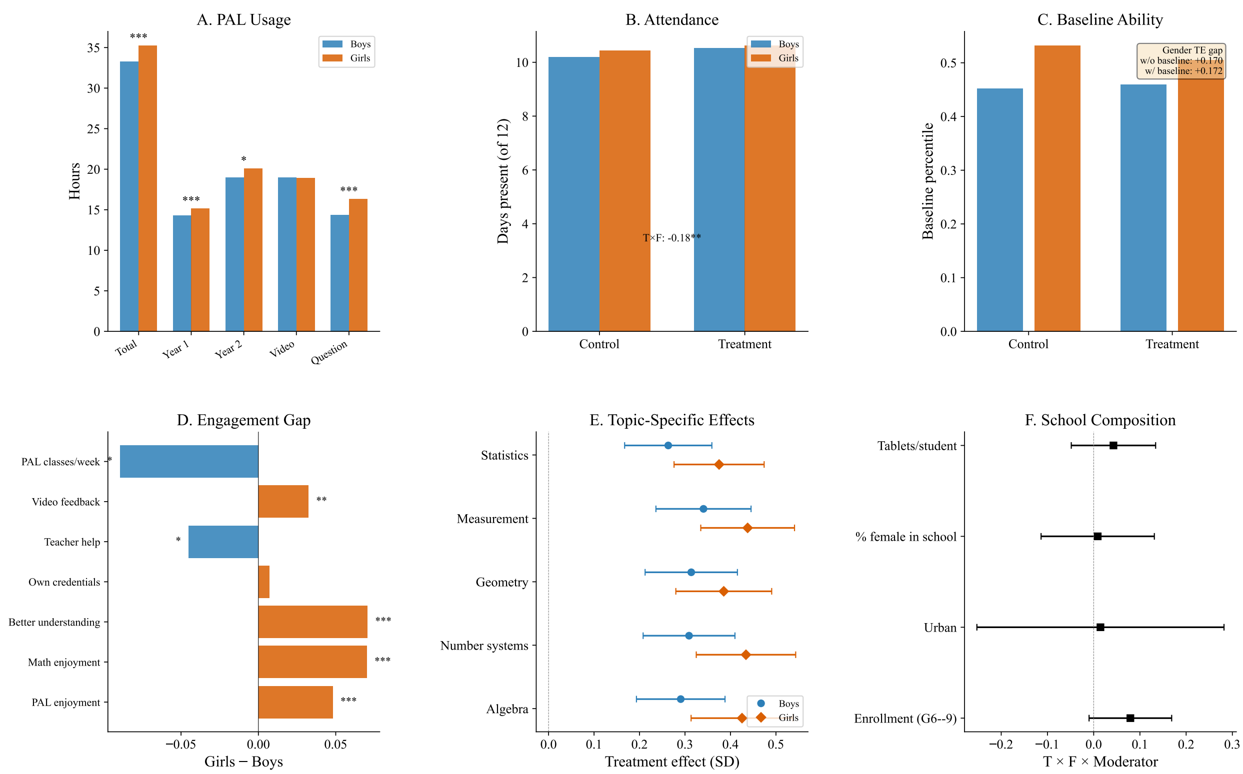

FIGURE H2. GENDER DIAGNOSIS: FULL MECHANISMS PANEL

Notes: Panel A: PAL usage by gender (treatment group). *Panel B*: Attendance by gender and treatment status. *Panel C*: Baseline ability by gender and treatment status. *Panel D*: Engagement gap (girls − boys, treatment group). *Panel E*: Topic-specific treatment effects by gender with 95% CIs. *Panel F*: School composition moderators (T×F×Moderator). All regressions include strata FE and school-clustered SE.

topic-specific content effects.

See Table D1 for the full regression results underlying each panel.

TABLE H4—PREDICTORS OF PAL USAGE (TREATMENT GROUP ONLY)

	Bivariate (1)	Joint (2)
Female	1.95***	1.27*
	(0.74)	(0.76)
Age	2.81***	3.03***
	(0.43)	(0.43)
SES index	-0.28	-0.30
	(0.28)	(0.28)
Baseline ability	1.70	1.35
	(1.42)	(1.39)
Enrollment (G6–9)	-0.06***	-0.07***
	(0.01)	(0.01)
Functional tablets	-0.08***	0.00
	(0.02)	(0.02)
R^2		0.273
Observations		6,742

Notes: Dependent variable: total PAL usage hours (2 years). Treatment group only. Column (1): bivariate regressions (one per row). Column (2): joint regression with all predictors. All regressions include strata FE and school-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C. *Descriptive Usage Statistics*

Baseline ability and age are the strongest predictors of PAL usage, while SES is not ($p = 0.32$; Table H4). Algebra accounts for the largest share of usage across all grades, consistent with the assessment blueprint weighting (Figure H3).

TABLE H5—PAL DOSAGE: TARGET VS. ACTUAL USAGE

	Year 1 (AY 2023–24)	Year 2 (AY 2024–25)	Total (2-year students)
School calendar			
Calendar weeks	31	33	64
Non-instructional weeks	9	12	21
PAL-eligible weeks	22	21	43
Target usage (hours)			
Scheduled (80 min/week)	29.3	28.0	57.3
Effective (60 min/week)	22.0	21.0	43.0
Actual usage (hours, treatment group mean)			
All students	14.7	19.5	34.2
2-year students (G7–9)	19.0	19.3	38.2
% of effective target			
2-year students (G7–9)	86%	92%	89%

Notes: Each school is allocated one 80-minute PAL period per week. The effective target of 60 minutes accounts for student transit to the computer lab and device setup time. Non-instructional weeks include holidays, formative assessment (FA) weeks, summative assessment (SA) weeks, and pre-exam revision weeks, during which PAL sessions are suspended. Year 1 ran late September 2023–April 2024; lab setup was staggered, with 49 schools starting in September and 11 in October. Year 2 ran July 2024–February 2025 (all 60 schools; capped at endline assessment start). Two-year students are grades 7–9 at endline, who had PAL in both academic years.

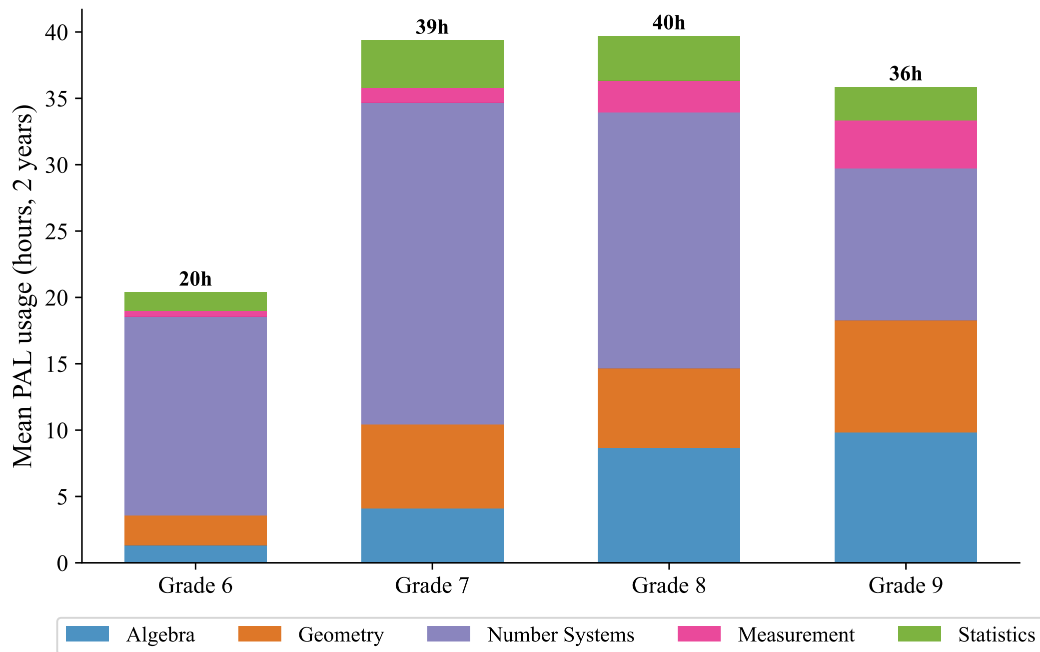

FIGURE H3. PAL USAGE BY MATH TOPIC AND GRADE

Notes: Mean PAL usage hours (cumulative over 2 academic years) by math topic and student grade. Treatment group only. Grade 6 had 1 year of PAL exposure; grades 7–9 had 2 years. Algebra accounts for the largest share of usage across all grades, consistent with the assessment blueprint weighting.

TABLE H6—TREATMENT EFFECTS ON SECONDARY OUTCOMES

	Treatment effect	SE	Control mean
Panel A: Subject enjoyment (standardized)			
Math	0.190^{***}	(0.025)	1.44
English	0.131^{***}	(0.026)	1.40
Telugu	0.105^{***}	(0.032)	1.34
Hindi	0.095	(0.065)	1.63
Science	0.144^{***}	(0.029)	1.45
Social science	0.140^{***}	(0.035)	1.53
Panel B: Attendance			
Days present (SD)	0.104^{***}	(0.035)	10.31
Days absent	-0.22^{***}	(0.07)	1.7
Observations		14,215	

Notes: Panel A: enjoyment rating for each subject, reverse-coded and standardized to control mean and SD (original scale: 1=A lot, 2=Somewhat, 3=Not at all, 4=Did not have; positive coefficients = treatment students enjoy the subject more). Strong ceiling effects — nearly all students report high enjoyment regardless of treatment. Panel B: days present out of 12 school days in last 2 weeks, self-reported by students during endline assessment ("In the last 2 weeks, how many days were you absent from school, excluding Sundays and holidays?"). Covers a 2-week window, not the full academic year. Days absent = $12 -$ days present (raw count). All regressions include controls, strata FE, grade FE, and school-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D. Secondary Outcomes and Item-Level Texture

PAL increases math enjoyment by 0.19 SD, with positive spillovers to other subjects (0.10–0.15 SD; Table H6, Panel A). Treatment students also report 0.22 fewer days absent over the last two weeks ($p = 0.003$; Panel B), though strong ceiling effects in enjoyment and the self-reported nature of attendance warrant caution in interpretation.

Treatment effects are remarkably uniform across all five math topics (0.31–0.38 SD; Table H7; Figure H4), indicating that PAL improves foundational math skills broadly rather than producing topic-specific gains. This uniformity is consistent with PAL's curriculum-aligned design, which covers all topics proportionally (Figure H3).

We examine treatment effects at the individual item level. Treatment effects are slightly larger on harder items (slope $= 0.10$; Figure H5), suggesting PAL helps students reach questions they would otherwise get wrong rather than reinforcing already-known material. Treatment students outperform control students at every content grade level, with the largest gains on grade 4–6 content (7–10 pp; Figure H6). Both groups show the expected decline in accuracy as content difficulty increases.

TABLE H7—TREATMENT EFFECTS BY MATH TOPIC

	Pooled (1)	Grade 6 (2)	Grade 7 (3)	Grade 8 (4)	Grade 9 (5)
Algebra	0.352***	0.345***	0.458***	0.327***	0.294***
	(0.049)	(0.067)	(0.070)	(0.059)	(0.054)
Geometry	0.346***	0.353***	0.441***	0.317***	0.286***
	(0.049)	(0.068)	(0.067)	(0.063)	(0.056)
Number systems	0.366***	0.397***	0.471***	0.320***	0.294***
	(0.050)	(0.069)	(0.065)	(0.061)	(0.058)
Measurement	0.383***	0.393***	0.493***	0.325***	0.322***
	(0.048)	(0.075)	(0.063)	(0.061)	(0.052)
Statistics	0.314***	0.417***	0.451***	0.245***	0.188***
	(0.045)	(0.070)	(0.066)	(0.056)	(0.053)
Controls	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Grade FE	Yes				
Observations	14,215	3,313	3,354	3,623	3,925

Notes: Each row shows the ITT estimate for a different math topic. Topic scores are constructed from IRT subscores on items mapped to each topic, standardized to control mean and SD. Topics are not equally weighted in the overall score — algebra has 2–3× more items than statistics. All regressions include controls, strata FE (and grade FE for pooled). School-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

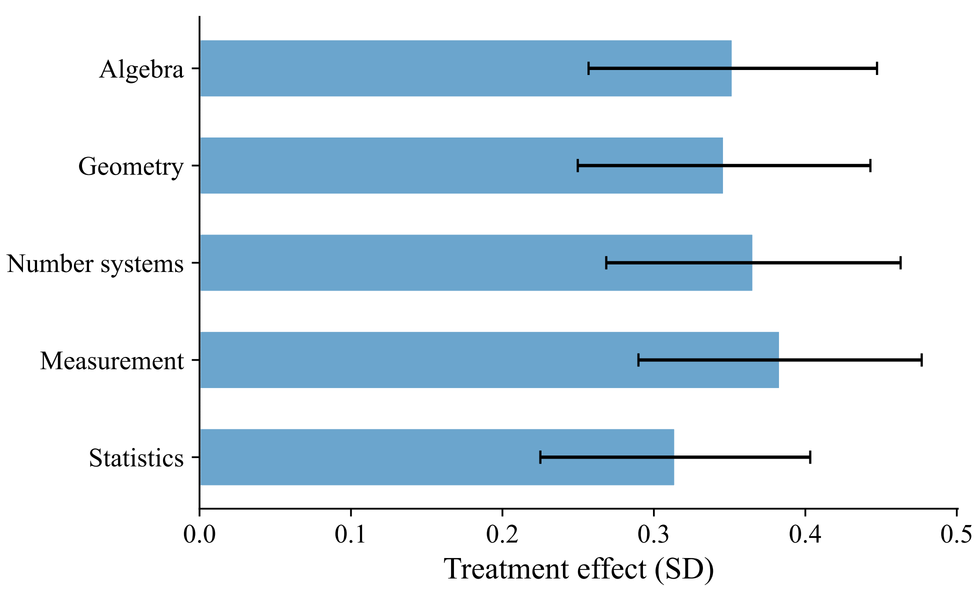

FIGURE H4. TREATMENT EFFECTS BY MATH TOPIC

Notes: Pooled ITT estimates for each of 5 math topics. Bars show point estimates; whiskers show 95% confidence intervals. Effects are remarkably uniform across topics (0.31–0.38 SD), indicating PAL improves foundational math skills broadly rather than topic-specifically. All regressions include controls, strata FE, grade FE, and school-clustered SE.

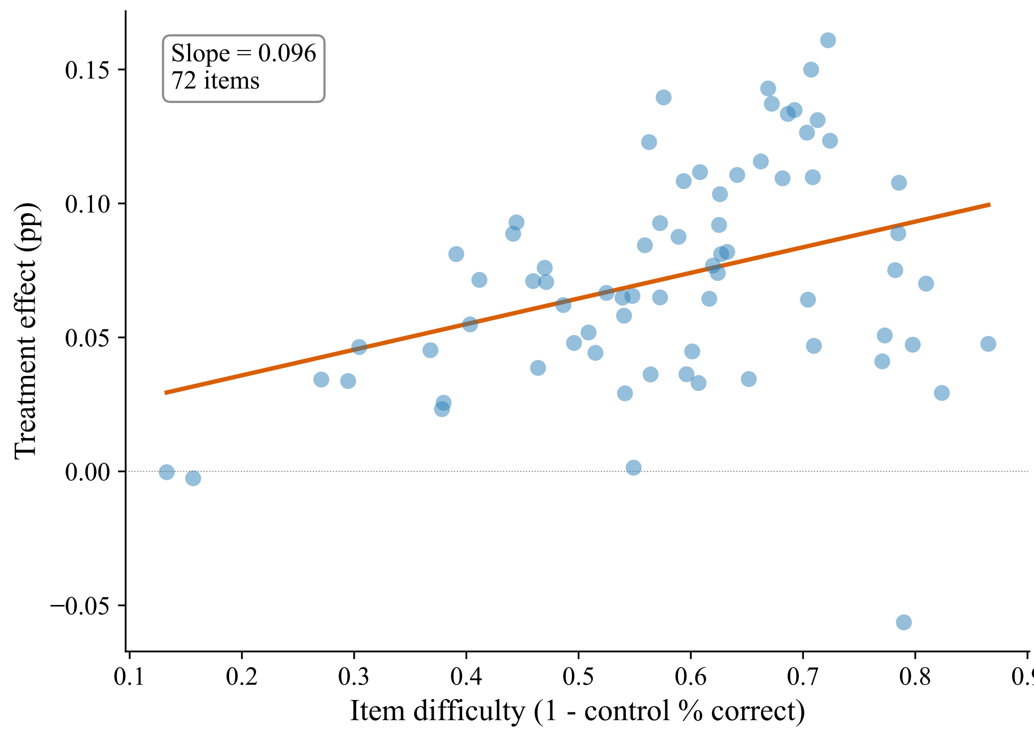

FIGURE H5. ITEM DIFFICULTY AND TREATMENT EFFECTS

Notes: Each dot represents one of 72 endline assessment items. The *x*-axis shows item difficulty (1 minus control group percent correct); the *y*-axis shows the treatment effect on that item (percentage point difference, T−C). The fitted line shows the relationship between difficulty and treatment effect magnitude. All regressions include strata FE and school-clustered SE.

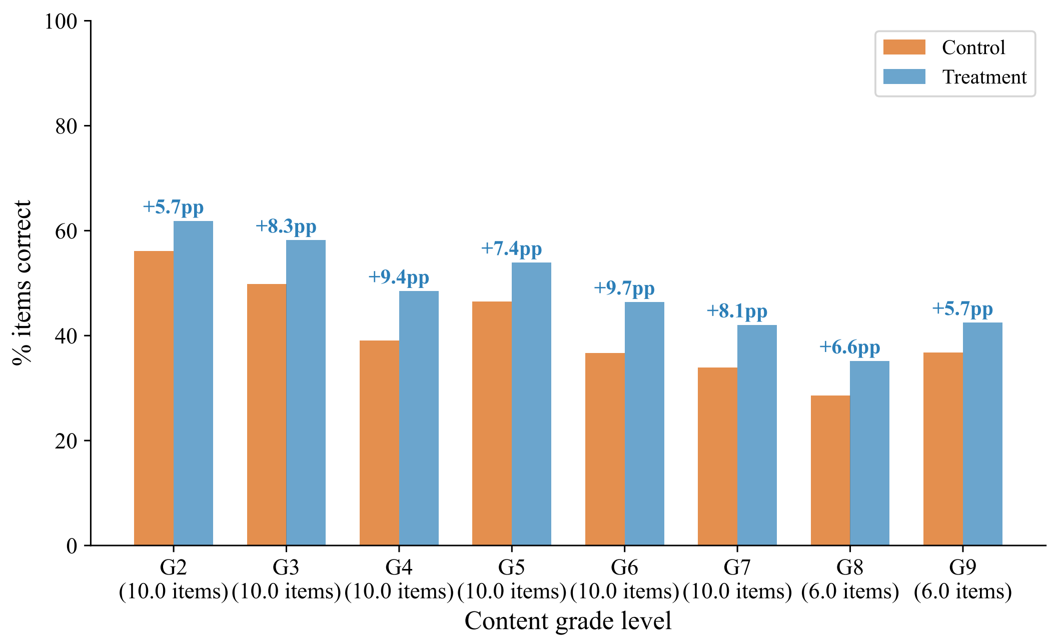

FIGURE H6. CONTENT MASTERY BY GRADE LEVEL: TREATMENT VS. CONTROL

Notes: Percent of items answered correctly by content grade level (grade 2 = easiest, grade 9 = hardest). Items are organized by the grade-level curriculum they test. Treatment students outperform control students at every content grade level, with the largest gaps on grade 4–6 content (7–10 pp). Both groups show the expected decline in accuracy as content difficulty increases.

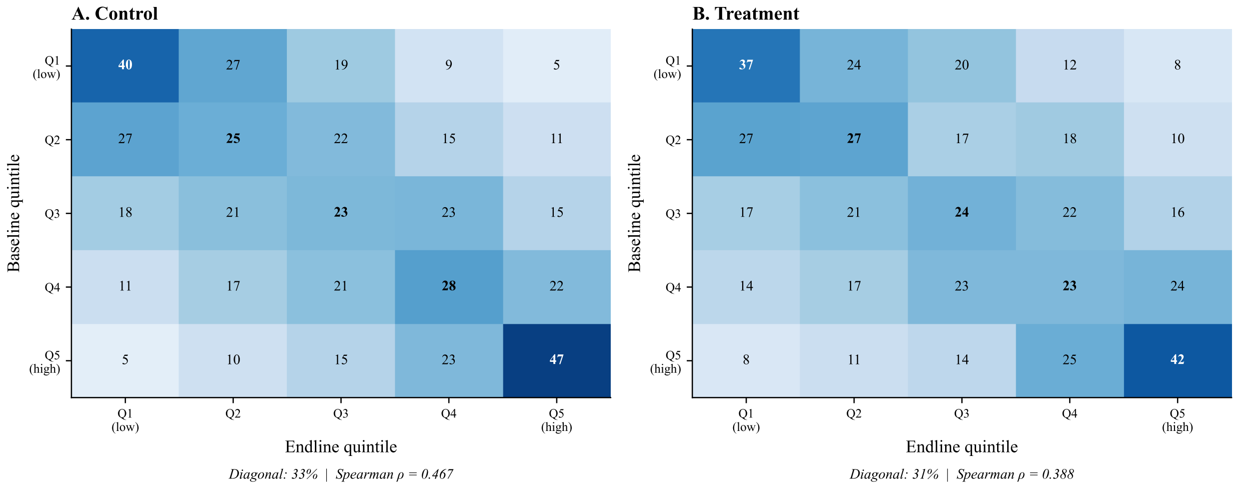

FIGURE H7. RANK INVARIANCE CHECK: BASELINE TO ENDLINE TRANSITION MATRICES

Notes: Each cell shows the percentage of students in a baseline quintile (row) who end up in a given endline quintile (column). Bold diagonal entries show the percentage staying in the same quintile. If rank invariance held perfectly, all mass would be on the diagonal. Treatment students are more reshuffled (diagonal average 31%) than control (33%), with a lower Spearman rank correlation (0.36 vs 0.47). This violation of rank invariance motivates the conditional quantile analysis in Figure H8. Sample restricted to students with non-missing baseline scores (72% of sample).

E. Rank Invariance and Conditional Quantile Effects

Treatment students are more reshuffled across the ability distribution than control students (Figure H7), violating rank invariance and motivating the conditional quantile analysis. Conditioning on baseline quintile, treatment effects increase from the 10th to 75th percentile of the endline distribution within every quintile, then flatten at the 90th (Figure H8)—indicating that PAL creates within-group variance in addition to the across-quintile gradient documented in Table 4.

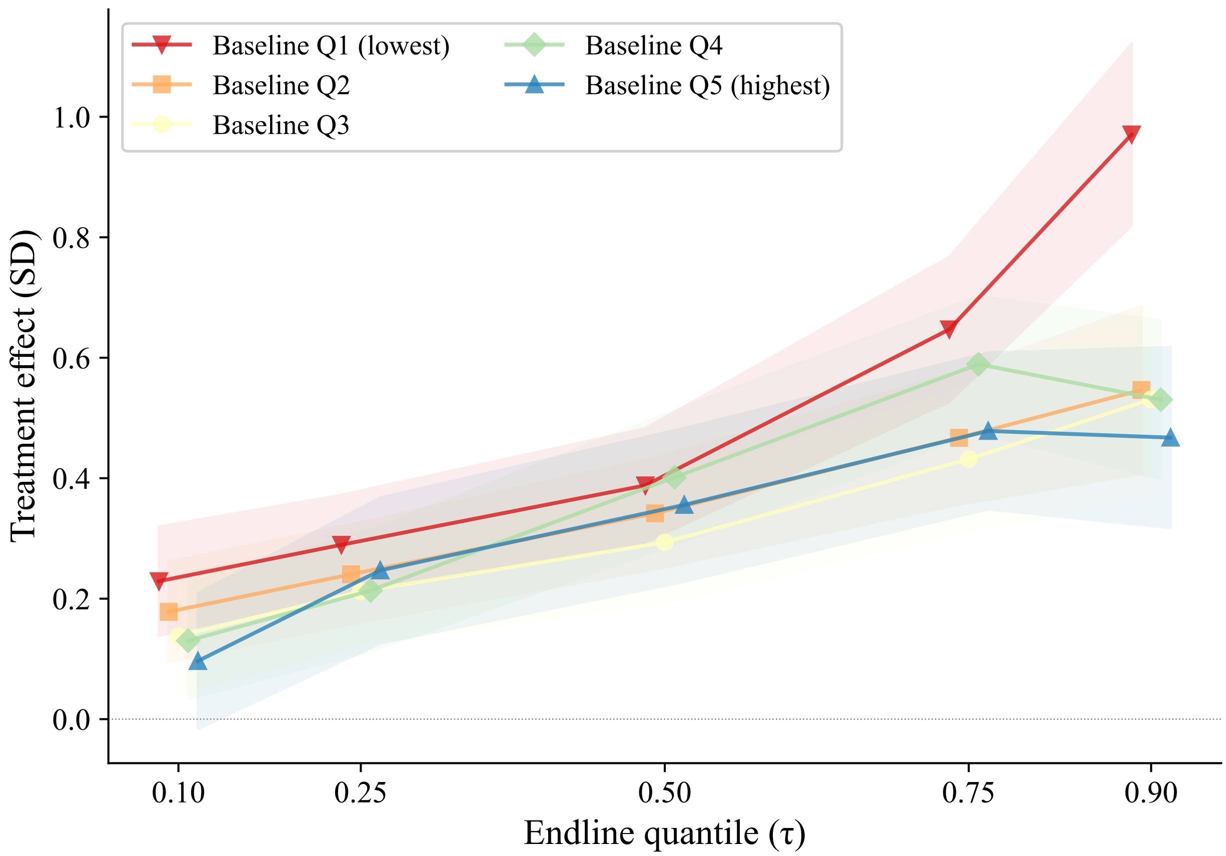

FIGURE H8. CONDITIONAL QUANTILE TREATMENT EFFECTS BY BASELINE ABILITY

Notes: Each line shows quantile treatment effects within a baseline ability quintile, addressing the rank-invariance violation in unconditional quantile regressions (Figure H7). Within every baseline quintile, effects increase from the 10th to the 75th percentile of the endline distribution, then flatten or decline at the 90th. Lines for lower-baseline quintiles sit above those for higher-baseline quintiles, consistent with the across-quintile gradient in Table 4. All regressions include controls, strata FE, and grade FE. Shaded bands show 95% confidence intervals.